Brain-age research summary: cohorts, experiments, the best model and what mattered

📄 **PDF version of this notebook** (figures, tables and formulas, no code; regenerate with `scripts/notebook_to_pdf.py notebooks/brain_age_research_summary.ipynb`).

One notebook for the whole EEG brain-age study on TUH (2026-10-05 → 10-16) and TUH + HEEDB (2026-11), in four phases:

- **Phase 1, TUAB only.** About 50 fine-tunes of LaBraM-base on the TUH Abnormal corpus (TUAB), aimed at memorization: schedule, regularizers, mixup/EMA, loss shape, codebook regularization, heads, training from scratch (`docs/age_overfitting_experiments.md`).
- **Phase 2, full TUH.** The same recipe on cohorts drawn from the whole TUH EEG corpus (TUEG, about 60k recordings), plus data, input and head changes (`docs/data/tuh_cohort_eval_v1.csv`).
- **Phase 3, cross-validation.** 5-fold patient-grouped CV of the three strongest phase-2 configurations (`notebooks/tuh_cv_summary.ipynb`, `docs/data/cv5_eval_v2.csv`).
- **Phase 4, TUH + Harvard EEG Database (2026-11-02 → 11-06).** The phase-2 recipe trained on HEEDB alone and on TUH + HEEDB (about 85K training recordings), with and without clinical inputs, and a longer schedule (Tables 2.11–2.12, Figure 2.2; `docs/heedb_onboarding_plan.md`). The epoch is selected on validation only.

It answers four questions:

1. Which datasets and cohorts were used, and how they differ (Section 1).
2. Which experiments ran and which came out near the best (Section 2).
3. What exactly the best experiment does, step by step (Section 3).
4. What changed relative to LaBraM as published (Section 4), and which of those changes measurably moved the error (Section 5).

Contents. 0 Metrics and evaluation sets · 1 Data · 2 Experiments · 3 The best experiment · 4 Changes versus the LaBraM recipe · 5 Which changes mattered · 6 Conclusions.

Inputs.

- In git: the registry, the two CSVs, `docs/data/line_removal_ab.json`, the configs, `docs/data/brain_age_clearml_runs.csv` (each run's ClearML experiment) and `scripts/submit_tuh_cohorts.sh`. Every phase-2 run's configuration is rebuilt from `labram/eval/research_summary.py::PHASE2` with the same config loader the trainer uses. `tests/test_research_summary.py` checks that catalog against the submit script.

- Optional, on the TUH data volume (`LABRAM_TUH_ROOT` , default `/data/datasets/EEG-public/TUH_EEG`):
  - the cohort manifests behind Section 1's counts;
  - the TUH recording index and NEDC's clinical sheet behind the TUEG specification (Tables 1.6–1.13);
  - the per-recording CV predictions behind the bootstraps (`LABRAM_CV_EVAL`).

  They stay out of git under the TUH data-use agreement. Without them those cells print a note and skip, and the outputs show only aggregate counts.

0. Metrics and evaluation sets

Every error is computed **per recording**. A recording's predicted age is the mean of its window predictions (Eq. (3.7)), in years. For a set S of recordings with true age y_i and predicted age $\hat{y}_i$:

$$\mathrm{MAE}(S) = \frac{1}{|S|}\sum_{i\in S}|\hat{y}_i - y_i| \tag{0.1}$$

$$R^2(S) = 1 - \frac{\sum_{i\in S}(\hat{y}_i - y_i)^2}{\sum_{i\in S}(y_i - \bar{y}_S)^2} \tag{0.2}$$

$$\mathrm{bias}(S) = \frac{1}{|S|}\sum_{i\in S}(\hat{y}_i - y_i) \tag{0.3}$$

where $\bar{y}_S$ is the mean true age of S (`labram/eval/cv_compare.py::metrics`). The effect of a change from a baseline run b to a variant run v is

$$\Delta\mathrm{MAE} = \mathrm{MAE}_v(S) - \mathrm{MAE}_b(S), \tag{0.4}$$

so **negative is better**. Section 5 defines when a Δ is larger than run-to-run noise.

All errors are before bias correction. Every MAE in this notebook scores the raw prediction $\hat{y}_i$. The exceptions are Table 3.5, which compares raw and bias-corrected errors, and the `cal.` columns of Tables 2.5–2.7 (a leak-free correction, Eq. (3.11)).

The **bias slope** β measures regression to the mean, the typical failure of an age model: old patients are predicted too young and young ones too old. It is the least-squares slope of the error $e_i = \hat{y}_i - y_i$ against the true age, with population moments over S (`labram/utils/regression_metrics.py` , `age_bias_slope`):

$$\beta(S) = \frac{\mathrm{Cov}_S(y, e)}{\mathrm{Var}_S(y)} \tag{0.5}$$

- $\beta = 0$: the error does not depend on age.

- $\beta = -1$: the prediction ignores age entirely; it is a constant.
- $\beta = -0.4$: a patient 10 years older than average is predicted about 4 years too young.

Evaluation sets.

- **TUAB test:** TUAB's official evaluation set, 274 recordings of 251 patients (149 normal, 125 abnormal). It is the same 274 recordings in every phase (checked in Section 1) and no training split contains any of its patients, so it is the one yardstick every run shares.
- **Val:** each run's own validation split. It differs between cohorts, so it compares runs only within a cohort.
- **CV test / TUEG out-of-fold:** each recording of the CV pool scored by the one fold model that never saw its patient.

No test label in the target normalization. The age is z-scored with the mean and sd of the training windows only (Eq. (0.6), Table 0.1).

Caveat. Phases 1 and 2 compared many configurations on the TUAB test set, which makes their best single-split numbers optimistically selected. The CV numbers (Phase 3) and the TUEG out-of-fold error were not used to choose anything.

```
registry: 45 phase-1 runs; cohort CSV: 24 phase-2 runs; CV CSV: 4 experiments x 5 folds
TUH data volume: found; per-recording CV predictions: found
```

Table 0.1. The age normalization uses training ages only

Training regresses a z-scored age, and predictions are mapped back to years before any metric (`labram/data/window_selection.py::_train_target_stats` , `losses/regression.py`):

$$z = \frac{y - \mu}{\sigma}, \qquad \hat{y} = \mu + \sigma\,\hat{z}, \qquad \mu = \frac{1}{N}\sum_{w \in \text{train}} y_w, \quad \sigma^2 = \frac{1}{N-1}\sum_{w \in \text{train}} (y_w - \mu)^2$$

where w runs over the N **selected training windows** of the run (after trimming), y_w is the age of the window's recording, and $\hat{z}$ is the model output. If μ or σ included a test recording, the test ages would leak into training and into every test prediction.

The check (`scripts/check_target_norm_leak.py` , `docs/data/target_norm_leak_check.csv`) rebuilds each training setup exactly as the trainer does (cohort, CV fold, window selection) twice. The first build is as recorded; the second has every **test** recording's age shifted by +40 years. Columns:

- `train` / `test recordings` and `train ∩ test recordings` (must be 0);
- `mean` / `std` : μ, σ of Eq. (0.6) as the run uses them, in years;
- `mean` / `std, test ages +shift` : the same after the test ages were shifted. They must not change;
- `= train windows' own stats` : μ, σ equal the training windows' own mean and sample sd;

- `test targets moved by`: the median change of the test targets, 40, confirming the shift took effect;
- `normalization uses test ages`: "no" when the two (mean, std) pairs are identical.

Phase 1 is checked on the local TUAB v3.0.1 copy (the phase-1 runs used v3.0.0, with the same split and code). The CV rows check fold 0 in both CV schemes. The other folds use the same code on other recordings.

setup	train record-ings	test record-ings	train ∩ test recordings	mean	std	mean, test ages +shift	std, test ages +shift	= train win-dows' own stats	test tar-gets moved by	normaliza-tion uses test ages
phase 1: TUAB (D recipe)	2145	274	0	48.88	17.82	48.88	17.82	True	40.00	no
phase 2: cohort all_both_v3 (Exp4*)	25613	274	0	52.34	17.59	52.34	17.59	True	40.00	no
phase 3: CV fold 0, test = fold (cvbase/cvctrl/cvaug)	17607	5663	0	52.49	17.28	52.49	17.28	True	40.00	no
phase 3: CV fold 0, test = TUAB held out (cvaugval)	23372	274	0	52.74	17.45	52.74	17.45	True	40.00	no

1. Data: every dataset and cohort

All data comes from the TUH EEG corpus. TUAB (TUH Abnormal, about 3k recordings with a normal/abnormal label) is a subset of TUEG (the full TUH EEG corpus, about 60k recordings). The age label is parsed from each EDF header (`labram/data/tuh_metadata.py`). Every dataset uses the same signal format:

- 23 referential channels at 200 Hz in µV, 0.1–75 Hz band-pass;
- 10 s windows;
- the first and last 60 s of each recording dropped;
- evaluation on the first 5 minutes after the trim (`labram/data/window_selection.py`).

The datasets differ in which recordings they contain, how mains interference is removed, and how patients are split.

Reading order.

- Table 1.1: the two **raw corpora**, TUAB and TUEG, as downloaded.
- Table 1.2: why each removed recording was removed.
- Table 1.3: how each training dataset is built from them, one inclusion rule at a time, with the recordings and patients each rule removes.
- Tables 1.4–1.5: the resulting datasets and their splits; Figure 1.1: the final training set by source (TUAB / TUEG / total) and status.
- Tables 1.6–1.13 and Figure 1.2: a closer look at TUEG.

Lineage.

- `TUAB v3.0.0` (phase 1) → `TUAB v3.0.1 interp` (phase-2 control): same recordings and split, mains removed by 50 + 60 Hz spectrum interpolation instead of a 50 Hz notch.
- `normal_TUEG_v1` , `normal_both_v1` : normal recordings only (TUEG alone; TUAB + TUEG).

- `all_both_v1` → `v2` → `v3`: every recording (normal, abnormal, unlabelled), with TUAB patients kept in their TUAB split.
  - `v2` gives TUEG one patient split shared by every cohort, so validation is comparable across cohorts.
  - `v3` adds the TUEG recordings that lack only the T1/T2 electrodes.
- `v3_compact` / `v3_cvcompact`: byte-for-byte subsets of `v3` that store only the windows a run reads (`dataset_maker/make_compact_cohort.py`). `cvcompact` keeps both training and evaluation windows of every recording, so any CV fold can use any recording.

Tables 1.6–1.13 and Figure 1.2 at the end of this section specify TUEG itself: size, subsets, labels and age.

All cohorts are built by `dataset_maker/make_cohort_dataset.py`, which drops every TUEG recording of a TUAB-test patient and asserts that no patient is in two splits.

Table 1.1. Raw data: TUAB and TUEG, and what the final cohort selects

One row per part of the raw data, split by status (`rs.raw_summary`). TOTAL counts every recording once:

- **TUAB train**, **TUAB eval (test)**: the two official splits of TUAB v3.0.1 (2,993 recordings), with TUAB's own normal / abnormal label.
- **TUEG (not in TUAB)**: the rest of TUEG v2.0.2. Every TUAB recording is also in TUEG, so these rows hold only what TUEG adds. Its status comes from NEDC's clinical report sheet; recordings without a sheet entry have **no label**.
- **TOTAL**: all of TUEG, 70,830 recordings, each counted once.

Columns, left to right:

- `patients`, `recordings`: the raw counts. A recording is one EDF file, a "case". A patient with recordings of different status counts in each status row, so the status rows' patients add up to more than the subtotal.
- `readable EDF, ≥ 60 s`: the recordings converted to the npy format (Section "step process" below).
- `age known`: of those, the ones with a valid age in the EDF header (TUH redacts 90+).
- `T1/T2 missing (selected)`: selected recordings that lack only the T1/T2 electrodes. They are kept, and the model leaves the two channels out per sample.
- `selected recordings`, `selected patients`: what the final cohort `all_both_v3` keeps. That is the dataset of the best models: train 25,613 + val 3,422 + test 274 = 29,309 recordings.
- `removed`: `recordings` − `selected recordings`. Table 1.2 gives the reason for each removed recording, and Table 1.3 follows the rules step by step for every dataset.

Phase 1 trains on TUAB only and selects differently; Table 1.3 gives that flow too.

part	status	patients	recordings	readable EDF, ≥ 60 s	age known	T1/T2 missing (selected)	selected recordings	selected patients	removed
TUAB train	normal	1237	1371	1370	1365	0	1258	1167	113
	abnormal	893	1346	1345	1337	0	1092	812	254
	all	2076	2717	2715	2702	0	2350	1944	367
TUAB eval (test)	normal	148	150	150	149	0	149	147	1
	abnormal	105	126	126	125	0	125	104	1
	all	253	276	276	274	0	274	251	2
TUEG (not in TUAB)	normal	6403	25340	20680	20289	2055	10197	5506	15143
	abnormal	8132	32373	27503	27045	3474	14298	7407	18075
	no label	808	10124	9212	9122	302	2190	662	7934
	all	13859	67837	57395	56456	5831	26685	12457	41152
TOTAL		14987	70830	60386	59432	5831	29309	13801	41521

Table 1.2. Why each raw recording was removed

Table 1.1's `removed` column, broken down by reason for the final cohort `all_both_v3` (`rs.removal_reasons`). Every raw recording counts once, under the **first** rule it fails, in the order the builder applies them (`dataset_maker/make_cohort_dataset.py`). The rows sum to the `raw` row, and `selected` matches Table 1.1. The rules:

- **EDF shorter than 60 s** / **EDF unreadable**: never converted to npy. One unreadable TUEG EDF has none of the 23 channels.
- **age unknown**: no valid age in the EDF header. TUH redacts ages of 90 and over.
- **age outside the cohort's range**: the TUEG cohorts use 18–89 years, so children and teenagers are out, including TUAB's.
- **recording shorter than the minimum length**: under 5 minutes. Too little EEG is left after the 60 s trims for the 5-minute evaluation.
- **channels other than the allowed ones missing**: the recording lacks a channel other than T1/T2, e.g. A1/A2 in the `*_a` montages. Recordings lacking only T1/T2 are kept.
- **patient is in the TUAB test set**: all other recordings of the 251 test patients are kept out of training and validation.
- **over the per-patient cap**: at most 10 train/val recordings per patient, drawn with a fixed seed.
  - It removes recordings, not patients: every patient with a usable recording keeps up to 10.
  - Some patients were recorded hundreds of times.
  - The cap counts a patient's TUAB and TUEG recordings together. TUAB patients who also have many TUEG recordings can therefore lose TUAB recordings to the draw.

`patients` counts the patients with at least one recording removed for that reason. A patient can appear under several reasons.

	TUAB train		TUAB eval (test)		TUEG (not in TUAB)		TOTAL	
reason	recordings	patients	recordings	patients	recordings	patients	recordings	patients
EDF shorter than 60 s (not converted)	0	0	0	0	10441	2471	10441	2471
EDF unreadable (not converted)	2	2	0	0	1	1	3	3
age unknown (missing, or 90+ redacted)	13	13	2	2	939	322	954	329
age outside the cohort's range	30	30	0	0	1543	760	1573	779
recording shorter than the minimum length	0	0	0	0	6662	2638	6662	2638
channels other than the allowed ones missing	0	0	0	0	5114	716	5114	716
patient is in the TUAB test set	0	0	0	0	417	74	417	74
over the per-patient cap	322	164	0	0	16035	756	16357	756
selected	2350	1944	274	251	26685	12457	29309	13801
raw	2717	2076	276	253	67837	13859	70830	14987

How the datasets are built: the step process and the data split

Steps, in order. Table 1.3 counts each one.

1. **Index.** Both corpora are scanned into one recording index, one row per EDF (`dataset_maker/tuh_index.py`). Age, sex and the TUAB label come from the EDF headers and folders; the report finding comes from NEDC's clinical sheet (`dataset_maker/tuh_clinical.py`).

2. **Convert** (`dataset_maker/make_TUH_npy.py`). Every readable EDF of at least 60 s becomes one npy file:

   - the 23 model channels in a fixed order; a missing channel is zero-filled and flagged in a channel mask;
   - 0.1–75 Hz band-pass and mains removal (50 + 60 Hz spectrum interpolation);
   - resampled to 200 Hz, in µV.

3. **Select per dataset** (`dataset_maker/make_cohort_dataset.py`):

   - age 18–89 years;
   - recording at least 5 minutes long;
   - channel rule: all 23 channels, or in v3 only T1/T2 may be missing;
   - for the `normal_*` cohorts only: report-normal, no neurological diagnosis, no sedating medication.

4. **Protect the test set.** Every recording of a patient in TUAB's eval split is removed from training and validation. TUAB's eval split itself (274 recordings with a known age) is the test set of every dataset.

5. **Split by patient**, never by recording:

   - TUAB patients keep the split of phase 1: TUAB's train split divided into train and a seeded 20 % val, patient-disjoint;
   - TUEG patients: `v1` draws a seeded 10 % patient val per cohort. `v2` / `v3` use one shared patient split for all cohorts (`dataset_maker/tueg_patient_split.py`), so a model can be validated on another cohort's val without patient leakage.

   The builder asserts that no patient is in two splits.

6. **Cap per patient.** At most 10 train/val recordings per patient, drawn with a fixed seed, so that patients recorded hundreds of times do not dominate.

7. **Windows.** Each training recording contributes at most 20 minutes of 10 s windows. Val and test score the first 5 minutes after 60 s trims at both ends (Section 0).

8. **Phase 3, cross-validation.** The train + val of `all_both_v3` (29,035 recordings) is split into 5 patient-grouped folds. Fold k tests, fold $k+1$ validates, and the other three train. The TUAB test set stays outside every fold.

Table 1.3. From the raw corpora to the training datasets: inclusion flow

Each row is the set of recordings left after one more inclusion rule. Columns as in Table 1.1, plus `removed`, the recordings that rule took out. The rows are replayed from each dataset's recorded build parameters with the builder's own functions:

- `rs.tuab_flow` for TUAB (`data/age_splits.py`, `make_TUAB.py`);
- `rs.cohort_flow` for the TUEG cohorts (`dataset_maker/make_cohort_dataset.py`).

Each flow's final counts are checked against the dataset's split files, so the table is exact.

The datasets:

- **TUAB (phase 1):** TUAB only. Ages 1–89, all 23 channels. TUAB's train split is divided patient-disjointly into train and val; its eval split is the test set.
- **TUEG cohorts (phase 2–3):** built from the TUEG npy export.
  - Shared rules (first block): readable EDF of at least 60 s, age 18–89, recording at least 5 minutes.
  - Per-cohort rules (later blocks): channel completeness and, for the `normal_*` cohorts, the normal filter (report-normal, no neurological diagnosis, no sedating medication).
  - Then every recording of a TUAB-test patient is removed, so no test patient is ever trained on.
  - At most 10 train/val recordings are kept per patient.
  - The test set is always TUAB's eval split.
  - `all_both_v1` and `v2` contain the same recordings and differ only in how TUEG patients are split into train and val. `all_both_v3` is the cohort of the best models, and its train + val (29,035 recordings) is the CV pool of phase 3.

dataset	cohort	step	recordings	patients	normal	abnormal	no label	all 23 channels	only T1/T2 missing	other channels missing	removed
TUAB (phase 1)		TUAB recordings (TUH index)	2993	2329	1521	1472	0	2990	3	0	–
		age known (1–89 y; 90+ redacted)	2978	2314	1515	1463	0	2975	3	0	15
		all 23 channels	2975	2313	1515	1460	0	2975	0	0	3
		TUAB train split = train + val	2701	2062	1366	1335	0	2701	0	0	–
		of which train	2145	1650	1075	1070	0	2145	0	0	–
		of which val (patient-disjoint, seeded)	556	412	291	265	0	556	0	0	–
		test: TUAB eval split	274	251	149	125	0	274	0	0	–
TUEG cohorts	shared rules	TUEG recordings (TUH index)	70830	14987	26861	33845	10124	55182	8479	7169	–
		converted to npy (readable EDF, ≥ 60 s)	60386	14979	22200	28974	9212	46247	7576	6563	10444
		age known, 18–89 y	57857	14019	21045	27909	8903	44645	6918	6294	2529
		recording ≥ 5 min	51195	14001	18203	25062	7930	40065	6016	5114	6662
		all 23 channels	40065	10553	14886	19126	6053	40065	0	0	11130
	normal_TUEG_v1	normal only: report-normal, no neurological diagnosis, not sedated	8798	3585	8798	0	0	8798	0	0	31267
		without TUAB patients (TUEG only)	5485	2389	5485	0	0	5485	0	0	3313
		≤ 10 recordings per patient = train + val	4158	2389	4158	0	0	4158	0	0	1327
		test: TUAB eval recordings with a known age	0	0	0	0	0	0	0	0	–
	normal_both_v1	all 23 channels	40065	10553	14886	19126	6053	40065	0	0	11130
		normal only: report-normal, no neurological diagnosis, not sedated	8798	3585	8798	0	0	8798	0	0	31267
		without recordings of TUAB-test patients	8592	3453	8592	0	0	8592	0	0	206
		≤ 10 recordings per patient = train + val	6293	3453	6293	0	0	6293	0	0	2299
		test: TUAB eval recordings with a known age	274	251	149	125	0	274	0	0	–
	all_both_v1 = v2 (same recordings)	all 23 channels	40065	10553	14886	19126	6053	40065	0	0	11130
		without recordings of TUAB-test patients	39420	10304	14591	18895	5934	39420	0	0	645
		≤ 10 recordings per patient = train + val	23273	10304	9458	11920	1895	23273	0	0	16147
		test: TUAB eval recordings with a known age	274	251	149	125	0	274	0	0	–
	all_both_v3	all 23 channels, or only T1/T2 missing	46081	13799	16998	22690	6393	40065	6016	0	5114
		without recordings of TUAB-test patients	45392	13550	16679	22448	6265	39420	5972	0	689
		≤ 10 recordings per patient = train + val	29035	13550	11455	15390	2190	23204	5831	0	16357
		test: TUAB eval recordings with a known age	274	251	149	125	0	274	0	0	–

Table 1.4. The training datasets (cohorts): contents, rules and splits

One row per dataset. Names show the corpora in capitals (TUAB, TUEG); the directories on disk under `$LABRAM_TUH_ROOT/cohorts/` use the same names in lower case (e.g. `normal_tueg_v1`).

column	meaning
`phase`	the phase that used the dataset
`contents`	which recordings it holds
`line removal`	how mains interference is removed (from the dataset's `manifest.json`)
`rules`	inclusion rules: `age` range in years; `≥ min` minimum recording length in minutes; `≤ k/patient` at most *k* train/val recordings per patient; `T1/T2 may be missing`
`size (train / val / test)`	recordings per split, from the dataset's `age_split.json` (`rs.split_sizes`); the CV runs split train + val into 5 folds
`TUEG split`	how TUEG patients are split; TUAB patients always keep their TUAB split
`used by`	the runs of Tables 2.2–2.7 trained on it

dataset	phase	contents	line removal	rules	size (train / val / test)	TUEG split	used by
TUAB v3.0.0	1	TUAB, all recordings (normal + abnormal)	notch:50	age 1–89 (TUAB)	2,145 / 556 / 274	TUAB: seeded 20 % val, official test	phase-1 runs (Table 2.2)
TUAB v3.0.1 interp	2	TUAB, all recordings	interp:50,60	age 1–89 (TUAB)	2,145 / 556 / 274	TUAB: seeded 20 % val, official test	Control
normal_TUEG_v1	2	TUEG normal, patients outside TUAB; no test split	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient	3,763 / 395 / 0	seeded 10 % val	Exp2
normal_both_v1	2	TUAB normal + TUEG normal	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient	5,450 / 843 / 274	seeded 10 % val	Exp1, Exp1b, Exp1t, Exp1f
all_both_v1	2	TUAB + TUEG, every recording (normal / abnormal / no label)	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient	20,368 / 2,905 / 274	seeded 10 % val	Exp3, Exp3b, Exp3q, Exp3c
all_both_v2	2	as v1, TUEG patients on the shared split	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient	20,470 / 2,803 / 274	shared patient split v2	Exp3v2, Exp3cond, Exp3clscond
all_both_v3	2	v2 + TUEG recordings missing only T1/T2	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient, T1/T2 may be missing	25,613 / 3,422 / 274	shared patient split v2	Exp4drop, Exp4base, Exp4ctrl, Exp4aug, Exp4aug15
all_both_v3_compact	2	v3, only the windows Exp4base reads	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient, T1/T2 may be missing	25,613 / 3,422 / 274	shared patient split v2	Exp4basec
all_both_v3_cvcompact	3	v3, training + evaluation windows of every recording (CV pool = train + val)	interp:50,60	age 18–89, ≥ 5 min, ≤ 10/patient, T1/T2 may be missing	25,613 / 3,422 / 274 (CV pool 29,035)	shared patient split v2	cvbase, cvctrl, cvaug

Table 1.5. Size and composition per split

For each dataset and split:

- `recordings` and `patients`;
- `from TUAB`: recordings that are also in TUAB;
- `normal` / `abnormal` / `no label`: the recording's report status (TUEG recordings without a normal/abnormal report have no label);
- `age mean` and `age sd`: the true age's mean and sample standard deviation (years).

Computed from each dataset's `cohort.parquet` / `manifest.json`, or from `age_split.json` + `age_metadata.json` for the TUAB-only datasets (`labram/eval/research_summary.py::cohort_summary`, `tuab_split_summary`).

dataset	split	recordings	patients	from TUAB	normal	abnormal	no label	age mean	age sd
TUAB v3.0.0	train	2145	1650	2145	1075	1070	0	49.2	17.7
	val	556	412	556	291	265	0	48.4	15.9
	test	274	251	274	149	125	0	50.4	18.1
TUAB v3.0.1 interp	train	2145	1650	2145	1075	1070	0	49.2	17.7
	val	556	412	556	291	265	0	48.4	15.9
	test	274	251	274	149	125	0	50.4	18.1
normal_TUEG_v1	train	3763	2150	0	3763	0	0	48.6	17.6
	val	395	239	0	395	0	0	49.9	16.9
normal_both_v1	train	5450	2995	775	5450	0	0	47.2	17.2
	val	843	458	196	843	0	0	46.6	15.7
	test	274	251	274	149	125	0	50.4	18.1
all_both_v1	train	20368	9069	1879	8204	10610	1554	52.0	17.5
	val	2905	1235	478	1254	1310	341	51.3	16.3
	test	274	251	274	149	125	0	50.4	18.1
all_both_v2	train	20470	9087	1879	8277	10617	1576	51.9	17.5
	val	2803	1217	478	1181	1303	319	51.6	16.5
	test	274	251	274	149	125	0	50.4	18.1
all_both_v3	train	25613	12017	1868	10067	13714	1832	52.4	17.5
	val	3422	1533	482	1388	1676	358	52.1	16.8
	test	274	251	274	149	125	0	50.4	18.1
all_both_v3_compact	train	25613	12017	1868	10067	13714	1832	52.4	17.5
	val	3422	1533	482	1388	1676	358	52.1	16.8
	test	274	251	274	149	125	0	50.4	18.1
all_both_v3_cvcompact	train	25613	12017	1868	10067	13714	1832	52.4	17.5
	val	3422	1533	482	1388	1676	358	52.1	16.8
	test	274	251	274	149	125	0	50.4	18.1

```
TUAB test: 274 recordings; identical in every dataset with a test split: True
{'TUAB v3.0.0': True, 'TUAB v3.0.1 interp': True, 'normal_both_v1': True, 'all_both_v
1': True, 'all_both_v2': True, 'all_both_v3': True, 'all_both_v3_compact': True, 'all_b
oth_v3_cvcompact': True}
```

Reading Table 1.5.

- **Scale:** training grows from 2,145 TUAB recordings (phase 1) to 25,613 recordings of about 12k patients (`all_both_v3`).
- **Composition:** in the `all_both` cohorts about half the training recordings are abnormal and about 7 % have no label.
- **Normal-only cohorts:** `normal_*` hold only normal recordings, but their test split is still the full TUAB test set, abnormal recordings included.
- **Test set:** the 274 TUAB test recordings are identical across datasets (the check above), so every TUAB test number in Sections 2–5 is on the same recordings.

Figure 1.1. Composition and age of the training set

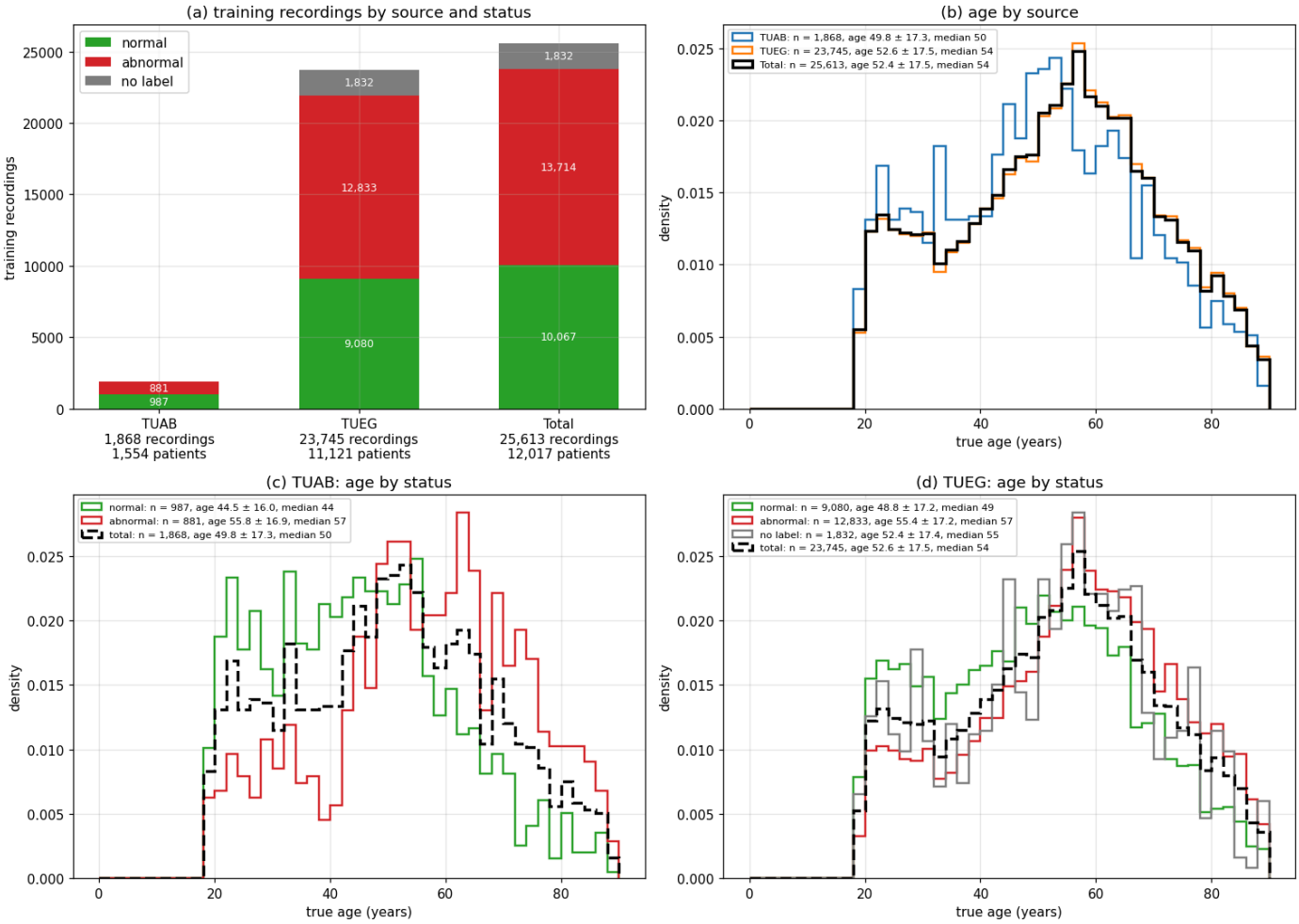
The training set of the final cohort `all_both_v3` , the one the best models use: its train split, 25,613 recordings. It is split by source:

- **TUAB:** recordings that are in TUAB;
- **TUEG:** the TUEG recordings that are not in TUAB;
- **Total:** both.

Status is the recording's normal / abnormal / no-label status (Eq. (1.2)). The other datasets of Table 1.4 are subsets or reshuffles of the same recordings, so they are not drawn again. Table 1.5 gives each dataset's counts per split.

- (a) Training recordings per source, stacked by status. Each segment shows its count, and the bar's label gives the total and the number of patients.
- (b) Distribution of true age (years) per source, as a density (step histogram, 2-year bins).
- (c) TUAB by status, as in (b).
- (d) TUEG by status, as in (b).

Each legend gives n (recordings), the mean ± sample standard deviation of age and its median.



TUEG, the source corpus. Every cohort above draws from TUEG v2.0.2, the full TUH EEG corpus (`$LABRAM_TUH_ROOT/tueg/v2.0.2`). Tables 1.6–1.13 and Figure 1.2 specify it from two files:

- the TUH recording index, one row per EDF over all TUH corpora (`index/tuh_index.parquet` , `dataset_maker/tuh_index.py`);
- NEDC's clinical sheet, one row per session, joined on patient + session (`index/nedc/metadata_v12.xlsm.xlsx` , `dataset_maker/tuh_clinical.py`).

`labram/eval/research_summary.py::tueg_recordings` builds the joined table. For a set S of the $N = |\mathrm{TUEG}|$ recordings, with recording i lasting d_i seconds:

$$H(S) = \frac{1}{3600} \sum_{i \in S} d_i, \qquad \%(S) = 100 \, \frac{|S|}{N} \tag{1.1}$$

A recording's **status** is the TUAB folder label where the recording is in TUAB. Otherwise it is the report finding of the clinical sheet: abnormal when the sheet's "abnormal due to" field is filled. Recordings whose session is not in the sheet have no label (`dataset_maker/make_cohort_dataset.py::label_of`, the label the `all_both` cohorts use):

$$\text{status}_i = \begin{cases} \text{TUAB label} & i \in \text{TUAB} \\ \text{abnormal if the report gives a cause, else normal} & i \notin \text{TUAB}, \text{ session} \\ \text{no label} & \text{otherwise} \end{cases}$$

Ages are in years, from the EDF header. TUH writes 999 for patients aged 90 or older (a HIPAA redaction), and those recordings count as age-unknown. Age statistics use the recordings with a known age a_i: mean $\bar{a}$, sample standard deviation and quartiles Q_1, median, Q_3:

$$\bar{a} = \frac{1}{n} \sum_{i=1}^{n} a_i, \qquad s_a = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (a_i - \bar{a})^2} \tag{1.3}$$

Data volume required.

Table 1.6. TUEG at a glance: all recordings and the selected ones

One row per quantity (`rs.tueg_overview`), in two columns:

- `all TUEG`: every TUEG recording, 70,830, TUAB included;
- `selected (all_both_v3)`: the recordings the final cohort keeps, train + val + test (Tables 1.1–1.2).

Rows:

- `size`: recordings, patients, sessions and hours (Eq. (1.1)).
  - A **session** is one visit of a patient (`<patient>_sNNN`). It can hold several recordings (EDF files `t000`, `t001`, ...), so there are more recordings than sessions.
  - The last size row splits the selected sessions into train / val / test.
- `recording`: length, year, sampling rate and montage. The year range omits a few EDF headers with impossible dates.
- `channels`: whether the recording has all 23 channels the models read.
- `patient`: sex and age (Eq. (1.3)), counted over recordings, so a patient with several recordings counts several times.

group	quantity	all TUEG	selected (all_both_v3)
size	recordings (EDF files)	70,830	29,309
	patients	14,987	13,801
	sessions	26,845	21,783
	sessions in train / val / test	–	18,923 / 2,586 / 274
	hours of EEG	27,021	15,940
	recordings per patient, median (max)	1 (713)	1 (10)
recording	length (min), median [IQR]	10.0 [5.0, 20.3]	20.1 [10.0, 22.6]
	year, median [5 %, 95 %]	2014 [2003, 2016]	2012 [2003, 2016]
	sampling rate (Hz)	256: 52%, 250: 37%, 1000: 4%	250: 60%, 256: 30%, 1000: 7%
	montage	01_tcp_ar: 73%, 02_tcp_le: 18%, 03_tcp_ar_a: 9%	01_tcp_ar: 70%, 02_tcp_le: 30%, 03_tcp_ar_a: 0%
channels	all 23 channels	55,182 (77.9 %)	23,478 (80.1 %)
	missing only T1/T2	8,479 (12.0 %)	5,831 (19.9 %)
	missing other channels	7,169 (10.1 %)	0 (0.0 %)
patient	sex F / M / unknown (recordings)	36,519 / 33,785 / 526	15,072 / 14,183 / 54
	age known (1–89 y)	69,747 (98.5 %)	29,309 (100.0 %)
	age 90+ (redacted as 999)	714 (1.0 %)	0 (0.0 %)
	age missing	369 (0.5 %)	0 (0.0 %)
	age (y), mean ± sd; median [IQR]	51.0 ± 18.1; 53 [38, 64]	52.4 ± 17.4; 53 [39, 65]

Table 1.7. Subsets of TUEG

The other TUH corpora are subsets of TUEG: the same EDF files, each corpus adding its own label. One recording can be in several of them, so the rows overlap (`rs.tueg_subsets`). The last four rows are the subsets the brain-age cohorts select from:

- `TUEG only` : in no other corpus;
- `in the NEDC clinical sheet` : has a report status (Eq. (1.2));
- `adult age (18–89 y)` : the age range of the cohorts;
- `normative` : the pool of the `normal_*` cohorts (`dataset_maker/tuh_clinical.py::normative_mask`).

Columns:

- `label counts (recordings)` : the subset's label per category, where it is per-recording and categorical:
  - TUAB: normal / abnormal;
  - TUEP: epilepsy / no epilepsy, a patient-level diagnosis;
  - clinical sheet: the report finding;
  - `normative` : the status (Eq. (1.2)). TUSZ, TUAR and TUSL annotate *events* inside a recording (seizures, artifacts, slowing) rather than one category per recording; their annotation files are not on the data volume, hence "–". Age is numeric.
- `recordings` , `% of TUEG` and `hours` (Eq. (1.1)), and `patients` .

The `normative` subset holds 75 recordings that are report-normal but in TUAB's *abnormal* folder. The `normal_*` cohorts drop these too (`make_cohort_dataset.py::select`). TUEV is not listed: none of its recordings is in TUEG v2.0.2.

subset	label it carries	label counts (recordings)	record-ings	% of TUEG	pa-tients	hours
in TUAB	normal / abnormal (TUAB folder)	normal 1,521 / abnormal 1,472	2993	4.2	2329	1141.0
in TUSZ	seizure events	–	8140	11.5	675	1480.7
in TUEP	epilepsy / no epilepsy (patient)	epilepsy 2,244 / no epilepsy 564	2808	4.0	200	635.5
in TUAR	artifact events	–	310	0.4	213	95.2
in TUSL	slowing events	–	112	0.2	38	26.8
TUEG only	–	–	57836	81.7	13448	24024.5
in the NEDC clinical sheet	report finding, 15 diagnosis flags, medications	report normal 26,846 / abnormal 33,426	60272	85.1	14508	24956.2
adult age (18–89 y)	age	–	68077	96.1	14025	25795.6
normative	report-normal, no neurological diagnosis, not se-dated, 18–89 y	normal 14,268 / abnormal 75	14343	20.2	4773	4569.7

Table 1.8. TUEG status labels and age per label

One row per status (Eq. (1.2)) and label source (`rs.tueg_status`):

- `TUAB folder` : TUAB's normal/abnormal label;
- `NEDC report` : the clinical sheet's finding;
- `all` : both sources;
- `total` : all of TUEG.

Columns:

- `recordings` , `% of TUEG` (Eq. (1.1));
- `patients` : a patient whose recordings differ in status counts under each status, so the rows sum to more than the total;
- `age n` : recordings with a known age;
- `age mean` , `age sd` (Eq. (1.3)), and `age Q1` , `age median` , `age Q3` in years.

Where TUAB and the report both label a session, they agree on about 95 % of its recordings.

status	source	recordings	% of TUEG	patients	age n	age mean	age sd	age Q1	age median	age Q3
	NEDC report	25340	35.8	6403	24921	48.2	17.9	34.0	50.0	62.0
normal	**TUAB folder**	1521	2.1	1385	1515	44.0	16.1	31.0	44.0	55.0
	all	26861	37.9	7348	26436	47.9	17.8	34.0	49.0	62.0
	NEDC report	32373	45.7	8132	31855	53.3	18.1	41.0	56.0	67.0
abnormal	**TUAB folder**	1472	2.1	998	1463	54.4	17.2	44.0	55.0	67.0
	all	33845	47.8	8664	33318	53.4	18.0	41.0	56.0	67.0
no label	**–**	10124	14.3	808	9993	51.1	17.7	39.0	53.0	63.0
total	**all**	70830	100.0	14987	69747	51.0	18.1	38.0	53.0	64.0

Figure 1.2. TUEG age distribution per status and in total

True age (years) of the TUEG recordings with a known age ($n =$ `age n` of Table 1.8), in 2-year bins, per status (Eq. (1.2)) and for all of TUEG:

- (a) Recordings per bin, stacked by status. The stack's outline is the total.
- (b) The same distributions as densities (each integrates to 1), so their shapes compare despite the different group sizes. The vertical ticks mark each group's median.

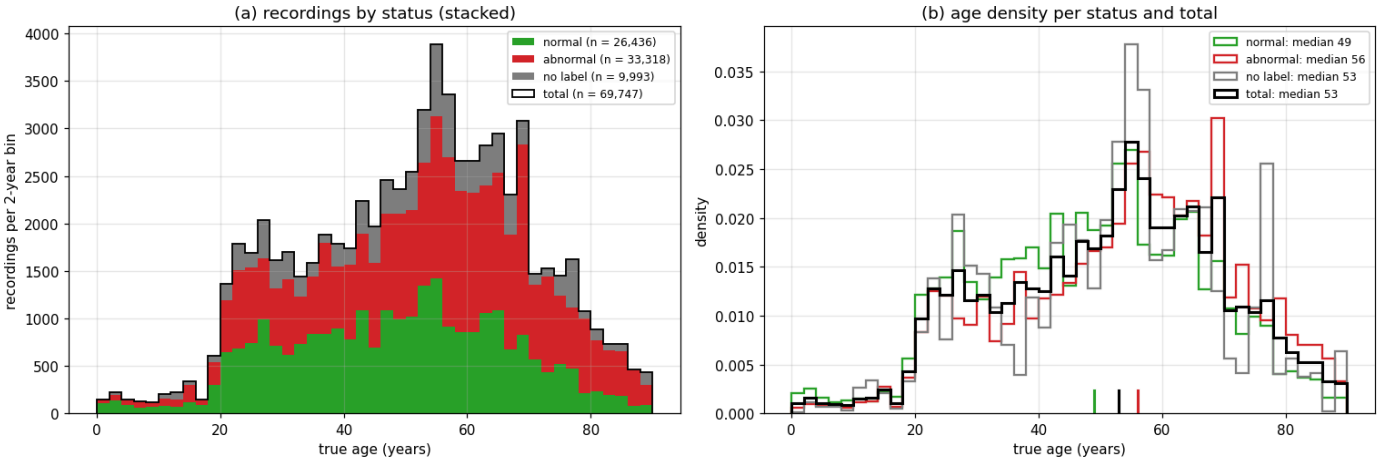


Table 1.9. TUEG diagnosis flags

The clinical sheet's 15 diagnosis flags and two derived flags over the TUEG recordings in the sheet, most frequent first (`rs.tueg_diagnoses`):

- `any neurological diagnosis` : seizure (clinical or electrographic), epilepsy, stroke, Parkinson's, Alzheimer's, TBI, dementia or MS;
- `sedating medication` : an anaesthetic, benzodiazepine, barbiturate or opioid on the medication list.

How the flags relate to the status label. Table 1.10 gives where each field comes from, and Table 1.11 explains, per kind of flag, why a flagged recording is not necessarily abnormal. The flags are **not** the abnormal label, and they are **not one-hot**:

- **Independent of the status.** Each flag is the patient's diagnosis as recorded in the clinical sheet. The normal / abnormal status (Eq. (1.2)) is the neurologist's reading of *this* EEG. A flagged recording can be normal: an epilepsy patient often has a normal EEG between seizures. The `normal` / `abnormal` columns show the split for every flag, and many abnormal recordings carry no flag at all (Table 1.12).
- **Multi-hot.** Every flag is its own yes/no column, so a recording can carry several at once, or none (Table 1.12). The condition vector of the `path_label` runs (Eq. (3.6)) encodes them the same way.

Columns:

- `recordings` and `patients` with the flag;
- `normal` / `abnormal` : those recordings by status (Eq. (1.2));
- `% of sheet` : Eq. (1.1) relative to the sheet's recordings instead of N;
- `age mean` : Eq. (1.3), in years;
- `% abnormal` : the share with status abnormal (Eq. (1.2)).

The last row, **Total**, counts every TUEG recording in the sheet, flagged or not. The flags are multi-hot and many recordings carry none, so the flag rows do not add up to it.

The 14 pathology groups of the condition vector (Eq. (3.6)) come from these flags.

flag	recordings	patients	normal	abnormal	% of sheet	age mean	% abnormal
sedating medication	23067	4727	9135	13932	38.3	52.3	60.4
any neurological diagnosis	11560	1736	3901	7659	19.2	51.1	66.3
Clinical seizure	6980	1092	2922	4058	11.6	48.5	58.1
Epilepsy	4660	306	1991	2669	7.7	47.0	57.3
Electrographic seizure	3015	66	328	2687	5.0	54.5	89.1
Stroke	2590	539	405	2185	4.3	61.3	84.4
Substance abuse / dependence	2083	725	592	1491	3.5	45.7	71.6
TBI	720	202	147	573	1.2	48.6	79.6
Bipolar disorder	502	240	243	259	0.8	42.7	51.6
Dementia	293	50	62	231	0.5	72.3	78.8
Depression	213	77	137	76	0.4	44.7	35.7
Schizophrenia	191	43	130	61	0.3	52.5	31.9
Anxiety	182	72	110	72	0.3	40.5	39.6
Alzheimer's	166	6	3	163	0.3	68.3	98.2
Multiple sclerosis	108	37	58	50	0.2	48.0	46.3
ADHD	78	37	47	31	0.1	25.4	39.7
Parkinson's	9	3	7	2	0.0	55.4	22.2
Total (all sheet recordings)	60272	14508	26775	33497	100.0	50.9	55.6

Table 1.10. Where the labels come from

Every status and flag in this notebook comes from human-written clinical records. None is computed from the EEG signal by an algorithm; the only signal-derived features in the repo are our own (`labram/features/`). The NEDC sheet (`metadata_v12` , sheet `TUEG` , one row per session) has review columns (`Reviewed spell check` , `Reviewed medication` , `Reviewed abnormal due`), so NEDC staff evidently extracted its fields from each session's EEG report and reviewed them. The documentation on disk does not state this explicitly.

field	where it comes from	what it describes	level
status, TUAB recordings	TUAB's normal / abnormal folders	the neurologist's reading of this EEG: the report's impression, curated into TUAB	recording
status, other TUEG recordings	NEDC sheet `abnormal due to` : filled = abnormal, empty = normal; session not in the sheet = no label	the neurologist's reading of this EEG, i.e. why the report calls it abnormal	session (its recordings share it)
diagnosis flags (15)	NEDC sheet, one yes/no column per diagnosis	the patient's clinical background (history, reason for the EEG), not the EEG	session
medications	NEDC sheet `Major Medications`	drugs the patient was taking	session
FIRDA / OIRDA / TIRDA / GPD / LPD	NEDC sheet	EEG patterns the neurologist saw in the signal: signal findings, read by a human	session

A clinical EEG report usually holds both kinds of information:

- **the clinical history and medications**, from the referral: the source of the diagnosis flags;
- **the neurologist's description and impression of the recording**: the source of the status.

The two can therefore disagree: a patient with epilepsy in the history can have a normal EEG (Table 1.11).

Table 1.11. Why a diagnosis flag is not always abnormal

A flag describes the patient. The status describes whether *this* 20–30-minute EEG shows a specific abnormality (slowing, epileptiform discharges, asymmetry). `% abnormal` is the share of the flag's recordings with status abnormal (Table 1.9).

kind of flag	% abnormal	why
EEG findings and dementias: electrographic seizure, Alzheimer's, dementia	89, 98, 79	Electrographic seizures are seen on the EEG itself. Dementia causes persistent diffuse slowing. Alzheimer's comes from only 6 patients.
structural brain injury: stroke, TBI	84, 80	A lesion shows on the EEG when it is large or cortical. Small, deep or old lesions can leave a normal EEG.
episodic conditions: epilepsy, clinical seizure	57, 58	Seizures come and go. Between seizures, a single routine EEG of an epilepsy patient is often normal. "Clinical seizure" also covers events that turned out non-epileptic (e.g. syncope or psychogenic attacks).
substance use: substance abuse / dependence	72	Intoxication, withdrawal and the drugs given for them can slow or otherwise change the EEG.
psychiatric / developmental: depression, schizophrenia, anxiety, ADHD, bipolar	36, 32, 40, 40, 52	These conditions usually do not change the resting EEG. When it is abnormal, medication or another condition is a frequent cause.
other: multiple sclerosis, Parkinson's	46, 22	MS lesions lie mostly in white matter, and the EEG is often normal. Parkinson's has only 9 recordings.

Two further reasons the match is loose:

- **Different levels.** The flags are per session, while the status comes from TUAB's label for TUAB recordings and from the report otherwise.
- **Noisy labels.** Both are noisy clinical records. TUAB's label and the report finding disagree on about 5 % of shared recordings (Table 1.8).

Table 1.12. Diagnosis flags per recording

How many of the 15 diagnosis flags each TUEG recording in the clinical sheet carries (`rs.flags_per_recording`), with those recordings by status (Eq. (1.2)) and as a share of the sheet's recordings. A count above 1 shows that the flags are multi-hot. The count-0 row shows how many normal and abnormal recordings have no recorded diagnosis.

diagnosis flags per recording	recordings	normal	abnormal	% of sheet
0	46407	22102	24305	77.0
1	8307	2799	5508	13.8
2	3582	1404	2178	5.9
3	1624	338	1286	2.7
4	316	99	217	0.5
5	33	33	0	0.1
6	3	0	3	0.0

Table 1.13. Shared diagnosis flags and status: pairwise counts

Over the 60,272 TUEG recordings in the clinical sheet, entry (row, column) is the number of recordings that carry **both** the row's and the column's item (`rs.flag_matrix`). The items are the status (`normal` , `abnormal` , Eq. (1.2)) and the 15 diagnosis flags, most frequent first. The diagonal is each item's total (Table 1.9). The matrix is symmetric. `normal` × `abnormal` is 0, because a recording has one status. Flag names are shortened: `el.` / `cl. seizure` = electrographic / clinical seizure, `substance` = substance abuse or dependence, `MS` = multiple sclerosis.

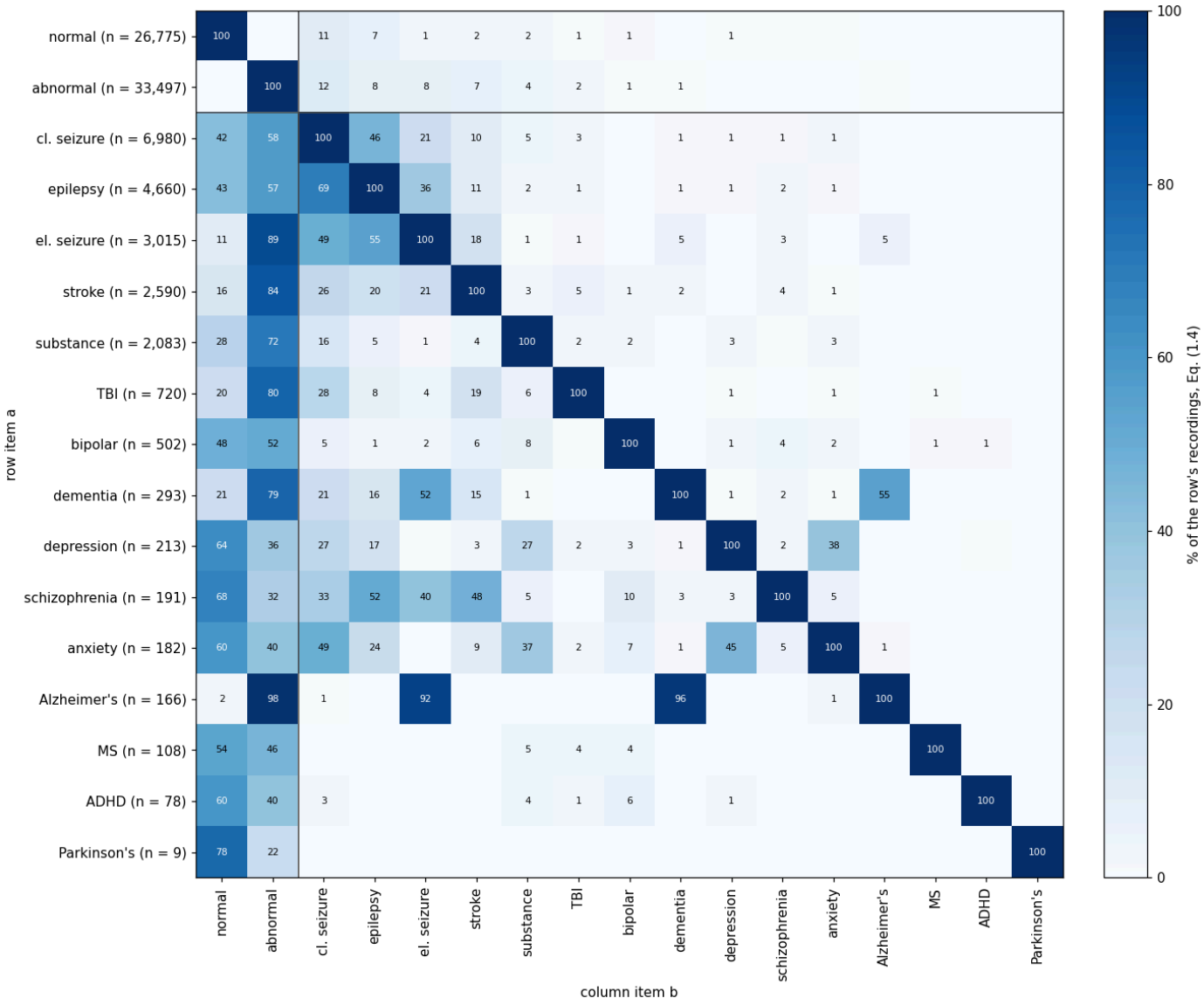
	normal	abnormal	cl. seizure	epilepsy	el. seizure	stroke	substance	TBI	bipolar	dementia	depression	schizophrenia	anxiety	Alzheimer's	MS	ADHD	Parkinson
normal	26775	0	2922	1991	328	405	592	147	243	62	137	130	110	3	58	47	
abnormal	0	33497	4058	2669	2687	2185	1491	573	259	231	76	61	72	163	50	31	
cl. seizure	2922	4058	6980	3222	1486	676	324	200	27	61	58	63	90	1	0	2	
epilepsy	1991	2669	3222	4660	1660	510	104	60	7	46	37	99	43	0	0	0	
el. seizure	328	2687	1486	1660	3015	549	18	27	9	153	1	77	0	153	0	0	
stroke	405	2185	676	510	549	2590	81	134	32	43	7	91	17	0	0	0	
substance	592	1491	324	104	18	81	2083	41	41	2	57	9	67	0	5	3	
TBI	147	573	200	60	27	134	41	720	2	1	4	0	4	0	4	1	
bipolar	243	259	27	7	9	32	41	2	502	0	7	19	12	0	4	5	
dementia	62	231	61	46	153	43	2	1	0	293	2	6	2	160	0	0	
depression	137	76	58	37	1	7	57	4	7	2	213	5	82	0	0	1	
schizophrenia	130	61	63	99	77	91	9	0	19	6	5	191	9	0	0	0	
anxiety	110	72	90	43	0	17	67	4	12	2	82	9	182	2	0	0	
Alzheimer's	3	163	1	0	153	0	0	0	0	160	0	0	2	166	0	0	
MS	58	50	0	0	0	0	5	4	4	0	0	0	0	0	108	0	
ADHD	47	31	2	0	0	0	3	1	5	0	1	0	0	0	0	78	
Parkinson's	7	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Figure 1.3. Shared diagnosis flags and status: share of the row

The counts of Table 1.13 as shares of the row's total:

$$s(a, b) = 100 \, \frac{n(a \wedge b)}{n(a)} \tag{1.4}$$

where $n(a \wedge b)$ is the number of recordings carrying both a and b. Read along a row: for example, the `epilepsy` row gives the % of epilepsy recordings that are normal, abnormal, or also carry each other flag. The matrix is not symmetric, because rows are normalized by their own totals. Cells are annotated with the rounded %. The diagonal is 100 %.



Figures 1.4–1.5. Age per diagnosis flag

True age (years) of the TUEG recordings in the clinical sheet that carry each flag. Each panel is a histogram in 5-year bins, stacked by status: green normal, red abnormal. Its title gives the number of recordings and patients and the median age.

Counts are **recordings**, and the title gives the patients. A spike usually means one or a few patients recorded many times: electrographic seizure has 46 recordings per patient on average. Recordings with an unknown age are left out, so `n` can be slightly below Table 1.9.

- Figure 1.4: the neurological flags, panels (a)–(i), and (j) the recordings with **no** diagnosis flag as a reference.
- Figure 1.5: the psychiatric and other flags, panels (a)–(f).

Figure 1.4. Age per diagnosis flag: neurological flags

As described above; panels (a)–(i) one neurological flag each, (j) recordings with no flag.

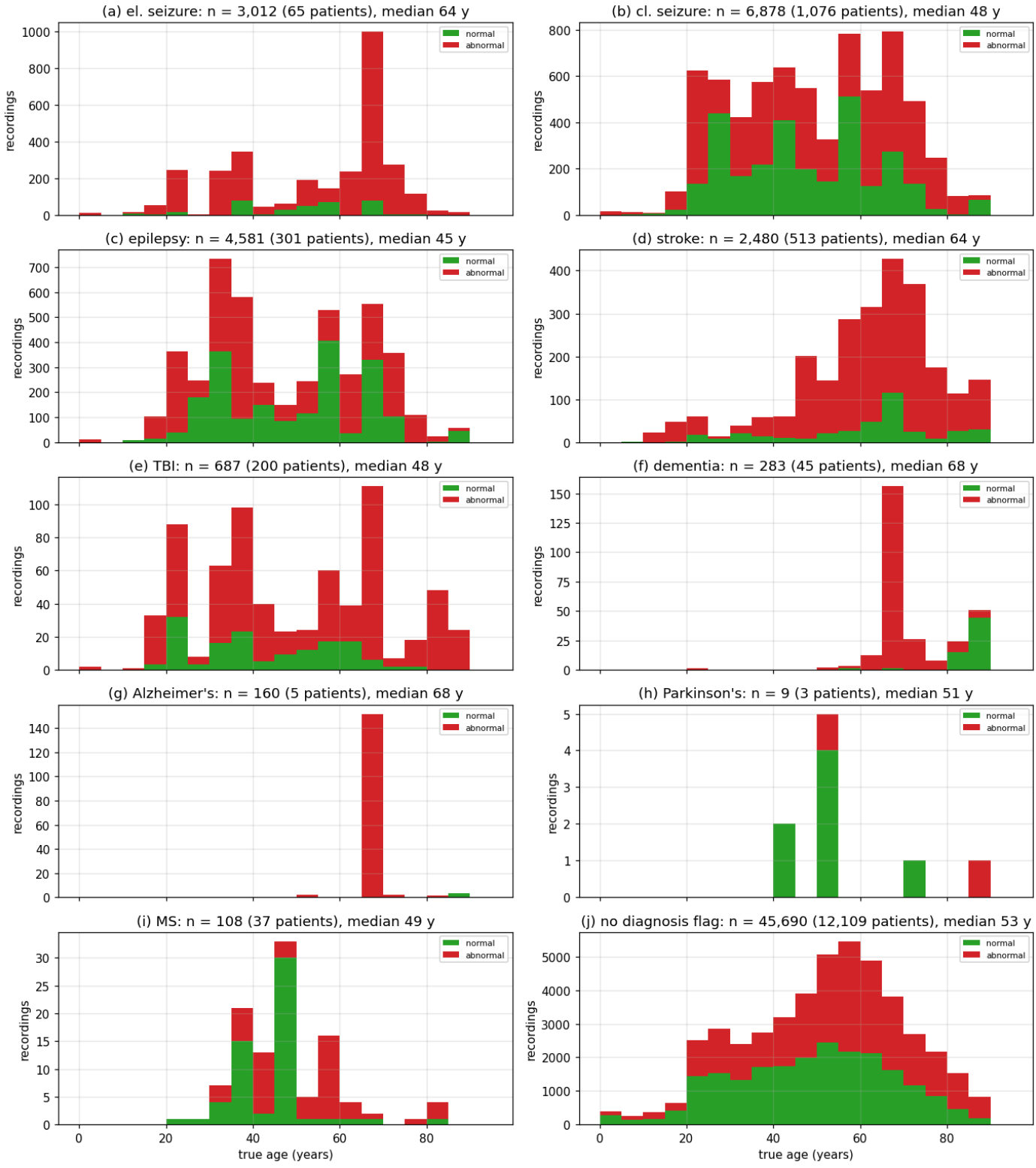
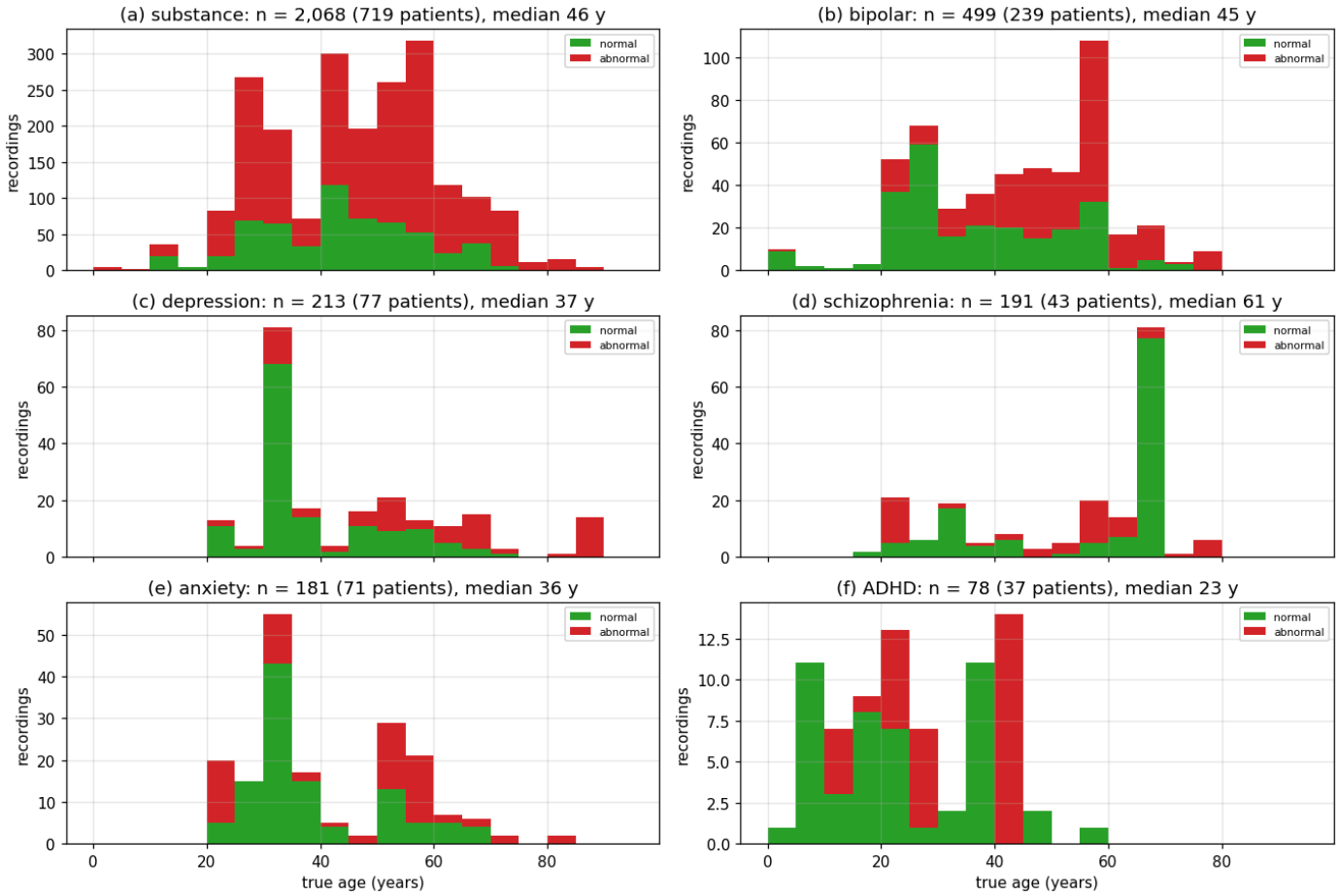


Figure 1.5. Age per diagnosis flag: psychiatric and other flags

As described above; panels (a)–(f) one flag each.



Reading Table 1.13 and Figures 1.3–1.5.

- **Seizure flags cluster.** 69 % of epilepsy recordings are also flagged clinical seizure, and 55 % of electrographic-seizure recordings are also flagged epilepsy. The three flags describe one patient group from different angles: the diagnosis, the reported event and the EEG finding.
- **Status follows the kind of flag** (the `normal` / `abnormal` columns of Figure 1.3, as in Table 1.9). Electrographic seizure, Alzheimer's and stroke are mostly abnormal. Psychiatric flags are mostly normal.
- **Small flags come from few patients.** Alzheimer's has 166 recordings from 6 patients. Almost all of them also carry dementia (160) and electrographic seizure (153), so its co-occurrences describe those 6 patients rather than Alzheimer's disease in general (Table 1.9 gives patients per flag).
- **Psychiatric flags overlap each other,** and in this hospital population often co-occur with neurological flags too. 38 % of depression recordings also carry anxiety. Schizophrenia recordings carry epilepsy in 52 % and stroke in 48 % of cases, and anxiety recordings carry clinical seizure in 49 %. Few patients carry each psychiatric flag, so these shares rest on few patients.
- **Age differs strongly by flag** (Figures 1.4–1.5). Dementia and Alzheimer's sit at 65–90 years, and stroke and electrographic seizure peak in the 60s. ADHD sits mostly under 40

(median 23). Epilepsy, clinical seizure, bipolar and substance use are spread over adulthood.
- **Abnormal versus normal age within a flag.** For 10 of the 15 flags the abnormal recordings are older (median), most clearly for depression (55 vs 33), anxiety (50 vs 31) and bipolar (52 vs 32). Epilepsy is the opposite (37 vs 50), and so is schizophrenia (58 vs 66). Stroke shows no difference. Dementia and Alzheimer's have their normal recordings at the oldest ages, from a handful of patients.

Reading Tables 1.6–1.13 and Figure 1.2.

- **Size:** TUEG holds about 70.8k recordings (27k hours) of about 15k patients. The cohorts use 25k–29k of them, after the age, length, channel and per-patient rules of Table 1.3.
- **Recordings per patient:** most patients have a single recording, but a few long-monitored patients have hundreds. This is why the cohorts cap train/val recordings per patient and every split is by patient.
- **Labels:** only 4 % of TUEG is in TUAB. The status of nearly all other recordings comes from the NEDC report: about half are abnormal, and 14 % have no label (Table 1.8).
- **Age:** abnormal recordings are older than normal ones by about 5 years in mean and 7 in median (Table 1.8, Figure 1.2(b)). A model trained on mixed statuses can therefore partly read age from pathology, which is one reason Section 5 reports normal and abnormal test recordings separately.

2. Experiments

Phase 1 (TUAB only) starts from scenario **D**, the default `finetune_tuab_age.json` (Table 2.1 lists how it differs from LaBraM's own fine-tuning recipe):

- LaBraM-base with a **common average reference (CAR)** on the input (Eq. (3.1));
- Huber loss on the z-scored age;
- lr 10^{-4}, 15 epochs.

CAR re-references the EEG. At every time sample it subtracts the mean over all channels from each channel, so every channel is measured against the average of all electrodes instead of the recording's own reference. This removes what all electrodes share: the activity of the reference electrode(s) and noise common to every channel. It also puts the two TUH montages on one reference, average reference (`tcp_ar` , about 80 % of the recordings) and linked ears (`tcp_le`). LaBraM as published feeds the signal on each recording's own reference. CAR is the only part of the LaBraM++ input block D uses: there is no per-patch z-scoring (Table 2.1). In Table 5.2, row P1 tests it.

E2 shortens it to 4 epochs / 1 warmup epoch, and **M1** = E2 + mixup $\alpha = 0.4$. Every **phase-2** run is M1 plus the change named in its row (`labram/eval/research_summary.py::M1`).
Phase 3 cross-validates the three strongest phase-2 configurations.

Loss. Every run trains on a loss from the family of Eqs. (2.1)–(2.11) below, and Tables 2.2, 2.5 and 2.7 name each run's loss in their `loss` column.

ClearML. Tables 2.2, 2.5, 2.6, 2.7 and 2.9 link every run to its ClearML experiment; the link text is the experiment's name in ClearML (`docs/data/brain_age_clearml_runs.csv`, written from the ClearML server by `labram/eval/research_summary.py::fetch_clearml_runs`). A CV run links to the ClearML project that holds its five fold experiments.

Table 2.1. Base configuration D versus the LaBraM fine-tuning recipe

Every configuration field on which LaBraM's TUAB fine-tuning recipe as published and the base configuration D differ:

- `LaBraM paper`: `labram/configs/defaults/finetune_tuab_paper.json`, the normal/abnormal fine-tune of LaBraM-base from the paper and the README;
- `D`: `labram/configs/defaults/finetune_tuab_age.json` as is, without overrides.

Both are rebuilt by `rs.resolve_config` and compared by `labram/eval/research_summary.py::config_diff`. Excluded:

- bookkeeping fields (paths, logging, infrastructure);
- classification-only fields a regression run never reads (`INERT_FOR_REGRESSION`, e.g. label smoothing);
- fields of a section that is off in both.

`group` is the part of the pipeline and `meaning` what the change does. Every other field is the paper's: LaBraM-base from `labram-base.pth`, batch 64, AdamW with weight decay 0.05, layer decay 0.65, drop path 0.1, mean pooling, cosine schedule. Changes that live in the data rather than the configuration (mains-interference removal, splits, labels) are in Table 4.2; how the best configuration differs from the paper's recipe is Table 4.1.

group	field	LaBraM paper	D	meaning
task	data.dataset	TUAB	TUAB_AGE	age regression on TUAB (age from the EDF header) instead of normal/abnormal
	model.task	classification	regression	scalar regression head, Huber loss on the z-scored age (Eqs. (3.2), (3.4))
data	data.data_format	pickle	npy	storage only: one float32 npy per recording, windows bit-identical to the pickles
input	labram_plus.enabled	False	True	LaBraM++ input block on: common average reference, CAR (Eq. (3.1))
	labram_plus.z_score_patches	True	False	LaBraM++ block with CAR only, no per-patch z-scoring
windows	data.eval_minutes	0	5	score each recording on its first 5 minutes after the trim
	data.trim_end_sec	0	60	drop the last 60 s of every recording
	data.trim_start_sec	0	60	drop the first 60 s of every recording
scoring	evaluation.agg_windows	none	mean	one prediction per recording = mean of its windows (Eq. (3.7))
optimizer	optimizer.lr	0.0005	0.0001	peak lr 1e-4 instead of 5e-4
schedule	optimizer.warmup_epochs	5	2	2 warmup epochs instead of 5
	trainer.epochs	50	15	15 epochs instead of 50

Loss functions. The loss is what the optimizer minimizes on every training batch. LaBraM as published fine-tunes TUAB as a binary normal/abnormal classifier. The head emits one logit s

per window with label $c \in \{0, 1\}$, and the loss is the binary cross-entropy (`nn.BCEWithLogitsLoss`, `labram/losses/classification.py`):

$$\ell_{\mathrm{BCE}}(s, c) = -\big[c \log \sigma(s) + (1 - c) \log(1 - \sigma(s))\big], \qquad \sigma(s) = \frac{1}{1 + e^{-s}} \tag{2.1}$$

For brain age the head instead emits a scalar $\hat{z}$ per 10 s window. The target is the z-scored age $z = (y - \mu)/\sigma$ (Eq. (3.2), with μ and σ from the training windows). The batch loss averages a per-window loss ℓ of the residual over the B windows of the batch (`labram/losses/regression.py::build_regression_criterion`):

$$\mathcal{L} = \frac{1}{B} \sum_{b=1}^{B} \ell(r_b), \qquad r_b = \hat{z}_b - z_b \tag{2.2}$$

Per-window loss. The default for D and every later run unless stated otherwise is the **Huber** loss with $\delta = 1$. It is quadratic for small residuals and linear for large ones, so a few recordings with a wrong or extreme age do not dominate the gradient. DX2 uses $\delta = 2$:

$$\ell_\delta(r) = \begin{cases} \frac{1}{2} r^2 & |r| \le \delta \\ \delta\,(|r| - \frac{1}{2}\delta) & \text{otherwise} \end{cases} \tag{2.3}$$

D2 uses the **L1** loss, the absolute error, which directly matches the MAE of Eq. (0.1):

$$\ell_1(r) = |r| \tag{2.4}$$

Mixup (M1 and every phase-2 run). Each window b of a batch is blended with a partner window $\pi(b)$ drawn from the same batch, usually from another recording with a different age. π permutes the **samples of the batch** (`torch.randperm` over the batch dimension), not the channels. The two windows are averaged point by point, channel with the same channel and time sample with the same time sample, and their targets with the same weight. For example, $\lambda = 0.7$ blends a 30-year-old's window with a 70-year-old's into target $0.7 \cdot 30 + 0.3 \cdot 70 = 42$. One $\lambda \sim \mathrm{Beta}(0.4, 0.4)$ is drawn per batch, usually near 0 or 1, so one window dominates. The loss of Eq. (2.2) is computed on the blended targets (`labram/train/mixup.py::mixup_batch`). A channel missing from either partner (T1/T2 in the `channel_drop` runs) is zeroed rather than blended:

$$\bar{x}_b = \lambda x_b + (1 - \lambda)\, x_{\pi(b)}, \qquad \bar{z}_b = \lambda z_b + (1 - \lambda)\, z_{\pi(b)} \tag{2.5}$$

Plain mixup draws the partner $\pi(b)$ uniformly at random. **C-Mixup** (M1c) prefers partners of similar age, so the blended target stays plausible. With $\sigma_m = 5$ years:

$$P\big(\pi(b) = j\big) \propto \exp\Big(-\frac{(y_b - y_j)^2}{2\sigma_m^2}\Big), \quad j \neq b \tag{2.6}$$

Soft labels (M4). The head emits a softmax p_k over age bins with centres c_k = 1, 2, …, 89 years. The target is a Gaussian q_k around the true age with $\sigma_s = 2.5$ years, the loss is the KL

divergence, and the prediction is the expected age (`SoftLabelRegressionLoss`). Centres and σ_s are z-scored like z:

$$q_k = \frac{\exp\!\big(-(c_k - z)^2 / 2\sigma_s^2\big)}{\sum_{k'} \exp\!\big(-(c_{k'} - z)^2 / 2\sigma_s^2\big)}, \qquad \ell_{\mathrm{KL}} = \sum_k q_k \log \frac{q_k}{p_k}, \qquad \hat{z} = \sum_k p_k \, c_k \tag{2.7}$$

Age-balanced loss (B1, B2). Rare ages (children, the very old) are up-weighted. ρ_k is the density of training windows in 1-year age bin k, smoothed by a Gaussian with $\sigma = 2$ years. The weight is $\rho^{-1/2}$ (B1) or ρ^{-1} (B2), capped at 10× the smallest weight and normalized to mean 1 over the training windows (`age_balance_weights` , `BalancedRegressionLoss`):

$$\mathcal{L} = \frac{1}{B} \sum_b w_{k(y_b)} \, \ell_\delta(r_b), \qquad w_k \propto \rho_k^{-1/2} \text{ or } \rho_k^{-1} \tag{2.8}$$

Codebook regularization: the spectral reconstruction and quantization terms (DX1, DX4, DX4b only). LaBraM's tokenizer (VQNSP) was pre-trained to encode each 1 s patch into a codeword and to decode from it the patch's Fourier amplitude and phase. These runs reuse its quantizer and decoder, frozen, on top of the fine-tuned encoder. Extra loss terms then penalize the encoder if its patch tokens drift away from what the frozen decoder can read back (`model.codebook_reg` , `labram/losses/codebook_regularized.py`). The age head still reads the mean patch token, as in D.

For each patch token n (one channel × one 1 s patch of 200 samples):

- $\tilde{A}_{n,k}$: the std-normalized FFT amplitude of the patch at frequency bin k (bin k = k Hz);
- $\hat{A}_{n,k}$: the decoder's reconstruction of it;
- h_n: the encoder token;
- e_n: its nearest codeword; sg is stop-gradient.

The terms are averaged over all tokens (and bins or feature dimensions) of the batch:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_\delta + w_A \, \mathcal{L}_{\mathrm{amp}} + w_P \, \mathcal{L}_{\mathrm{phase}} + w_Q \, \mathcal{L}_{\mathrm{q}}, \\ \mathcal{L}_{\mathrm{amp}} &= \mathrm{mean}_{n,\, k \le K} \left(\hat{A}_{n,k} - \tilde{A}_{n,k}\right)^2, \\ \mathcal{L}_{\mathrm{q}} &= \beta_q \, \mathrm{mean}_n \left(\mathrm{sg}(e_n) - h_n\right)^2, \qquad \beta_q = 1 \end{aligned} \tag{2.9}$$

$\mathcal{L}_{\mathrm{phase}}$ is the same MSE on the std-normalized FFT phase. K is the number of frequency bins kept: `loss.freq_fraction` × 200. The weights each run used, from its ClearML configuration (`model.codebook_reg.*_weight`):

- **DX1:** $w_A = 1$, $w_P = 0$, $w_Q = 0$, so amplitude reconstruction only. `freq_fraction` = 0.5 keeps $K = 100$ bins, 0–99 Hz. That is the whole non-redundant half of the spectrum: for a real signal, the upper 100 bins mirror the lower ones.

- **DX4:** $w_A = 0$, $w_P = 0$, $w_Q = 1$, so the quantization (commitment) term only. No reconstruction.
- **DX4b:** as DX4 with $w_Q = 50$.
- **Phase** (w_P) is 0 in all three. The phase reconstruction loss was never applied in any fine-tune of this study.
- **Every other run** has `model.codebook_reg.enabled=false`. The model then has no quantizer or decoder, and all three terms are absent. The `loss.amplitude_weight` / `phase_weight` / `embedding_weight` fields that read 1.0 in every run's ClearML configuration belong to the VQNSP tokenizer's own training and are never read by a fine-tune.

None of the three helped. Compared with D, the test MAE changes by +0.24 (DX1), +0.14 (DX4) and +0.36 (DX4b), all worse but all within seed noise, τ = 0.38 (Eq. (5.2)). Table 5.2 tests DX1 and DX4 in rows P17 and P18.

Multi-task (MT1). A second output predicts the normal/abnormal label c with the BCE of Eq. (2.1), weighted 0.5 (`MultiTaskAgeCriterion`). Selection still uses the age MAE:

$$\mathcal{L} = \mathcal{L}_\delta + 0.5\,\frac{1}{B}\sum_b \ell_{\mathrm{BCE}}(s_b, c_b) \tag{2.10}$$

Channel-dropout consistency (Exp4aug, Exp4aug15, cvaug). With probability 0.2 per batch, 1–2 of the T1/T2 channels are dropped from every sample. The age loss is computed on that dropped pass. The feature f the head reads should not change: it is pulled towards the feature f^{full} of a frozen all-channels pass, with $\lambda = 0.1$, over the samples A that lost a channel and the feature dimensions $d = 1..D$ (`labram/train/channel_drop.py`):

$$\mathcal{L} = \mathcal{L}_\delta^{\mathrm{drop}} + \lambda\,\frac{1}{|A|D}\sum_{b\in A}\sum_{d=1}^{D}\left(f_{b,d}^{\mathrm{full}} - f_{b,d}^{\mathrm{drop}}\right)^2 \tag{2.11}$$

Here $\mathcal{L}_\delta$ is Eq. (2.2) with the Huber loss of Eq. (2.3). Changes that do not touch the loss (EMA evaluation, layer decay, dropout, heads, input conditioning) keep Huber $\delta = 1$, with mixup wherever the run's `loss` column says so.

Table 2.2. Phase 1: TUAB-only runs

The experiment registry's master table (`docs/age_overfitting_experiments.md`, Table 12.1).

Columns:

- `ClearML`: the run's ClearML experiment, linked by its name.
- `family` and `change`: the question the run belongs to, and what it changes relative to D unless it names another run. Abbreviations:
  - CAR common average reference; ep epochs; lr learning rate;

    - ld layer decay; dp drop path; drop dropout; wd weight decay; clip gradient clipping at norm 1;
    - EMA evaluation with an exponential moving average of the weights;
    - LoRA r16 low-rank adapters of rank 16 instead of full fine-tuning;
    - CLS the class token; MLP a two-layer head.
- `loss`: the run's training loss, read from its ClearML configuration (Eqs. (2.2)–(2.11)).
- `train recs`: the number of training recordings, from the run's recorded `data_split.json` (`docs/data/brain_age_run_splits.csv`). It is the same 2,145 for every phase-1 run (the check below the table); only the windows per recording differ (A: whole recordings; C: 30 s windows).
- `val` / `test`: MAE (Eq. (0.1), years) at the epoch with the lowest validation MAE. Phase 1 is **TUAB only**, in training and evaluation:
    - `val`: TUAB's validation split, 556 recordings (291 normal, 265 abnormal);
    - `test`: TUAB's official evaluation set, 274 recordings.
- `Norm` / `Ab`: the same TUAB test MAE restricted to its 149 normal and 125 abnormal recordings.
- `β`: the bias slope of Eq. (0.5) on the 274 TUAB test recordings. Every model regresses towards the mean age. The fine-tuned runs sit between −0.46 and −0.33, and the from-scratch runs S1–S3 at −0.49 to −0.52. The collapsed runs (C2, C2-head, C2-last, DX3, F2; test MAE above 13) reach −0.8 to −1.0, meaning they predict almost a constant age.
- `e*/E`: the selected epoch $e^\star$ and the number of epochs E.

Same data split in every run. Every run records the recordings of its train, val and test splits (the `data_split` artifact of its ClearML experiment), saved here as SHA-1 digests of the sorted recording lists (`docs/data/brain_age_run_splits.csv`, `rs.split_fingerprint`, regenerated by `rs.fetch_run_splits`). The check below the table compares them: all phase-1 runs train, validate and test on identical recordings, with no patient in two splits. Only the number of windows differs, where the run changes the windows: A scores whole recordings, and C uses 30 s windows. Phase 2's `Control` uses the same split.

Why some `Norm` / `Ab` are missing ("–"). The per-cohort test metrics were added to the trainer on 2026-10-07 06:47 UTC (commit `62b159d`). 16 of the 21 runs without them started before that. The other five (E2, DX3, E1, E1c, E2c) are SageMaker jobs started 07:06–08:30 the same day whose job code did not include the change yet. All 21 logged only the overall val/test MAE. They saved no per-recording predictions, so the split by status cannot be recomputed from their logs; it needs a new inference pass of their checkpoints. Their `test` column is on the same 274 recordings as everyone else's. Table 2.4 fills these gaps: it re-scores every run's saved checkpoint, so `Norm` / `Ab` exist there for all of them.

The pre-study baseline (LaBraM fine-tuned with the original recipe adapted to regression) scored val 8.31 / test 9.04 (`docs/age_training_scenario_D.md`, Section 1).

run	ClearML	family	change	loss	train recs	val	test	Norm	Ab	β	e*/E
A	age_A_ep15_lr1e-4_fullrec_20261005_043322_643	Evaluation protocol	B with whole-recording val/test	Huber δ 1 (2.3)	2145	8.29	9.03	–	–	-0.46	2/12
B	age_B_eval5min_20261005_064356_435	Evaluation protocol	CAR off, 15 ep, lr 1e-4, 5-min eval (pre-D baseline)	Huber δ 1 (2.3)	2145	8.40	8.91	–	–	-0.43	2/6
D	age_D_eval5min_car_20261005_013250_973	Baseline	B + CAR (finetune_tuab_age.json)	Huber δ 1 (2.3)	2145	7.77	8.30	–	–	-0.40	1/15
D-npy	age_D_eval5min_car_npy_file_20261006_185443_758	Baseline (repro)	D on the npy format	Huber δ 1 (2.3)	2145	7.77	8.30	–	–	-0.40	1/15
D-cohorts	age_D_npy_cohorts_20261007_064710_242	Baseline (repro)	D, local, cohort metrics	Huber δ 1 (2.3)	2145	7.77	8.31	6.57	10.38	-0.40	1/15
E2	age_E2_npy_ep4_warm1_20261007_070635_244	Schedule	D, 4 ep / 1 warmup	Huber δ 1 (2.3)	2145	7.68	8.25	–	–	-0.37	3/4
L1	age_L1_ep12_mixup_ema_20261009_095540_331	Schedule	M12 on 12 ep / 2 warmup	Huber δ 1 + mixup (2.3, 2.5)	2145	7.51	8.47	6.65	10.63	-0.42	3/12
C2	age_C2_eval5min_regularized_20261005_012652_087	Classic regularizers	B + ld 0.5, dp 0.2, drop 0.1, wd 0.1, clip 1	Huber δ 1 (2.3)	2145	12.47	13.52	–	–	-0.81	14/15
DX3	age_DX3_npy_regularized_20261007_070603_726	Classic regularizers	D + C2 bundle	Huber δ 1 (2.3)	2145	12.83	14.90	–	–	-1.00	0/15
F1	age_F1_short_ld05_20261008_215552_344	Classic regularizers	E2 + layer decay 0.5	Huber δ 1 (2.3)	2145	7.74	8.36	6.42	10.68	-0.36	3/4
F2	age_F2_short_drop01_20261008_230105_724	Classic regularizers	E2 + dropout 0.1	Huber δ 1 (2.3)	2145	13.25	15.17	13.37	17.32	-0.97	0/4
F3	age_F3_short_dp02_20261008_232500_075	Classic regularizers	E2 + drop path 0.2	Huber δ 1 (2.3)	2145	7.70	8.41	6.31	10.91	-0.35	3/4
F4	age_F4_short_wd01_20261008_232504_307	Classic regularizers	E2 + weight decay 0.1	Huber δ 1 (2.3)	2145	7.66	8.24	6.49	10.32	-0.37	3/4
G1	age_G1_short_init1_20261008_230259_560	Head init	E2 + init_scale 1.0	Huber δ 1 (2.3)	2145	7.76	8.38	6.64	10.45	-0.37	3/4
H1	age_H1_short_ld075_20261009_003609_895	Layer decay	E2 + layer decay 0.75	Huber δ 1 (2.3)	2145	7.71	8.67	6.64	11.09	-0.40	1/4
H2	age_H2_short_ld085_20261009_003617_662	Layer decay	E2 + layer decay 0.85	Huber δ 1 (2.3)	2145	7.86	8.54	6.74	10.69	-0.42	3/4
C2-head	age_C2_eval5min_regularized_headonly_20261004_220639_025	Partial fine-tuning	C2, head only	Huber δ 1 (2.3)	2145	12.77	14.78	–	–	-0.99	5/8
C2-last	age_C2_eval5min_regularized_lastblock_head_20261005_000707_645	Partial fine-tuning	C2, last block + head	Huber δ 1 (2.3)	2145	12.50	13.64	–	–	-0.87	6/15
M3	age_M3_short_lora16_20261008_215551_826	Partial fine-tuning	E2 + LoRA r16, lr 1e-3, ld 1.0	Huber δ 1 (2.3)	2145	7.68	9.24	7.41	11.42	-0.44	3/4
M1	age_M1_short_mixup04_20261007_231716_154	Mixup / EMA	E2 + mixup 0.4	Huber δ 1 + mixup (2.3, 2.5)	2145	7.46	8.45	6.60	10.65	-0.41	3/4
M1c	age_M1c_short_cmixup04_s5_20261008_001050_123	Mixup / EMA	E2 + C-Mixup 0.4, σ 5 y	Huber δ 1 + C-Mixup σ 5 y (2.3, 2.5, 2.6)	2145	7.54	8.23	6.50	10.29	-0.39	3/4
M2	age_M2_short_ema_20261008_212509_991	Mixup / EMA	E2 + EMA eval (0.9995)	Huber δ 1 (2.3)	2145	7.58	8.17	6.35	10.35	-0.36	2/4
M12	age_M12_short_mixup_ema_20261008_204214_769	Mixup / EMA	E2 + mixup 0.4 + EMA eval	Huber δ 1 + mixup (2.3, 2.5)	2145	7.48	8.45	6.57	10.69	-0.42	3/4
D2	age_D2_eval5min_l1_20261005_000636_646	Loss	B + L1 loss	L1 (2.4)	2145	8.36	9.32	–	–	-0.46	6/15
DX2	age_DX2_huber_delta2_20261005_185121_984	Loss	D + Huber δ 2	Huber δ 2 (2.3)	2145	7.79	8.41	–	–	-0.38	3/15
M4	age_M4_short_softlabel_20261008_215551_240	Loss	E2 + soft labels σ 2.5 y	soft-label KL, σ 2.5 y (2.7)	2145	7.67	8.59	6.92	10.57	-0.40	3/4
B1	age_B1_short_balance_sqrtinv_20261009_124653_371	Loss	E2 + age-balanced, sqrt-inverse	Huber δ 1, age weights ρ^-1/2 (2.8)	2145	7.83	8.48	6.66	10.63	-0.37	2/4
B2	age_B2_short_balance_inv_20261009_134321_186	Loss	E2 + age-balanced, inverse	Huber δ 1, age weights ρ^-1 (2.8)	2145	7.96	8.57	6.70	10.80	-0.33	1/4
DX1	age_DX1_amp_freq05_20261006_022243_393	Codebook regularization	D + amplitude reconstruction (½ spectrum)	Huber δ 1 + amplitude reconstruction, w 1 (2.9)	2145	7.72	8.54	–	–	-0.39	4/15
DX4	age_DX4_quant_20261005_160655_990	Codebook regularization	D + quantization term, weight 1	Huber δ 1 + quantization, w 1 (2.9)	2145	7.67	8.44	–	–	-0.39	3/15
DX4b	age_DX4b_quant_w50_20261005_215817_973	Codebook regularization	D + quantization term, weight 50	Huber δ 1 + quantization, w 50 (2.9)	2145	7.78	8.66	–	–	-0.37	4/15
C	age_C_eval5min_win30s_20261004_235942_658	Augmentation / input	B + 30-s windows	Huber δ 1 (2.3)	2145	8.24	9.35	–	–	-0.42	8/15
D-clip1	age_D_npy_clip1_20261007_003433_521	Augmentation / input	D + clip 1.0	Huber δ 1 (2.3)	2145	7.70	8.36	–	–	-0.41	1/15
E1	age_E1_npy_random_crop_20261007_070629_845	Augmentation / input	D + random crop	Huber δ 1 (2.3)	2145	7.82	8.56	–	–	-0.40	3/15

run	ClearML	family	change	loss	train recs	val	test	Norm	Ab	β	e*/E
E1c	age_E1c_npy_random_crop_clip1_20261007_070930_530	Augmentation / input	D + random crop + clip 1.0	Huber δ 1 (2.3)	2145	7.72	8.13	–	–	-0.38	1/15
E2c	age_E2c_npy_ep4_warm1_clip1_20261007_083022_212	Augmentation / input	E2 + clip 1.0	Huber δ 1 (2.3)	2145	7.63	8.24	–	–	-0.38	3/4
D-cls	age_D_npy_cls_head_20261007_003458_637	Head	D, CLS token instead of mean pooling	Huber δ 1 (2.3)	2145	7.90	8.24	–	–	-0.38	2/15
D-clsmean	age_D_npy_cls_mean_head_20261007_003450_736	Head	D, [mean patch; CLS] concat	Huber δ 1 (2.3)	2145	7.80	8.35	–	–	-0.41	1/15
N2L	age_N2L_ep12_mixup04_clip1_attnhead_20261009_232420_478	Head	D + mixup 0.4 + clip 1 + attention head, 12 ep / 4 warmup	Huber δ 1 + mixup (2.3, 2.5)	2145	7.63	8.49	6.68	10.64	-0.43	4/12
MT1	age_MT1_short_mixup04_multitask_20261010_234501_492	Head (multi-task)	M1 + age/abnormal multi-task head (seed 0)	Huber δ 1 + mixup + 0.5 BCE abnormal (2.3, 2.5, 2.10)	2145	7.54	8.38	6.73	10.35	-0.40	3/4
MT1-seed1	age_MT1_short_mixup04_multitask_seed1_local_20261011_003814_186	Head (multi-task)	MT1, seed 1	Huber δ 1 + mixup + 0.5 BCE abnormal (2.3, 2.5, 2.10)	2145	7.62	8.54	6.75	10.69	-0.41	3/4
S1	age_S1_scratch_ep15_20261010_084032_716	From scratch	D, no pretrained weights, ld 1.0	Huber δ 1 (2.3)	2145	8.79	9.42	7.75	11.42	-0.49	14/15
S2	age_S2_scratch_d6_mlp_ep15_20261010_092116_377	From scratch	scratch, 6 blocks + MLP head, tuned (seed 0)	Huber δ 1 + mixup (2.3, 2.5)	2145	8.53	9.30	7.48	11.48	-0.50	12/15
S2-seed1	age_S2_scratch_d6_mlp_ep15_seed1_local_20261010_142248_698	From scratch	S2, seed 1	Huber δ 1 + mixup (2.3, 2.5)	2145	8.58	9.44	7.84	11.35	-0.50	14/15
S3	age_S3_scratch_d6_mlp_aug_ep15_local_20261010_215949_465	From scratch	S2 + label-preserving augmentation + ±800 µV clip	Huber δ 1 + mixup (2.3, 2.5)	2145	8.93	9.59	7.91	11.59	-0.52	9/15

```
Split check: all 45 phase-1 runs use the same recordings: train 2,145 (1,650 patients), val 556 (412), test 274 (251). Window counts differ only for A, C. Same split outside phase 1: Control.
```

Table 2.3. Phase 1 short list: one row per configuration, relative to D

Table 2.2 condensed to the 41 distinct configurations, each described and measured against D (`rs.phase1_short_list`).

- **Same results merged:** D-npy (the npy data format) and D-cohorts (a local rerun that also logs the cohort metrics) reproduce D exactly, so they are one row. D's `Norm` / `Ab` come from D-cohorts.
- **Seeds averaged:** MT1 + MT1-seed1 and S2 + S2-seed1 are the same configuration run twice; their row is the mean.
- **Every other configuration is kept**, grouped by family as in Table 2.2.

Columns:

- `change vs D` : what the configuration changes relative to D. `4/1 ep` is E2's schedule (4 epochs, 1 warmup) instead of D's 15 / 2; `no CAR` means without the common average reference.
- `val` , `test` , `Norm` , `Ab` , `β` , `e*/E` : as in Table 2.2.
- `train recs / patients` : training recordings / training patients, in thousands (K = 1,000): 2.15K / 1.65K, i.e. 2,145 recordings of 1,650 patients, for every phase-1 configuration (Table 2.2).
- `Δ val` , `Δ test` : Eq. (0.4) against D (val 7.77, test 8.30); negative is better.

- `vs D`: `better` / `worse` when $|\Delta \text{test}|$ exceeds the single-run noise threshold $\tau = 0.38$ (Eq. (5.2)), otherwise `n.s.`. τ is conservative for the seed-averaged rows.
- `runs`: the runs merged into the row.

family	config	change vs D	train recs / patients	val	Δ val	test	Δ test	vs D	Norm	Ab	β	e*/E	runs
Evaluation protocol	**A**	no CAR; scored on whole recordings	2.15K / 1.65K	8.29	0.52	9.03	0.73	worse	–	–	-0.46	2/12	
	B	no CAR	2.15K / 1.65K	8.40	0.63	8.91	0.61	worse	–	–	-0.43	2/6	
Baseline	**D**	— (reference)	2.15K / 1.65K	7.77	0.00	8.30	0.00	reference	6.57	10.38	-0.40	1/15	= D-npy, D-cohorts (same results)
Schedule	**E2**	4/1 ep	2.15K / 1.65K	7.68	-0.09	8.25	-0.05	n.s.	–	–	-0.37	3/4	
	L1	mixup 0.4 + EMA eval, 12 ep / 2 warmup	2.15K / 1.65K	7.51	-0.26	8.47	0.17	n.s.	6.65	10.63	-0.42	3/12	
Classic regularizers	**C2**	no CAR + layer decay 0.5, drop path 0.2, dropout 0.1, wd 0.1, clip 1	2.15K / 1.65K	12.47	4.70	13.52	5.22	worse	–	–	-0.81	14/15	
	DX3	C2's regularizer bundle (with CAR)	2.15K / 1.65K	12.83	5.06	14.90	6.60	worse	–	–	-1.00	0/15	
	F1	4/1 ep + layer decay 0.5	2.15K / 1.65K	7.74	-0.03	8.36	0.06	n.s.	6.42	10.68	-0.36	3/4	
	F2	4/1 ep + dropout 0.1	2.15K / 1.65K	13.25	5.48	15.17	6.87	worse	13.37	17.32	-0.97	0/4	
	F3	4/1 ep + drop path 0.2	2.15K / 1.65K	7.70	-0.07	8.41	0.11	n.s.	6.31	10.91	-0.35	3/4	
	F4	4/1 ep + weight decay 0.1	2.15K / 1.65K	7.66	-0.11	8.24	-0.06	n.s.	6.49	10.32	-0.37	3/4	
Head init	**G1**	4/1 ep + head init scale 1.0	2.15K / 1.65K	7.76	-0.01	8.38	0.08	n.s.	6.64	10.45	-0.37	3/4	
Layer decay	**H1**	4/1 ep + layer decay 0.75	2.15K / 1.65K	7.71	-0.06	8.67	0.37	n.s.	6.64	11.09	-0.40	1/4	
	H2	4/1 ep + layer decay 0.85	2.15K / 1.65K	7.86	0.09	8.54	0.24	n.s.	6.74	10.69	-0.42	3/4	
Partial fine-tuning	**C2-head**	C2, only the head trained	2.15K / 1.65K	12.77	5.00	14.78	6.48	worse	–	–	-0.99	5/8	
	C2-last	C2, only the last block + head trained	2.15K / 1.65K	12.50	4.73	13.64	5.34	worse	–	–	-0.87	6/15	
	M3	4/1 ep + LoRA r16 instead of full fine-tuning (lr 1e-3, ld 1.0)	2.15K / 1.65K	7.68	-0.09	9.24	0.94	worse	7.41	11.42	-0.44	3/4	
Mixup / EMA	**M1**	4/1 ep + mixup 0.4	2.15K / 1.65K	7.46	-0.31	8.45	0.15	n.s.	6.60	10.65	-0.41	3/4	
	M1c	4/1 ep + C-Mixup (σ 5 y)	2.15K / 1.65K	7.54	-0.23	8.23	-0.07	n.s.	6.50	10.29	-0.39	3/4	
	M2	4/1 ep + EMA eval	2.15K / 1.65K	7.58	-0.19	8.17	-0.13	n.s.	6.35	10.35	-0.36	2/4	
	M12	4/1 ep + mixup 0.4 + EMA eval	2.15K / 1.65K	7.48	-0.29	8.45	0.15	n.s.	6.57	10.69	-0.42	3/4	
Loss	**D2**	no CAR + L1 loss	2.15K / 1.65K	8.36	0.59	9.32	1.02	worse	–	–	-0.46	6/15	
	DX2	Huber δ 2	2.15K / 1.65K	7.79	0.02	8.41	0.11	n.s.	–	–	-0.38	3/15	
	M4	4/1 ep + soft labels (σ 2.5 y)	2.15K / 1.65K	7.67	-0.10	8.59	0.29	n.s.	6.92	10.57	-0.40	3/4	
	B1	4/1 ep + age-balanced loss (sqrt-inverse)	2.15K / 1.65K	7.83	0.06	8.48	0.18	n.s.	6.66	10.63	-0.37	2/4	
	B2	4/1 ep + age-balanced loss (inverse)	2.15K / 1.65K	7.96	0.19	8.57	0.27	n.s.	6.70	10.80	-0.33	1/4	
Codebook regularization	**DX1**	codebook amplitude reconstruction (w 1)	2.15K / 1.65K	7.72	-0.05	8.54	0.24	n.s.	–	–	-0.39	4/15	
	DX4	codebook quantization term (w 1)	2.15K / 1.65K	7.67	-0.10	8.44	0.14	n.s.	–	–	-0.39	3/15	
	DX4b	codebook quantization term (w 50)	2.15K / 1.65K	7.78	0.01	8.66	0.36	n.s.	–	–	-0.37	4/15	
Augmentation / input	**C**	no CAR + 30 s windows	2.15K / 1.65K	8.24	0.47	9.35	1.05	worse	–	–	-0.42	8/15	
	D-clip1	gradient clipping 1.0	2.15K / 1.65K	7.70	-0.07	8.36	0.06	n.s.	–	–	-0.41	1/15	
	E1	random crop	2.15K / 1.65K	7.82	0.05	8.56	0.26	n.s.	–	–	-0.40	3/15	
	E1c	random crop + clipping 1.0	2.15K / 1.65K	7.72	-0.05	8.13	-0.17	n.s.	–	–	-0.38	1/15	
	E2c	4/1 ep + clipping 1.0	2.15K / 1.65K	7.63	-0.14	8.24	-0.06	n.s.	–	–	-0.38	3/4	
Head	**D-cls**	class token instead of mean pooling	2.15K / 1.65K	7.90	0.13	8.24	-0.06	n.s.	–	–	-0.38	2/15	
	D-clsmean	[mean; class token] concatenated	2.15K / 1.65K	7.80	0.03	8.35	0.05	n.s.	–	–	-0.41	1/15	
	N2L	mixup 0.4 + clip 1 + attention head, 12 ep / 4 warmup	2.15K / 1.65K	7.63	-0.14	8.49	0.19	n.s.	6.68	10.64	-0.43	4/12	
Head (multi-task)	**MT1**	4/1 ep + mixup 0.4 + age/abnormal multi-task head	2.15K / 1.65K	7.58	-0.19	8.46	0.16	n.s.	6.74	10.52	-0.41	3/4	mean of 2 seeds (MT1, MT1-seed1)
From scratch	**S1**	no pretrained weights, layer decay 1.0	2.15K / 1.65K	8.79	1.02	9.42	1.12	worse	7.75	11.42	-0.49	14/15	
	S2	no pretrained weights, 6 blocks + MLP head, mixup, tuned	2.15K / 1.65K	8.55	0.79	9.37	1.07	worse	7.66	11.41	-0.50	13/15	mean of 2 seeds (S2, S2-seed1)

family	config	change vs D	train recs / patients	val	Δ val	test	Δ test	vs D	Norm	Ab	β	e*/E	runs
	S3	S2 + label-preserving augmentation + ±800 µV clip	2.15K / 1.65K	8.93	1.16	9.59	1.29	worse	7.91	11.59	-0.52	9/15	

Reading Table 2.3.

- **Nothing in phase 1 beats D on test beyond noise.** The lowest test MAEs (E1c 8.13, M2 8.17, M1c 8.23) are 0.07–0.17 years below D, well inside τ.
- **What clearly hurts** (`worse`):
  - dropping the common average reference: A, B, C and D2 are all without CAR;
  - the strong regularizer bundle and dropout, which collapse the model (C2, DX3, F2, and C2-head / C2-last);
  - LoRA instead of full fine-tuning (M3);
  - training from scratch (S1–S3).
- **Everything else is n.s.**: schedule, mixup / EMA, loss shape, codebook terms, heads, clipping and cropping all land within ±0.37 years of D on test.
- **Validation tells a slightly different story.** The mixup / EMA runs gain 0.2–0.3 years on val, the largest val gains of phase 1, which is why M1 was carried into phase 2. Their test gain is not significant. Section 5 tests each change against its own baseline.

Table 2.4. Phase 1 re-evaluated in eval mode: train, val and test

Every phase-1 run's saved best checkpoint was scored again, **without any training**, on its train, val and test splits:

- **Code and jobs:** `scripts/eval_runs.py` , launched as SageMaker jobs by `scripts/submit_eval_runs.py` .
- **What a job does:** download the run's checkpoint from S3, rebuild the model and data from the run's own `run_config.yaml` on the TUAB v3.0.0 data it trained on, and predict every window in eval mode (no dropout, no mixup).
- **Results:** `docs/data/phase1_reeval_metrics.csv` and `docs/data/phase1_reeval_split_check.csv` . The per-recording predictions stay outside git under the TUH data-use agreement.

All 45 runs are re-scored. Their val and test MAE reproduce the registry (Table 2.2) to within 0.01 years. Rows are merged as in Table 2.3: same-result runs and seed repeats are averaged, and `runs` counts them. Metrics are recording-level (Section 0):

- **train**: scored on the run's own evaluation budget per recording (the first 5 minutes after the trim, like val and test), so the three splits compare directly.
- **train (all windows)**: MAE over every training window, as the model saw them in training.

The generalization gap is

$$\text{gap} = \text{MAE}_{\text{test}} - \text{MAE}_{\text{train}} \tag{2.12}$$

where a large gap is memorization. `split = recorded` is ✓ when every merged run's rebuilt train, val and test recordings are exactly those of its recorded `data_split.json`.

Columns: MAE (Eq. (0.1), years) and R^2 (Eq. (0.2)) per split; `test Norm` / `test Ab` over the 149 normal and 125 abnormal test recordings.

family	config	train MAE	train R²	train MAE (all windows)	val MAE	val R²	test MAE	test R²	test Norm MAE	test Norm R²	test Ab MAE	test Ab R²	gap (test − train)	split = recorded	runs
Evaluation protocol	A	3.54	0.92	3.54	8.29	0.57	9.03	0.57	7.02	0.69	11.43	0.33	5.49	✓	1
	B	4.00	0.90	3.54	8.40	0.55	8.91	0.60	6.94	0.71	11.26	0.37	4.91	✓	1
Baseline	D	6.05	0.80	5.79	7.77	0.61	8.30	0.65	6.57	0.74	10.37	0.45	2.25	✓	3
Schedule	E2	3.41	0.93	2.95	7.67	0.62	8.25	0.64	6.51	0.74	10.33	0.43	4.84	✓	1
	L1	4.51	0.88	4.21	7.51	0.63	8.47	0.62	6.65	0.72	10.63	0.40	3.96	✓	1
Classic regularizers	C2	13.29	0.15	13.36	12.47	0.06	13.52	0.14	12.58	0.13	14.65	-0.09	0.24	✓	1
	DX3	14.60	0.00	14.60	12.83	0.00	14.90	-0.01	14.39	-0.06	15.51	-0.24	0.30	✓	1
	F1	5.17	0.84	4.79	7.74	0.61	8.36	0.63	6.42	0.75	10.68	0.40	3.19	✓	1
	F2	14.91	-0.03	14.88	13.25	-0.04	15.17	-0.09	13.37	0.03	17.32	-0.51	0.27	✓	1
	F3	4.77	0.86	4.35	7.70	0.61	8.41	0.62	6.31	0.75	10.91	0.38	3.64	✓	1
	F4	3.41	0.93	2.95	7.66	0.62	8.24	0.64	6.49	0.74	10.32	0.43	4.83	✓	1
Head init	G1	3.36	0.93	2.91	7.76	0.60	8.38	0.63	6.64	0.73	10.45	0.43	5.02	✓	1
Layer decay	H1	4.06	0.90	3.67	7.71	0.61	8.67	0.60	6.64	0.71	11.09	0.38	4.62	✓	1
	H2	1.49	0.99	1.11	7.86	0.59	8.54	0.61	6.74	0.69	10.69	0.41	7.05	✓	1
Partial fine-tuning	C2-head	14.54	0.01	14.53	12.77	0.02	14.78	0.01	14.07	-0.02	15.62	-0.25	0.24	✓	1
	C2-last	13.86	0.08	13.79	12.50	0.07	13.64	0.18	13.24	0.10	14.12	0.03	-0.22	✓	1
	M3	1.61	0.99	1.23	7.68	0.60	9.24	0.57	7.41	0.65	11.42	0.35	7.63	✓	1
Mixup / EMA	M1	4.50	0.88	4.18	7.46	0.63	8.45	0.61	6.60	0.73	10.65	0.39	3.95	✓	1
	M1c	4.38	0.88	4.02	7.54	0.62	8.23	0.63	6.50	0.72	10.29	0.43	3.85	✓	1
	M2	4.11	0.90	3.70	7.58	0.63	8.17	0.65	6.35	0.75	10.35	0.44	4.07	✓	1
	M12	4.67	0.87	4.36	7.48	0.63	8.45	0.61	6.57	0.72	10.69	0.39	3.78	✓	1
Loss	D2	1.78	0.97	1.40	8.37	0.54	9.32	0.55	7.87	0.62	11.04	0.35	7.54	✓	1
	DX2	3.19	0.94	2.72	7.79	0.60	8.41	0.62	6.49	0.72	10.70	0.41	5.22	✓	1
	M4	2.25	0.96	1.93	7.67	0.61	8.59	0.61	6.92	0.68	10.57	0.42	6.34	✓	1
	B1	4.10	0.91	3.66	7.83	0.60	8.48	0.61	6.66	0.72	10.63	0.39	4.38	✓	1
	B2	5.56	0.84	5.10	7.96	0.58	8.57	0.60	6.70	0.72	10.80	0.36	3.01	✓	1
Codebook regularization	DX1	3.48	0.93	3.01	7.72	0.60	8.54	0.63	7.06	0.70	10.30	0.45	5.06	✓	1
	DX4	3.24	0.94	2.88	7.67	0.61	8.44	0.64	6.88	0.72	10.30	0.45	5.21	✓	1
	DX4b	3.15	0.94	2.64	7.78	0.60	8.66	0.61	6.89	0.72	10.77	0.39	5.51	✓	1
Augmentation / input	C	2.68	0.96	2.21	8.24	0.55	9.35	0.56	7.42	0.68	11.66	0.31	6.67	✓	1
	D-clip1	6.12	0.79	5.86	7.70	0.61	8.36	0.65	6.74	0.73	10.29	0.47	2.24	✓	1
	E1	3.49	0.93	3.04	7.82	0.60	8.56	0.61	6.65	0.71	10.85	0.40	5.08	✓	1
	E1c	6.25	0.78	5.93	7.72	0.61	8.13	0.64	6.31	0.73	10.29	0.45	1.87	✓	1
	E2c	3.56	0.92	3.10	7.63	0.62	8.24	0.64	6.49	0.73	10.33	0.43	4.68	✓	1
Head	D-cls	4.44	0.88	4.05	7.90	0.59	8.24	0.61	6.69	0.71	10.09	0.39	3.80	✓	1
	D-clsmean	6.03	0.80	5.78	7.80	0.61	8.35	0.64	6.62	0.73	10.41	0.45	2.32	✓	1
	N2L	4.47	0.88	4.22	7.63	0.62	8.49	0.61	6.68	0.71	10.64	0.40	4.01	✓	1
Head (multi-task)	MT1	4.59	0.88	4.26	7.58	0.63	8.46	0.62	6.74	0.72	10.52	0.40	3.87	✓	2
From scratch	S1	3.28	0.93	3.07	8.79	0.51	9.42	0.55	7.75	0.63	11.42	0.33	6.14	✓	1
	S2	5.38	0.84	5.16	8.56	0.54	9.37	0.54	7.66	0.63	11.42	0.32	4.00	✓	2
	S3	6.67	0.76	6.46	8.94	0.51	9.59	0.54	7.91	0.63	11.59	0.31	2.92	✓	1

Reading Table 2.4.

- **The splits are verified.** All 45 runs rebuild exactly their recorded train (2,145), val (556) and test (274) recordings, so every phase-1 comparison is on the same data.
- **The stored results are reproduced.** Re-scoring the saved checkpoints gives the logged val / test MAE to within 0.01 years. The duplicates merged in Table 2.3 (D, D-npy, D-cohorts) also agree on train.
- **Memorization is common.** Most fine-tuned configurations reach a train MAE of 3–6 years against 8.1–8.7 on test, a gap of about 2–5.5 years (Eq. (2.12)).

    - The extreme cases fit the training recordings almost perfectly yet do not improve test: H2 (layer decay 0.85: train 1.5, test 8.5), M3 (LoRA: 1.6 vs 9.2), D2 (L1: 1.8 vs 9.3) and M4 (soft labels: 2.3 vs 8.6).
    - The smallest gaps among the working runs belong to the D-schedule runs with clipping or cropping (E1c 1.9, D-clip1 2.2), and to D itself (2.3). All three select an early epoch (`e*/E` in Table 2.3).
- **The collapsed runs underfit, they do not overfit.** C2, DX3, F2, C2-head and C2-last have $R^2 \leq 0.15$ on train as well as test. They predict about the mean age for everyone.
- **Normal / abnormal for every run.** On test, R^2 is 0.62–0.75 on normal recordings and only 0.31–0.47 on abnormal ones for every configuration that did not collapse. Abnormal recordings stay the hard group throughout phase 1.

What the age head reads: the feature space (patch tokens, class token, pathology labels)

Tables 2.5–2.9 have a `feature space` column. It says which vector the linear age head reads, so EEG-only runs and runs given clinical labels are not compared unawares. Let $t_{c,a} \in \mathbb{R}^{200}$ be the final-layer token of channel c and 1-s patch a (C channels, A patches per window), and t_{cls} the class token, both after the final LayerNorm (`labram/models/neural_transformer.py`):

$$
\begin{aligned}
h_{\text{patch}} &= \frac{1}{CA}\sum_{c=1}^{C}\sum_{a=1}^{A} t_{c,a}, \qquad h_{\text{cls}} = t_{\text{cls}} \\
c &= \begin{bmatrix} s \\ p \end{bmatrix} \in \{0,1\}^{3+14} \\
\hat{z} &= w^{\top} h + b \qquad (\text{EEG only}) \\
\hat{z} &= w^{\top} \begin{bmatrix} h \\ W_c\, c \end{bmatrix} + b, \quad W_c \in \mathbb{R}^{32\times 17} \qquad (\text{with path label})
\end{aligned}
\tag{2.13}
$$

- **`patch_tokens`** : $h = h_{\text{patch}}$, the mean over all patch tokens. This is LaBraM's default (`model.use_mean_pooling=true`).
- **`cls_token`** : $h = h_{\text{cls}}$, the class token, which attends to every patch token in each layer (`model.use_mean_pooling=false`).
- **`+ path_label`** : the head also reads the recording's clinical **pathology-label vector** c, through a learned 32-d projection W_c (`model.session_features=head`). How c is built (`dataset_maker/make_condition_features.py` , saved per cohort as `conditions_v1.npz` , loaded by `labram/data/session_features.py`):

    1. **Status s, one-hot over 3:** normal / abnormal / not available. It is the TUAB folder label for a TUAB recording. For a TUEG session in the NEDC clinical sheet it is the

report finding: "abnormal due to" filled → abnormal, empty → normal. Otherwise it is not available (Eq. (1.2)).

2. **Pathology p, multi-hot over 14:** one entry per diagnosis group of the NEDC sheet (Table 1.9): clinical seizure, electrographic seizure, epilepsy, stroke, TBI, substance use, bipolar, anxiety, depression, schizophrenia, dementia (with Alzheimer's), multiple sclerosis, and other (Parkinson's, ADHD, under 50 recordings each). Entry 0 is "healthy or not available": it is set when no diagnosis is flagged or the session is not in the sheet. Several entries can be 1 at once.
3. **Concatenate** $c = [s; p]$, 17 numbers. They are categorical, so nothing is standardized. In training, mixup blends c along with the EEG and the age (Eq. (3.3)).

The labels are inputs, not the target. They hold no age, so they leak no age. But a `path_label` model needs the recording's clinical status and diagnoses at prediction time, also on TUAB test. It is therefore not an EEG-only model. Its MAEs compare with the EEG-only rows only with that in mind.

- **`+ feat_tokens`** (Exp1t): the top-10 handcrafted EEG features enter as extra tokens, standardized on the cohort's training split. **`features only (no EEG)`** (Exp1f): an MLP on handcrafted features alone.

Table 2.5. Phase 2: TUH cohort runs

One row per run of `docs/data/tuh_cohort_eval_v1.csv`, from `labram/eval/research_summary.py::PHASE2`:

- `exp`: the run, linked to its ClearML experiment (click the name);
- `cohort` (Table 1.4) and `train recs / patients`, its training recordings and patients, both in thousands with two decimals (2.15K / 1.65K = 2,145 recordings of 1,650 patients; the run's recorded split, `docs/data/brain_age_run_splits.csv`); `epochs` / `warmup`;
- `feature space`: what the age head reads (Eq. (2.13)): `patch_tokens`, `cls_token`, and `+ path_label` when it also receives the recording's status and pathology vector;
- `parent` and `change`: the run it is compared with in Section 5 and what differs;
- `loss`: the training loss (Eqs. (2.2)–(2.11)). Every phase-2 run is M1, Huber + mixup; the T1/T2 dropout runs add the consistency term.

Metrics:

- `test all` / `normal` / `abnormal`: MAE (Eq. (0.1)) on the **TUAB test set only**, all 274 recordings and its 149 normal / 125 abnormal ones. The phase-2 runs train on TUEG too, but none of them is tested on TUEG recordings: TUAB test is the one set every run shares (Section 0).
- `R² normal`: Eq. (0.2) on the normal TUAB test recordings;
- `TUEG val v2`: MAE on the TUEG part of the shared v2 validation split (2,325 recordings no v2/v3 model trained on), available only for runs scored on it. It is the only TUEG number

in the table. Each run's own validation split is not shown, because it differs from cohort to cohort.

- **The last three rows are phase 3**, for comparison on the same TUAB test set: each 5-fold CV experiment (Table 2.7) as the median of its 5 fold models (`ens.` , Eq. (3.8)). Their `train recs / patients` is the mean over the fold models (3 of the 5 folds of the 29,035-recording pool), and their `R² normal` is the ensemble's.
- `cal. all` / `cal. normal` / `cal. abnormal` : the TUAB test MAE after a **leak-free age calibration**, and `cal. a / b` its two coefficients. Per experiment r it takes three steps.
  1. **Fit set** F_r: normal recordings that r never trained on and that are not TUAB test. For a single-split run (every row above the CV rows) these are the normal recordings of its **own validation split**. For a CV row they are its out-of-fold predictions of normal recordings (each recording predicted by the fold model whose in-pool test fold held it out, Table 3.5).
  2. **Fit** the least-squares line from predicted to true age over F_r, with $\hat{y}_j$ the run's recording-level prediction and y_j the true age (years):

$$(a_r, b_r) = \arg\min_{a,b} \sum_{j \in F_r} \left(y_j - a\,\hat{y}_j - b\right)^2, \qquad a_r = \frac{\mathrm{Cov}_{F_r}(\hat{y}, y)}{\mathrm{Var}_{F_r}(\hat{y})}, \quad b_r = \bar{y}_{F_r} - a_r\,\bar{\hat{y}}_{F_r}$$

$$\hat{y}_i^{\,\mathrm{cal}} = a_r\,\hat{y}_i + b_r, \qquad \mathrm{MAE}^{\mathrm{cal}}(S) = \frac{1}{|S|} \sum_{i \in S} \left|\hat{y}_i^{\,\mathrm{cal}} - y_i\right|$$

  3. **Apply** it to each TUAB test prediction $\hat{y}_i$ and score the result with Eq. (0.1) on S = all, normal or abnormal TUAB test recordings.

The TUAB test true age y_i appears only in the final MAE, exactly as in every raw MAE. It is never used to fit a_r, b_r. The check below the table confirms that no fit set shares a recording or a patient with TUAB test. Eq. (2.14) is Eq. (3.11) of Section 3. Caveat for the single-split runs: their validation split also chose the best epoch, so their fit set was used once for model selection. It still holds no test data. The single-split numbers come from re-scoring each run's checkpoint on its val and test splits (`scripts/eval_runs.py` ; `docs/data/phase2_calibration.csv`). The re-score reproduces every run's raw test MAE within 0.01 except Exp3q (7.37 vs 7.34): it applies Exp3q's signal-quality filter to the test windows (8,210 of 8,220), which the table's number did not. Exp2's cohort has no test split, so it is scored on the same 274 TUAB test recordings through `normal_both_v1` , with its own training statistics. Table 3.5 gives the other corrections, including the gap correction that uses the test ages.

- No `β` : the cohort evaluation (`docs/data/tuh_cohort_eval_v1.csv`) stores MAE, RMSE and R² but not the bias slope, so β is reported for phase 1 only.

Same split or not. The check below the table groups the runs by their recorded train/val/test recordings, as for Table 2.2:

- Runs on the same cohort share one split. Exp1f's `normal_both_v1_meta` holds the same recordings as `normal_both_v1`, and `all_both_v3_compact` the same as `all_both_v3`.
- **Exp3q is the exception** within `all_both_v1`. Its signal-quality filter drops windows beyond ±800 µV and windows with a flat channel, and recordings left without windows drop out. That leaves 19,815 instead of 20,368 train recordings and 2,836 instead of 2,905 val recordings, and its test loses 10 windows but keeps all 274 recordings.
- Splits differ across cohorts by design (Table 1.4): `v1`, `v2` and `v3` assign TUEG patients differently, so their val MAEs do not compare.
- **Exp4anymiss** trains on `all_both_v3_anymiss` (`dataset_maker/extend_cohort.py`). That is `all_both_v3` plus 2,003 training recordings of 545 training patients that lack A1/A2 or other channels: 1,115 without A1/A2, 876 without A1/A2/T1/T2, and 12 others. Its val and test recordings are identical to Exp4aug15's (same digests in `docs/data/brain_age_run_splits.csv`), so its own split differs only in training.
- **Exp4noear** uses Exp4anymiss's cohort and split (identical train, val and test digests) with A1, A2, T1 and T2 removed from every recording (`data.drop_channels`, 19 channels), no channel augmentation, and 8 / 2 epochs. It was trained locally.
- **Exp4noearEEG** is Exp4noear without the status / pathology input (`model.session_features=off`), an EEG-only model. It used the same split and was trained on SageMaker.
- Every run with a test split tests on the same 274 TUAB recordings. Exp2 (`normal_TUEG_v1`) has no test split.

exp	cohort	train recs / patients	feature space	test all	normal	abnormal	R^2 normal	TUEG val v2	cal. all	cal. normal	cal. abnormal	cal. a / b	epochs	parent	change
Control	TUAB_v3.0.1_interp50-60	2.15K / 1.65K	patch_tokens	8.23	6.42	10.39	0.75	9.29	8.12	6.30	10.29	1.09 / -3.8	4 / 1		M1 on TUAB only (the TUAB baseline of phase 2)
Exp1	normal_both_v1	5.45K / 3.00K	patch_tokens	8.03	5.99	10.47	0.78	8.68	7.96	5.95	10.36	1.01 / 0.4	4 / 1	Control	train on normal recordings only: TUAB normal + TUEG normal
Exp2	normal_TUEG_v1	3.76K / 2.15K	patch_tokens	8.00	6.06	10.31	0.78	8.75	7.91	5.95	10.25	1.10 / -2.0	4 / 1	Exp1	TUEG normal only (no TUAB in training)
Exp1b	normal_both_v1	5.45K / 3.00K	patch_tokens	8.11	6.09	10.52	0.77	8.66	8.15	6.13	10.54	0.96 / 2.0	10 / 3	Exp1	10 epochs / 3 warmup instead of 4 / 1
Exp1t	normal_both_v1	5.45K / 3.00K	patch_tokens + feat_tokens	8.80	7.17	10.73	0.70	–	9.00	7.35	10.97	0.93 / 2.9	4 / 1	Exp1	classical EEG features as extra transformer tokens
Exp1f	normal_both_v1_meta	5.45K / 3.00K	features only (no EEG)	8.13	6.21	10.41	0.77	–	8.21	6.28	10.51	0.93 / 2.9	4 / 1	Exp1	classical EEG features only, MLP head, no EEG encoder
Exp3	all_both_v1	20.37K / 9.07K	patch_tokens	7.26	5.57	9.28	0.80	–	7.23	5.53	9.26	1.03 / -1.6	4 / 1	Control	add every TUEG recording (normal, abnormal, unlabelled): ~10x training data
Exp3b	all_both_v1	20.37K / 9.07K	patch_tokens	7.70	6.17	9.51	0.75	–	7.48	5.96	9.29	1.06 / -2.3	10 / 3	Exp3	10 epochs / 3 warmup instead of 4 / 1
Exp3q	all_both_v1	19.82K / 8.97K	patch_tokens	7.34	5.69	9.31	0.79	–	7.29	5.67	9.22	1.03 / -1.4	4 / 1	Exp3	drop windows beyond ±800 µV or with a flat channel
Exp3c	all_both_v1	20.37K / 9.07K	patch_tokens	7.29	5.66	9.22	0.79	–	7.24	5.61	9.19	1.03 / -1.5	4 / 1	Exp3	clip the input to ±800 µV inside the model
Exp3v2	all_both_v2	20.47K / 9.09K	patch_tokens	7.42	5.46	9.75	0.80	8.00	7.33	5.37	9.68	1.04 / -1.7	4 / 1	Exp3	cohort v2: one TUEG patient split shared by every cohort (data bookkeeping only)
Exp3cond	all_both_v2	20.47K / 9.09K	patch_tokens + path_label	7.28	5.39	9.53	0.81	7.90	7.15	5.30	9.35	1.04 / -1.6	4 / 1	Exp3v2	status + pathology vectors concatenated to the pooled feature
Exp3clscond	all_both_v2	20.47K / 9.09K	cls_token + path_label	7.25	5.36	9.51	0.81	7.98	7.18	5.27	9.45	1.04 / -1.3	4 / 1	Exp3cond	head on the class token instead of the mean patch token
Exp4drop	all_both_v3	25.61K / 12.02K	patch_tokens + path_label	7.28	5.26	9.68	0.82	7.76	7.10	5.04	9.55	1.07 / -3.0	4 / 1	Exp3cond	cohort v3 (+ recordings missing only T1/T2) on 21 channels (T1/T2 dropped everywhere)
Exp4base	all_both_v3	25.61K / 12.02K	patch_tokens	7.36	5.62	9.44	0.81	7.86	7.29	5.56	9.36	1.03 / -1.1	6 / 2	Exp3v2	cohort v3, 23 channels with missing T1/T2 left out per sample;

exp	cohort	train recs / patients	feature space	test all	normal	abnormal	R² normal	TUEG val v2	cal. all	cal. normal	cal. abnormal	cal. a / b	epochs	parent	change	
															speed settings; 6 / 2 epochs	
Exp4basec	all_both_v3_compact	25.61K / 12.02K	patch_tokens	7.42	5.52	9.68	0.81	7.85	7.31	5.41	9.57	1.05 / -2.0	6 / 2	Exp4base	compact copy of the cohort, same recorded split (reproduction check)	
Exp4ctrl	all_both_v3	25.61K / 12.02K	cls_token + path_label	7.30	5.35	9.63	0.81	7.92	7.16	5.18	9.51	1.06 / -2.3	10 / 3	Exp4base	class token + condition vectors, 10 / 3 epochs with early stopping	
Exp4aug	all_both_v3	25.61K / 12.02K	cls_token + path_label	7.29	5.45	9.49	0.82	7.85	7.13	5.26	9.36	1.07 / -3.1	6 / 2	Exp4ctrl	T1/T2 batch dropout (p 0.2) + consistency loss (λ 0.1); 6 / 2 epochs	c
Exp4aug15	all_both_v3	25.61K / 12.02K	cls_token + path_label	7.17	5.38	9.29	0.81	7.91	7.00	5.17	9.18	1.07 / -2.7	15 / 5	Exp4aug	15 / 5 epochs with early stopping (stopped at 11)	c
Exp4anymiss	all_both_v3_anymiss	27.62K / 12.19K	cls_token + path_label	7.21	5.47	9.30	0.82	7.75	7.13	5.36	9.24	1.04 / -1.7	15 / 5	Exp4aug15	+2,003 training recordings missing A1/A2 or other channels; any 1-3 channels dropped (p 0.2) from epoch 4 (stopped at 12)	c
Exp4noear	all_both_v3_anymiss	27.62K / 12.19K	cls_token + path_label	7.41	5.55	9.64	0.81	8.02	7.31	5.42	9.58	1.07 / -2.6	8 / 2	Exp4anymiss	A1/A2/T1/T2 removed from every recording (19 channels), no channel augmentation; 8 / 2 epochs	
Exp4noearEEG	all_both_v3_anymiss	27.62K / 12.19K	cls_token	7.48	5.75	9.54	0.81	7.99	7.45	5.74	9.50	1.02 / -0.5	8 / 2	Exp4noear	no status / pathology input (EEG only); otherwise as Exp4noear	
Exp4mt	all_both_v3_anymiss	27.62K / 12.19K	cls_token (EEG only; multi-task head)	7.62	5.93	9.64	0.79	8.32	7.50	5.73	9.60	1.05 / -3.2	8 / 2	Exp4noearEEG	multi-task head: + status and 13 diagnosis groups as targets (BCE over all known labels)	p
Exp4mtbal	all_both_v3_anymiss	27.62K / 12.19K	cls_token (EEG only; multi-task head)	7.59	5.86	9.65	0.79	8.33	7.51	5.76	9.59	1.04 / -1.8	8 / 2	Exp4mt	class-balanced pathology loss (groups present in the batch, equally weighted)	p
cvbase (CV ens.)	all_both_v3_cvcompact	17.42K / 8.13K	patch_tokens	7.44	5.75	9.46	0.80	–	7.28	5.58	9.31	1.09 / -4.0	5 / 1	Exp4base	5-fold CV of Exp4base	

exp	cohort	train recs / patients	feature space	test all	normal	abnormal	R^2 normal	TUEG val v2	cal. all	cal. normal	cal. abnormal	cal. a / b	epochs	parent	change
															(5 / 1 epochs)
cvctrl (CV ens.)	all_both_v3_cvcompact	17.42K / 8.13K	cls_token + path_label	7.10	5.31	9.23	0.82	–	6.98	5.19	9.11	1.06 / -2.7	10 / 3	cvbase	class token + condition vectors, 10 / 3 epochs (CV of Exp4ctrl)
cvaug (CV ens.)	all_both_v3_cvcompact	17.42K / 8.13K	cls_token + path_label	7.23	5.43	9.37	0.81	–	7.11	5.30	9.28	1.06 / -2.3	10 / 3	cvctrl	T1/T2 batch dropout + consistency loss λ 0.1 (CV of Exp4aug)
cvaugval (CV ens.)	all_both_v3_cvcompact	23.23K / 10.84K	cls_token + path_label	7.22	5.36	9.44	0.82	–	7.04	5.16	9.28	1.07 / -2.3	10 / 3	cvaug	val-only CV: val = fold k, the other 4 folds train (~23.2K recordings), test = TUAB

```
Calibration fit sets (val, normal recordings) of the single-split runs: 291–1,388 recordings; shared with TUAB test: 0 recordings, 0 patients.
Re-scored raw TUAB test MAE vs the table: max difference 0.03 years (Exp3q +0.03).
same split: Exp1, Exp1b, Exp1t, Exp1f  (cohort normal_both_v1, normal_both_v1_meta)
same split: Exp3v2, Exp3cond, Exp3clscond  (cohort all_both_v2)
same split: Control  (cohort TUAB_v3.0.1_interp50-60)
same split: Exp4drop, Exp4base, Exp4basec, Exp4ctrl, Exp4aug, Exp4aug15  (cohort all_both_v3, all_both_v3_compact)
same split: Exp4anymiss, Exp4noear, Exp4noearEEG, Exp4mt, Exp4mtbal  (cohort all_both_v3_anymiss)
same split: Exp3q  (cohort all_both_v1)
same split: Exp3, Exp3b, Exp3c  (cohort all_both_v1)
same split: Exp2  (cohort normal_TUEG_v1)
TUAB test recordings identical in every phase-2 run with a test split: True
```

Table 2.6. Phase 2 and CV: the best runs (normal < 5.8, abnormal < 9.8)

The rows of Table 2.5 whose raw TUAB test MAE (Eq. (0.1)) is below **5.8 years on normal** and below **9.8 years on abnormal** recordings, sorted by `test all`. The columns are those of Table 2.5. `exp` links to ClearML, and `feature space` is defined by Eq. (2.13). The `cal.` columns are the leak-free calibration of Eq. (2.14).

exp	cohort	train recs / patients	feature space	test all	nor-mal	abnor-mal	R^2 normal	cal. all	cal. normal	cal. abnormal
cvctrl (CV ens.)	all_both_v3_cvcompact	17.42K / 8.13K	cls_token + path_label	7.10	5.31	9.23	0.82	6.98	5.19	9.11
Exp4aug15	all_both_v3	25.61K / 12.02K	cls_token + path_label	7.17	5.38	9.29	0.81	7.00	5.17	9.18
Exp4anymiss	all_both_v3_anymiss	27.62K / 12.19K	cls_token + path_label	7.21	5.47	9.30	0.82	7.13	5.36	9.24
cvaugval (CV ens.)	all_both_v3_cvcompact	23.23K / 10.84K	cls_token + path_label	7.22	5.36	9.44	0.82	7.04	5.16	9.28
cvaug (CV ens.)	all_both_v3_cvcompact	17.42K / 8.13K	cls_token + path_label	7.23	5.43	9.37	0.81	7.11	5.30	9.28
Exp3clscond	all_both_v2	20.47K / 9.09K	cls_token + path_label	7.25	5.36	9.51	0.81	7.18	5.27	9.45
Exp3	all_both_v1	20.37K / 9.07K	patch_tokens	7.26	5.57	9.28	0.80	7.23	5.53	9.26
Exp4drop	all_both_v3	25.61K / 12.02K	patch_tokens + path_label	7.28	5.26	9.68	0.82	7.10	5.04	9.55
Exp3cond	all_both_v2	20.47K / 9.09K	patch_tokens + path_label	7.28	5.39	9.53	0.81	7.15	5.30	9.35
Exp3c	all_both_v1	20.37K / 9.07K	patch_tokens	7.29	5.66	9.22	0.79	7.24	5.61	9.19
Exp4aug	all_both_v3	25.61K / 12.02K	cls_token + path_label	7.29	5.45	9.49	0.82	7.13	5.26	9.36
Exp4ctrl	all_both_v3	25.61K / 12.02K	cls_token + path_label	7.30	5.35	9.63	0.81	7.16	5.18	9.51
Exp3q	all_both_v1	19.82K / 8.97K	patch_tokens	7.34	5.69	9.31	0.79	7.29	5.67	9.22
Exp4base	all_both_v3	25.61K / 12.02K	patch_tokens	7.36	5.62	9.44	0.81	7.29	5.56	9.36
Exp4noear	all_both_v3_anymiss	27.62K / 12.19K	cls_token + path_label	7.41	5.55	9.64	0.81	7.31	5.42	9.58
Exp3v2	all_both_v2	20.47K / 9.09K	patch_tokens	7.42	5.46	9.75	0.80	7.33	5.37	9.68
Exp4basec	all_both_v3_compact	25.61K / 12.02K	patch_tokens	7.42	5.52	9.68	0.81	7.31	5.41	9.57
cvbase (CV ens.)	all_both_v3_cvcompact	17.42K / 8.13K	patch_tokens	7.44	5.75	9.46	0.80	7.28	5.58	9.31
Exp4noearEEG	all_both_v3_anymiss	27.62K / 12.19K	cls_token	7.48	5.75	9.54	0.81	7.45	5.74	9.50

Table 2.7. Phase 3: five-fold cross-validation

- **Folds:** the pool is `all_both_v3_cvcompact` train + val (29,035 recordings), split into 5 patient-disjoint folds, the same partition for every experiment. Two schemes:
  - `cvbase`, `cvctrl`, `cvaug`: for fold k, test = fold k, val = fold $k+1 \bmod 5$, and the other three folds train.
  - `cvaugval` (val-only, `cross_validation.test_set=held_out`): it is `cvaug` with val = fold k and the other **four** folds training, with no in-pool test fold, so its `CV test` reads "–".

  The TUAB test set is in no fold, so each of the 5 fold models also scores it.
- **`exp`:** the CV experiment, linked to the ClearML project holding its five fold experiments (click the name).
- **`train recs / patients`:** training recordings / patients of a fold model, mean over the 5 folds, in thousands (each fold model trains on 3 of the 5 folds; `cvaugval` on 4).
- **`loss`:** the training loss (Eqs. (2.2)–(2.11)).
- **Per-fold columns:** mean ± sample sd over the 5 fold models of the MAE (Eq. (0.1)) on:
  - `CV val`: the fold's own val;
  - `CV test`: the fold's own test;
  - `TUAB test`, with `normal` and `abnormal`.
- **`ens.`:** the TUAB test MAE of the median of the 5 models' predictions (Eq. (3.8)). It needs the per-recording predictions.
- **`cal.`:** the same ensemble's TUAB test MAE after the leak-free bias correction of Eq. (3.11), all / normal / abnormal. The line age ≈ a · prediction + b is fitted on the

experiment's own out-of-fold predictions of normal recordings (the CV test folds, never TUAB test), then applied to each test prediction alone, so no TUAB test true age is used (Table 3.6). Table 3.5 compares it with the other corrections.

Source: `docs/data/cv5_eval_v2.csv`.

exp	train recs / patients	feature space	loss	CV val	CV test	TUAB test	nor-mal	ab-nor-mal	ens. TUAB test	ens. nor-mal	ens. abnor-mal	cal. TUAB test	cal. nor-mal	cal. abnor-mal
cvbase	17.42K / 8.13K	patch_tokens	Huber δ 1 + mixup (2.3, 2.5)	8.16 ± 0.11	8.18 ± 0.15	7.49 ± 0.09	5.81 ± 0.10	9.50 ± 0.11	7.44	5.75	9.46	7.28	5.58	9.31
cvctrl	17.42K / 8.13K	cls_token + path_label	Huber δ 1 + mixup (2.3, 2.5)	8.09 ± 0.12	8.08 ± 0.13	7.28 ± 0.14	5.53 ± 0.08	9.37 ± 0.25	7.10	5.31	9.23	6.98	5.19	9.11
cvaug	17.42K / 8.13K	cls_token + path_label	Huber δ 1 + mixup + 0.1 consistency (2.3, 2.5, 2.11)	8.06 ± 0.13	8.06 ± 0.15	7.33 ± 0.17	5.55 ± 0.17	9.44 ± 0.18	7.23	5.43	9.37	7.11	5.30	9.28
cvaug-val	23.23K / 10.84K	cls_token + path_label	Huber δ 1 + mixup + 0.1 consistency (2.3, 2.5, 2.11)	7.90 ± 0.14	–	7.26 ± 0.10	5.41 ± 0.17	9.48 ± 0.09	7.22	5.36	9.44	7.04	5.16	9.28

Reading Tables 2.5–2.7: bias correction.

- **Raw, the CV ensembles are on par with the best single runs.** Their raw TUAB test MAE is 7.10–7.44, against 7.17–7.42 for the phase-2 runs on the same cohort (Exp4*), although each fold model trains on 17.4K (`cvaugval`: 23.2K) instead of 25.6K recordings.
- **The leak-free correction gains 0.1 years.** Calibrated, the ensembles reach 6.98–7.28 overall, 5.16–5.58 on normal and 9.11–9.31 on abnormal recordings: 0.09–0.20 years better than raw on every cohort. cvctrl stays the best (6.98).
- **No test label is used.** The `cal.` columns are fitted on out-of-fold predictions and applied to the prediction alone (Table 3.6). The much lower "corrected" numbers of the literature, e.g. Gemein et al.'s 6.60 / 12.85, use a correction applied with the test recordings' true ages. Table 3.5 lists ours of that kind (†) for comparison only.
- **Four folds instead of three (`cvaugval` vs `cvaug`, Table 2.7).** Training each fold model on 23.2K instead of 17.4K recordings improves the single model a little: the fold-mean TUAB test MAE goes from 7.33 to 7.26 and the fold-to-fold sd from 0.17 to 0.10. The ensemble barely moves, from 7.23 to 7.22 (7.18 with each fold's last-epoch checkpoint instead of the one val picked, `docs/data/cv5_eval_cvaugval_final.csv`). Calibrated, it reaches 7.04 overall and 5.16 on normal recordings, the lowest normal MAE of any CV run. All these differences are within the fold-to-fold sd. More data per model helps each model, but five models already average most of the gain away.
- **Missing-channel recordings and any-channel dropout (`Exp4anymiss` vs `Exp4aug15`, same val and test).** Exp4anymiss adds 2,003 training recordings with missing channels (mostly A1/A2) and, from epoch 4, drops any 1–3 channels from 20 % of the batches (`channel_drop.start_epoch`). Its validation MAE is slightly lower (7.58 vs 7.63), its TUEG val v2 MAE too (7.75 vs 7.91), and its TUAB test MAE about the same (raw 7.21 vs 7.17; normal 5.47 vs 5.38, abnormal 9.30 vs 9.29; calibrated 7.13 vs 7.00). All of these differences are within the epoch-to-epoch noise. The model now also covers recordings without A1/A2 and other channels, at no measurable cost on full-channel recordings.

- **Without the ear and anterior temporal channels (`Exp4noear` vs `Exp4anymiss` , same split).** Dropping A1, A2, T1 and T2 (19 channels, no channel augmentation, 8 / 2 epochs) costs about 0.2 years. Raw TUAB test goes from 7.21 to 7.41 (normal 5.47 → 5.55, abnormal 9.30 → 9.64), calibrated from 7.13 to 7.31, validation from 7.58 to 7.84, and TUEG val v2 from 7.75 to 8.02. Most of the loss is on abnormal recordings. The comparison is not clean: Exp4noear trained for 8 epochs with a schedule that decays over 8, against Exp4anymiss's 13 epochs (15 / 5, early-stopped). At the same epoch (7), the two are about level (test 7.41 vs 7.47, validation 7.84 vs 7.70). A 15-epoch run would separate the channels' effect from the schedule's.
- **Without the status / pathology input (`Exp4noearEEG` vs `Exp4noear` , both 19 channels and 8 epochs, same split).** Taking the labels away changes little. Validation is 7.81 vs 7.84 and TUEG val v2 7.99 vs 8.02. TUAB test is 7.48 vs 7.41: normal recordings get worse (5.75 vs 5.55) and abnormal ones better (9.54 vs 9.64). Calibrated, it is 7.45 vs 7.31. These differences are of the order of the run-to-run noise. With these 19 channels, the status / pathology vector adds no measurable accuracy.
- **As an EEG-only model.** At 7.48 on TUAB test (calibrated 7.45), Exp4noearEEG trails the best EEG-only runs only a little: Exp3 7.26 (`all_both_v1` , 23 channels), Exp4base 7.36 (23 channels, T1/T2 left out where missing), and the cvbase ensemble 7.44. Losing A1/A2/T1/T2 together with the shorter 8-epoch schedule costs about 0.1–0.2 years against them, mostly on normal recordings.

Table 2.8. From D to the best model: short list

The path from the base configuration D to the best models, one row per distinct configuration, all scored on the same 274 TUAB test recordings:

- **D:** the phase-1 reference. D-npy and D-cohorts reproduce it on another data format or scoring path with the same results (Table 2.3).
- **M1:** the phase-1 configuration carried into phase 2 (lowest validation MAE).
- **Every phase-2 run** (Table 2.5). One duplicate is merged: Exp4basec is Exp4base trained again on a compact copy of the same data with the same split, so its row is the mean of the two.
- **The four CV configurations** (Table 2.7), each as the mean over its 5 fold models and as the 5-model median ensemble.

Columns:

- `phase` and `config` .
- `cohort` (Table 1.4) and `train recs / patients` : the run's training recordings / training patients, in thousands (K = 1,000). For a CV row this is the mean over its 5 fold models, each of which trains on 3 of the 5 folds (17,239–17,607 recordings).
- `feature space` (Eq. (2.13)): what the head reads. A row is EEG-only when it is `patch_tokens` or `cls_token` , without `path_label` .
- `change` : what the run changes relative to its parent (Table 2.5).

- `test`, `Norm`, `Ab`: TUAB test MAE (Eq. (0.1)) over all, normal and abnormal recordings.
- `Δ test vs D`: Eq. (0.4) against D (8.30); negative is better.
- `note`: merged runs.

`◆` marks the lowest `test` among the EEG-only rows of each phase.

phase	config	cohort	train recs / patients	feature space	change	test	Δ test vs D	Norm	Ab	note
1	◆ **D**	TUAB v3.0.0	2.15K / 1.65K	patch_tokens	base configuration (Table 2.1)	8.30	0.00	6.57	10.38	= D-npy, D-cohorts (same results)
	M1	TUAB v3.0.0	2.15K / 1.65K	patch_tokens	4 / 1 epochs + mixup 0.4 (carried into phase 2)	8.45	0.15	6.60	10.65	
	Control	TUAB_v3.0.1_interp50-60	2.15K / 1.65K	patch_tokens	M1 on TUAB only (the TUAB baseline of phase 2)	8.23	-0.07	6.42	10.39	
	Exp1	normal_both_v1	5.45K / 3.00K	patch_tokens	train on normal recordings only: TUAB normal + TUEG normal	8.03	-0.27	5.99	10.47	
	Exp2	normal_TUEG_v1	3.76K / 2.15K	patch_tokens	TUEG normal only (no TUAB in training)	8.00	-0.30	6.06	10.31	
	Exp1b	normal_both_v1	5.45K / 3.00K	patch_tokens	10 epochs / 3 warmup instead of 4 / 1	8.11	-0.19	6.09	10.52	
	Exp1t	normal_both_v1	5.45K / 3.00K	patch_tokens + feat_tokens	classical EEG features as extra transformer tokens	8.80	0.50	7.17	10.73	
	Exp1f	normal_both_v1_meta	5.45K / 3.00K	features only (no EEG)	classical EEG features only, MLP head, no EEG encoder	8.13	-0.17	6.21	10.41	
	◆ **Exp3**	all_both_v1	20.37K / 9.07K	patch_tokens	add every TUEG recording (normal, abnormal, unlabelled): ~10x training data	7.26	-1.04	5.57	9.28	
	Exp3b	all_both_v1	20.37K / 9.07K	patch_tokens	10 epochs / 3 warmup instead of 4 / 1	7.70	-0.60	6.17	9.51	
	Exp3q	all_both_v1	19.82K / 8.97K	patch_tokens	drop windows beyond ±800 µV or with a flat channel	7.34	-0.96	5.69	9.31	
	Exp3c	all_both_v1	20.37K / 9.07K	patch_tokens	clip the input to ±800 µV inside the model	7.29	-1.01	5.66	9.22	
	Exp3v2	all_both_v2	20.47K / 9.09K	patch_tokens	cohort v2: one TUEG patient split shared by every cohort (data bookkeeping only)	7.42	-0.88	5.46	9.75	
	Exp3cond	all_both_v2	20.47K / 9.09K	patch_tokens + path_label	status + pathology vectors concatenated to the pooled feature	7.28	-1.02	5.39	9.53	
	Exp3clscond	all_both_v2	20.47K / 9.09K	cls_token + path_label	head on the class token instead of the mean patch token	7.25	-1.05	5.36	9.51	
2	**Exp4drop**	all_both_v3	25.61K / 12.02K	patch_tokens + path_label	cohort v3 (+ recordings missing only T1/T2) on 21 channels (T1/T2 dropped everywhere)	7.28	-1.02	5.26	9.68	
	Exp4base	all_both_v3	25.61K / 12.02K	patch_tokens	cohort v3, 23 channels with missing T1/T2 left out per sample; speed settings; 6 / 2 epochs	7.39	-0.91	5.57	9.56	mean of Exp4base, Exp4basec (same data, compact copy)
	Exp4ctrl	all_both_v3	25.61K / 12.02K	cls_token + path_label	class token + condition vectors, 10 / 3 epochs with early stopping	7.30	-1.00	5.35	9.63	
	Exp4aug	all_both_v3	25.61K / 12.02K	cls_token + path_label	T1/T2 batch dropout (p 0.2) + consistency loss (λ 0.1); 6 / 2 epochs	7.29	-1.01	5.45	9.49	
	Exp4aug15	all_both_v3	25.61K / 12.02K	cls_token + path_label	15 / 5 epochs with early stopping (stopped at 11)	7.17	-1.13	5.38	9.29	
	Exp4anymiss	all_both_v3_anymiss	27.62K / 12.19K	cls_token + path_label	+2,003 training recordings missing A1/A2 or other channels; any 1-3 channels dropped (p 0.2) from epoch 4 (stopped at 12)	7.21	-1.09	5.47	9.30	
	Exp4noear	all_both_v3_anymiss	27.62K / 12.19K	cls_token + path_label	A1/A2/T1/T2 removed from every recording (19 channels), no channel augmentation; 8 / 2 epochs	7.41	-0.89	5.55	9.64	
	Exp4noearEEG	all_both_v3_anymiss	27.62K / 12.19K	cls_token	no status / pathology input (EEG only); otherwise as Exp4noear	7.48	-0.82	5.75	9.54	
	Exp4mt	all_both_v3_anymiss	27.62K / 12.19K	cls_token (EEG only; multi-task head)	multi-task head: + status and 13 diagnosis groups as targets (BCE over all known labels)	7.62	-0.68	5.93	9.64	
	Exp4mtbal	all_both_v3_anymiss	27.62K / 12.19K	cls_token (EEG only; multi-task head)	class-balanced pathology loss (groups present in the batch, equally weighted)	7.59	-0.71	5.86	9.65	
3	**cvbase (fold mean)**	all_both_v3_cvcompact	17.42K / 8.13K	patch_tokens	5-fold CV of Exp4base (5 / 1 epochs)	7.49	-0.81	5.81	9.50	mean of 5 fold models
	◆ **cvbase (ensemble)**	all_both_v3_cvcompact	17.42K / 8.13K	patch_tokens	5-fold CV of Exp4base (5 / 1 epochs)	7.44	-0.86	5.75	9.46	median of 5 fold models
	cvctrl (fold mean)	all_both_v3_cvcompact	17.42K / 8.13K	cls_token + path_label	class token + condition vectors, 10 / 3 epochs (CV of Exp4ctrl)	7.28	-1.02	5.53	9.37	mean of 5 fold models
	cvctrl (ensemble)	all_both_v3_cvcompact	17.42K / 8.13K	cls_token + path_label	class token + condition vectors, 10 / 3 epochs (CV of Exp4ctrl)	7.10	-1.20	5.31	9.23	median of 5 fold models
	cvaug (fold mean)	all_both_v3_cvcompact	17.42K / 8.13K	cls_token + path_label	T1/T2 batch dropout + consistency loss λ 0.1 (CV of Exp4aug)	7.33	-0.97	5.55	9.44	mean of 5 fold models
	cvaug (ensemble)	all_both_v3_cvcompact	17.42K / 8.13K	cls_token + path_label	T1/T2 batch dropout + consistency loss λ 0.1 (CV of	7.23	-1.07	5.43	9.37	median of 5 fold models

phase	config	cohort	train recs / patients	feature space	change	test	Δ test vs D	Norm	Ab	note
					Exp4aug)					
	cvaugval (fold mean)	all_both_v3_cvcompact	23.23K / 10.84K	cls_token + path_label	val-only CV: val = fold k, the other 4 folds train (~23.2K recordings), test = TUAB	7.26	-1.04	5.41	9.48	mean of 5 fold models
	cvaugval (ensemble)	all_both_v3_cvcompact	23.23K / 10.84K	cls_token + path_label	val-only CV: val = fold k, the other 4 folds train (~23.2K recordings), test = TUAB	7.22	-1.08	5.36	9.44	median of 5 fold models

Reading Table 2.8.

- **Phase 1 (TUAB, 2,145 training recordings) stays at about 8.3 test MAE.** No phase-1 change beats D beyond noise (Table 2.3).
- **More data is the big step.** Adding every TUEG recording (Exp3, about 20k training recordings) takes the test MAE to about 7.3, roughly 1 year better than D. The step is largest on abnormal recordings.
- **Everything after that moves the test MAE by a few tenths:** cohort versions, channel handling, condition vectors, epochs.
- **The lowest test MAE overall is the `cvctrl` 5-model ensemble (7.10)**, although each of its fold models trains on fewer recordings (about 17.4k) than the single-split v3 runs (25.6k).
- **EEG-only:** the single-split Exp3 (7.26) scores below the `cvbase` ensemble (7.44). But phase-2 single runs were compared on the test set itself (Section 0), so their numbers are optimistic, and `cvbase` is the unselected estimate.

Figure 2.1. TUAB test MAE of every run

Every run's TUAB test MAE (Eq. (0.1), years), in the order the runs were made, left to right. Dotted vertical lines separate phase 1 (Table 2.2), phase 2 (Table 2.5), the CV runs (Table 2.7) and phase 4 (TUH + HEEDB, Table 2.11).

- (a) All 274 test recordings.
- (b) The 149 normal test recordings.
- (c) The 125 abnormal test recordings.

The 21 phase-1 runs without cohort metrics (Table 2.2) have no point in (b) or (c).

Legend.

- **Colour = phase:** blue phase 1 (TUAB only), orange phase 2 (TUH cohorts), green phase 3 (cross-validation), red phase 4 (TUH + HEEDB; diamond markers).
- **Fill = what the head sees:**
  - filled marker: an EEG-only model;
  - hollow marker: the head also receives the recording's status and pathology vector (Eq. (3.6)), so the model is not EEG-only and its MAE is not comparable with published results.
- **Shape:**
  - circle: one model;

    - green circle with bars: the mean ± sample sd over the 5 CV fold models;
    - green star: the median ensemble of those 5 models (Eq. (3.8));
    - triangle: a collapsed run whose MAE is above the panel's top edge (11 years in (a) and (b), 14 in (c)). It is drawn at the edge so the axis stays readable; its true value is in Table 2.2.
- **Published results, black horizontal lines.** Both are the best published TUAB brain-age models. Their training data and splits differ from ours, so the lines are reference points, not a matched comparison.
    - Dashed: Gemein et al. 2024, a TCN ensemble trained on TUEG normal recordings only. On TUAB's evaluation set it scores 6.60 on normal recordings (b) and 12.85 on abnormal ones (c), **after** their age-bias correction, which is applied with the test recordings' true ages; their raw normal MAE was 6.48. Our points are raw. Table 3.5 gives our numbers after the same kind of correction (marked †) and after a leak-free one. Their abnormal error comes from a model that never saw an abnormal recording; ours do.
    - Dotted: Engemann et al. 2022, the best earlier result, 7.75 on TUAB normal recordings (b) (10-fold cross-validation).
    - Panel (a) has no line: neither paper reports one MAE over all TUAB recordings.

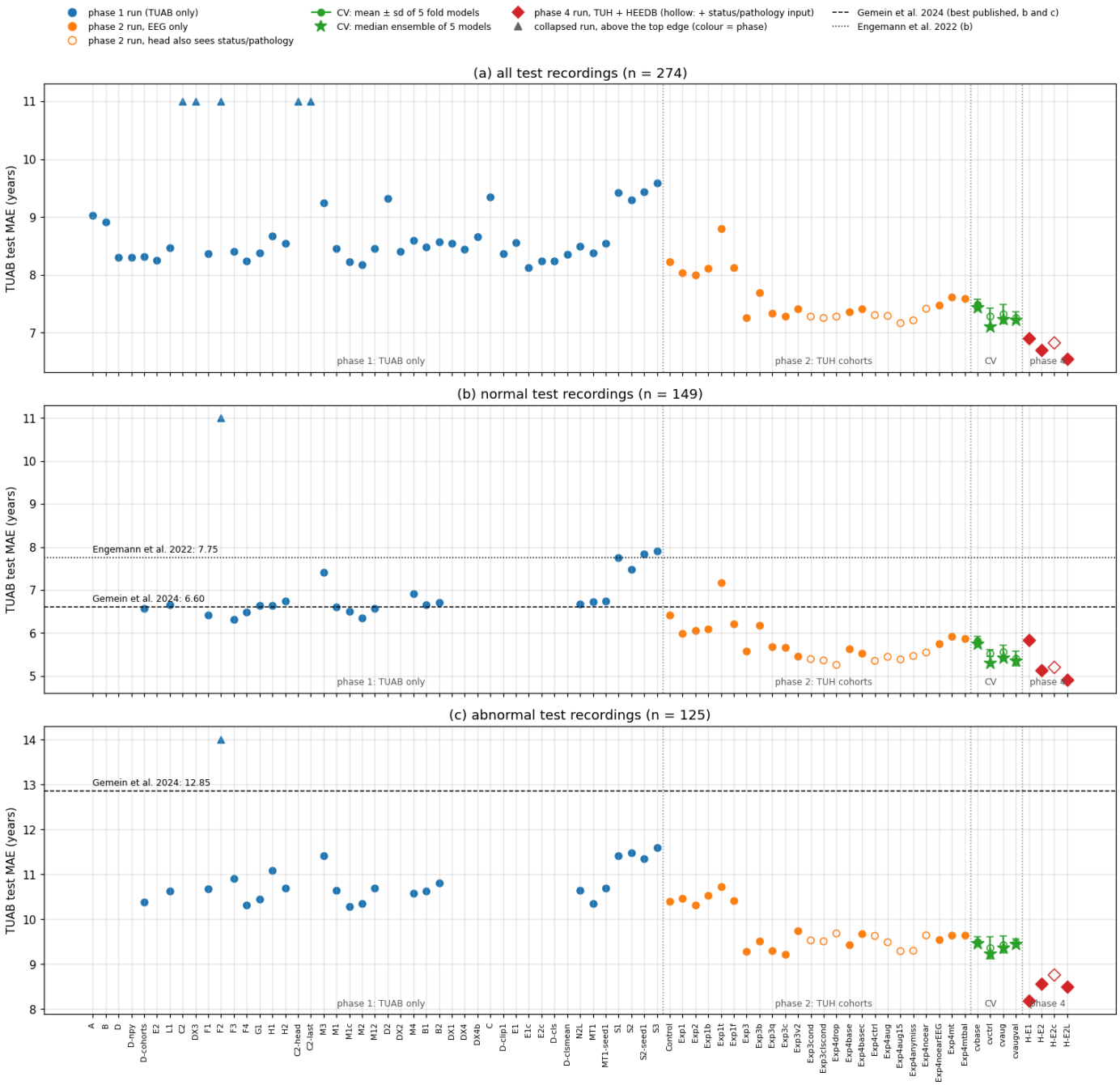


Table 2.9. The runs around the best result

The 12 runs with the lowest TUAB test MAE across all phases (Tables 2.2–2.7):

- `kind`:
  - `single split`: one run;
  - `CV mean`: mean over 5 fold models;
  - `CV ensemble`: median of 5 fold models.
- `run`: linked to its ClearML experiment (for CV: the project of its folds).
- `feature space` (Eq. (2.13)): what the head reads; EEG-only rows have no `path_label`.
- `test all` / `normal` / `abnormal`: MAE (Eq. (0.1)) on the **TUAB test set only**, all 274 recordings and its 149 normal / 125 abnormal ones. The same set is used in every kind of

row: for the CV rows it is the held-out TUAB test set, which no fold trains on (Table 2.7). The status is TUAB's own normal/abnormal label.

TUEG columns (MAE, Eq. (0.1), years, on **TUEG recordings only**, those outside TUAB, that the run never trained on):

- `TUEG all` / `TUEG normal` / `TUEG abnormal`: all of them, and by status (Eq. (1.2));
- one column per NEDC diagnosis group of the pathology vector (Eq. (2.13), Table 1.9). A recording counts in every group it is flagged with, so groups overlap: `cl. seiz.` / `el. seiz.` clinical / electrographic seizure, `epil.` epilepsy, `stroke`, `TBI`, `subst.` substance use, `bipol.` bipolar, `anx.` anxiety, `depr.` depression, `schiz.` schizophrenia, `dement.` dementia (with Alzheimer's), `MS`, `other` (Parkinson's, ADHD).

Where the recordings come from:

- **Single-split run:** its own **validation** split, re-scored from its checkpoint (`scripts/eval_runs.py`). That is about 1.2–2.9K TUEG recordings, depending on the cohort.
- **CV row:** the out-of-fold predictions of the whole CV pool (26,685 TUEG recordings): the test folds, or for `cvaugval` its val folds, which also chose its epoch. The `CV mean` and `CV ensemble` rows therefore show the same TUEG values.
- **Phase-1 run:** trained and validated on TUAB only, so it has no TUEG numbers ("–").
- A group with fewer than 20 recordings for a run is "–". Many diagnosis groups are that small in a single validation split, while the CV pool has 36–2,064 per group.
- The single-split validation sets differ between cohorts, so their TUEG columns compare only within a cohort. The CV columns compare across the CV experiments.

The table is shown in two parts with the same rows: the TUAB test and TUEG status columns first, then the TUEG diagnosis-group columns.

Rows: the 12 lowest `test all`, listed single-split runs first, then the CV rows, each block ranked on `test all`. The best EEG-only run of each kind is marked `◆`. Source: `docs/data/tueg_pathology_mae.csv`.

run	kind	phase	feature space	test all	normal	abnormal	TUEG all	TUEG normal	TUEG abnormal
Exp4aug15	single split	2	cls_token + path_label	7.17	5.38	9.29	7.84	7.26	8.24
Exp4anymiss	single split	2	cls_token + path_label	7.21	5.47	9.30	7.78	7.25	8.12
Exp3clscond	single split	2	cls_token + path_label	7.25	5.36	9.51	7.98	7.47	8.22
◆ Exp3	single split	2	patch_tokens	7.26	5.57	9.28	7.82	6.91	8.08
Exp4drop	single split	2	patch_tokens + path_label	7.28	5.26	9.68	7.72	6.97	8.27
Exp3cond	single split	2	patch_tokens + path_label	7.28	5.39	9.53	7.90	7.35	8.20
Exp3c	single split	2	patch_tokens	7.29	5.66	9.22	7.84	6.94	8.10
cvaugval	CV mean	3	cls_token + path_label	7.26	5.41	9.48	7.99	7.36	8.32
cvctrl	CV mean	3	cls_token + path_label	7.28	5.53	9.37	8.17	7.51	8.50
cvctrl	CV ensemble	3	cls_token + path_label	7.10	5.31	9.23	8.17	7.51	8.50
cvaugval	CV ensemble	3	cls_token + path_label	7.22	5.36	9.44	7.99	7.36	8.32
cvaug	CV ensemble	3	cls_token + path_label	7.23	5.43	9.37	8.14	7.48	8.50

run	kind	cl. seiz.	el. seiz.	epil.	stroke	TBI	subst.	bipol.	anx.	depr.	schiz.	dement.	MS	other
Exp4aug15	single split	7.07	–	8.46	7.21	8.53	8.42	7.48	–	–	–	–	–	–
Exp4anymiss	single split	7.02	–	8.35	7.08	8.84	7.60	7.46	–	–	–	–	–	–
Exp3clscond	single split	7.43	–	8.43	7.49	10.25	7.78	–	–	–	–	–	–	–
◆ Exp3	single split	6.70	8.58	5.93	9.79	10.49	9.20	–	–	–	–	–	–	–
Exp4drop	single split	7.03	–	9.08	7.76	11.09	8.32	6.77	–	–	–	–	–	–
Exp3cond	single split	7.13	–	8.49	7.89	10.52	7.33	–	–	–	–	–	–	–
Exp3c	single split	6.68	9.29	6.07	9.63	10.65	9.23	–	–	–	–	–	–	–
cvaugval	CV mean	7.31	8.36	7.85	8.69	10.03	7.38	5.98	6.17	7.16	8.39	7.88	10.26	7.14
cvctrl	CV mean	7.44	9.19	7.91	8.85	10.38	7.65	5.79	6.17	7.63	8.63	7.45	11.64	7.38
cvctrl	CV ensemble	7.44	9.19	7.91	8.85	10.38	7.65	5.79	6.17	7.63	8.63	7.45	11.64	7.38
cvaugval	CV ensemble	7.31	8.36	7.85	8.69	10.03	7.38	5.98	6.17	7.16	8.39	7.88	10.26	7.14
cvaug	CV ensemble	7.49	8.71	7.85	8.79	10.61	7.58	5.93	6.30	7.28	8.94	7.50	10.80	6.74

Table 2.10. Multi-task runs: normal/abnormal status and diagnosis groups predicted from the EEG

`Exp4mt` and `Exp4mtbal` are `Exp4noearEEG` (19 channels, EEG-only input, 8 / 2 epochs, same split) with a **multi-task head** (`model.head_type=multitask` , `data.multitask_targets`). The class-token feature h feeds 2 shared layers (256, 256) and then two heads of 3 layers each:

- the **age head** (128, 64, 1);
- the **classification head** (128, 64, 14): one logit for the normal/abnormal status and one per NEDC diagnosis group.

The labels are **targets only**: the model never receives them as input. Their source is `dataset_maker/make_multitask_targets.py` :

- status: the TUAB label, else the NEDC report;
- groups: the clinical sheet's diagnosis flags, as in the pathology vector of Eq. (2.13).

A label is unknown (NaN) when the recording has no status, or for the groups when its session is not in the sheet. An unknown label is never scored. With the age loss ℓ_{age} (Huber, Eq. (2.3)), the status logit s and the group logits $q_{i,k}$ (y the 0/1 target, BCE the binary cross-entropy with logits), over a batch B:

$$\begin{aligned} L &= \ell_{\text{age}} + 0.5\,\overline{\mathrm{BCE}}\big(s_i, y_i^{\text{st}}\big)_{i\in K_{\text{st}}} + 0.5\,L_{\text{path}}, \\ L_{\text{path}}^{\text{known}} &= \frac{1}{|K|}\sum_{(i,k)\in K} \mathrm{BCE}\big(q_{i,k}, y_{i,k}\big) \end{aligned} \tag{2.15}$$

where K_{st} are the batch's recordings with a known status and K the known (recording, group) pairs. `Exp4mt` uses $L_{\text{path}}^{\text{known}}$. In it the frequent groups and the many negatives dominate, because a rare group has only a handful of positives per epoch. `Exp4mtbal` uses the **class-balanced** form (`loss.pathology_reduction=present`):

$$\begin{aligned} L_{\text{path}}^{\text{present}} &= \frac{1}{|P|}\sum_{k\in P}\frac{1}{|K_k|}\sum_{i\in K_k}\mathrm{BCE}\big(q_{i,k}, y_{i,k}\big), \\ P &= \{k : \exists\, i \in K_k,\ y_{i,k} > 0\} \end{aligned} \tag{2.16}$$

Here K_k are the batch's recordings whose group-k label is known. Only the groups with at least one positive in the batch enter, each with the same weight. Model selection stays on the age MAE. `Exp4mt` ran on SageMaker and `Exp4mtbal` locally.

Columns: per target (status, the 13 groups, and their macro mean):

- `val positives / known`: positive and known recordings in the validation split;
- `prevalence %`: their ratio in percent; as a fraction it is the PR-AUC of a random predictor;
- ROC-AUC and PR-AUC per recording on validation (each recording's window probabilities averaged), for both runs;
- TUAB test ROC-AUC, where test has positives at all. TUAB test has 125 abnormal recordings but almost no diagnosis positives (10 substance, 7 bipolar, 1 stroke, 1 dementia), so the groups are judged on validation.

Source: `docs/data/multitask_classification.csv` (each run's best epoch, picked on age).

target	val positives / known	preva-lence %	val ROC-AUC Exp4mt	val ROC-AUC Exp4mtbal	val PR-AUC Exp4mt	val PR-AUC Exp4mtbal	test ROC-AUC Exp4mt	test ROC-AUC Exp4mtbal
status (abnormal)	1,676 / 3,064	54.70	0.82	0.82	0.82	0.82	0.92	0.92
clinical_seizure	185 / 2,991	6.19	0.78	0.79	0.18	0.20	–	–
electrographic_seizure	11 / 2,991	0.37	0.96	0.96	0.08	0.05	–	–
epilepsy	46 / 2,991	1.54	0.74	0.68	0.03	0.02	–	–
stroke	93 / 2,991	3.11	0.76	0.77	0.08	0.10	0.73	0.76
tbi	50 / 2,991	1.67	0.76	0.74	0.05	0.05	–	–
substance	103 / 2,991	3.44	0.62	0.61	0.06	0.06	0.51	0.53
bipolar	31 / 2,991	1.04	0.56	0.53	0.01	0.01	0.72	0.71
anxiety	10 / 2,991	0.33	0.74	0.77	0.03	0.04	–	–
depression	8 / 2,991	0.27	0.63	0.64	0.01	0.01	–	–
schizophrenia	9 / 2,991	0.30	0.55	0.70	0.00	0.01	–	–
dementia	10 / 2,991	0.33	0.85	0.89	0.01	0.02	0.90	0.93
ms	7 / 2,991	0.23	0.52	0.64	0.00	0.00	–	–
other	12 / 2,991	0.40	0.62	0.65	0.01	0.01	–	–
pathology macro	–	–	0.70	0.72	0.04	0.05	0.72	0.73

Reading Table 2.10.

- **Status is learnt well from the EEG.** Normal versus abnormal reaches ROC-AUC 0.82 on validation and 0.92 on TUAB test with either loss.
- **The frequent groups are partly predictable.** Clinical seizure, stroke and TBI reach ROC-AUC 0.74–0.79, and electrographic seizure 0.96. Their PR-AUCs are well above prevalence (clinical seizure 0.18–0.20 against 0.06, stroke 0.08–0.10 against 0.03), but still low in absolute terms. Psychiatric groups (substance, bipolar, depression) are close to chance (0.53–0.64).
- **The class-balanced loss helps the rare groups.** It raises the macro ROC-AUC from 0.70 to 0.72 on validation and from 0.72 to 0.73 on test. The gain is mostly on the rare groups: schizophrenia 0.55 → 0.70, MS 0.52 → 0.64, dementia 0.85 → 0.89, anxiety 0.74 → 0.77. Epilepsy (0.74 → 0.68) and TBI lose some. With 7–12 validation positives per rare group, a single group's AUC is noisy (± about 0.1), so only the overall direction is meaningful.

- **The auxiliary heads cost age accuracy.** Both runs predict age worse than the same model without them (`Exp4noearEEG` , Table 2.5): validation 8.12 vs 7.81, TUEG val v2 8.33 vs 7.99, TUAB test 7.62 / 7.59 vs 7.48, calibrated 7.50 / 7.51 vs 7.45. In training, the weighted status term (0.5 × BCE ≈ 0.24) outweighs the age loss (Huber ≈ 0.12 in z-units). The pathology term is small (0.5 × 0.07–0.11), so the status head, not the diagnosis groups, pulls the shared layers away from age. In phase 1, the TUAB-only multi-task run MT1 (status only) was within seed noise of M1 (test 8.38 and 8.54 for its two seeds against 8.45, Table 2.2). A smaller status weight, or stopping the classification gradient into the encoder, should keep the age model as good.

- **Per-target detection quality:** `pathology_prediction.ipynb` (PDF) adds, for every target, the best F1 with its bootstrap interval, the errors (false positives and negatives) at that threshold, and the ROC and precision-recall curves. It also covers two classification-only runs on the same split: without the age prediction (`Exp4cls`), and with the known age as an encoder input (`Exp4clsage`).

Phase 4: adding the Harvard EEG Database (HEEDB)

Phase 4 asks whether a second, larger clinical corpus improves the TUH model. It adds the **Harvard EEG Database** (HEEDB v4.1, BDSP, credentialed access) to the phase-2 data (`docs/heedb_onboarding_plan.md` , `notebooks/heedb_data_report.ipynb`).

Data:

- HEEDB adult (18–89) routine sessions, converted to the same 19-channel, 200 Hz npy format as TUH (71.8K recordings).
- Patient-level split: 10 % test, 10 % val by hashed BDSP patient id.
- The joint cohort `tuh_heedb_v1` adds the TUH cohort of `Exp4noearEEG` (`all_both_v3_anymiss`) with its own split. Its TUH test split is the same 274 TUAB recordings.

Runs. Every run starts from `Exp4noearEEG` , the phase-2 recipe of Table 2.5: 19 channels, EEG-only, class-token readout, Huber loss, mixup, lr $2 \cdot 10^{-4}$, batch 128, 8 epochs (2 warm-up), 5 minutes per recording per epoch. The H- prefix keeps their names apart from phase 1's E2 recipe.

- **H-E0:** `Exp4noearEEG` itself (trained on TUH only), scored on HEEDB, i.e. transfer without HEEDB training.
- **H-E1:** the recipe trained on HEEDB only.
- **H-E2:** the recipe trained on TUH + HEEDB.
- **H-E2c:** H-E2 with the status / pathology vector (Eq. (3.6)) as a head input, as in `cvctrl` . HEEDB's pathology flags come from its patients' ICD-10 codes. *Not EEG-only.*
- **H-E2L:** H-E2 trained longer and on a faster loader:
  - 16 epochs, 4 of them warm-up;

- epochs 1–6 at batch 512 / lr $4 \cdot 10^{-4}$;
- from epoch 7, batch 256 / lr $2.8 \cdot 10^{-4}$, on the same cosine schedule, after epoch 6 proved unstable;
- trained on the local A10G.

Selection. Every phase-4 run picks its epoch on its own **val** split, never on the TUAB test set. Its TUAB numbers are therefore not selected on test, unlike the phase-1/2 single runs (Section 0, caveat).

Sources (all in git, aggregates only):

- `docs/data/tuab_eval_heedb_comparison.csv` : TUAB test, raw;
- `docs/data/brain_age_bias_correction.csv` : HEEDB test and bias correction;
- `docs/data/heedb_epoch_metrics.csv` : per epoch, parsed from the training logs by `scripts/epoch_metrics_from_log.py` .

Table 2.11. Phase 4 runs: data, recipe and results

One row per run. MAE (Eq. (0.1)) in years, raw (before bias correction).

Columns:

- `train rec. / pat.` : training recordings / patients (thousands);
- `epochs (best)` : epochs trained and the epoch selected on val;
- `val` : MAE on the run's own validation split at the selected epoch. The splits differ: TUH val for `Exp4noearEEG` , HEEDB val for H-E1, the joint TUH + HEEDB val for H-E2*;
- `HEEDB test` : the held-out HEEDB test patients (7,338 recordings);
- `TUAB all / normal / abnormal` : the 274 / 149 / 125 TUAB test recordings;
- `bias` , `R²` : Eqs. (0.3) and (0.2) on all TUAB test recordings;
- `EEG only` : whether the model is EEG-only (no status / pathology input).

run	trained on	train rec. / pat. (k)	recipe	EEG only	epochs (best)	val	HEEDB test	TUAB all	TUAB normal	TUAB abnormal	bias	R²
Exp4noearEEG / H-E0	TUH	27.6 / 12.2	phase-2 recipe, 8 ep	yes	8 (5)	7.81	9.82	7.48	5.75	9.54	-0.64	0.71
H-E1	HEEDB	57.3 / 46.0	same recipe	yes	8 (6)	7.50	7.27	6.90	5.83	8.18	1.29	0.76
H-E2	TUH + HEEDB	84.9 / 58.2	same recipe	yes	8 (8)	7.43	7.27	6.70	5.13	8.56	-0.22	0.76
H-E2c	TUH + HEEDB	84.9 / 58.2	+ status/pathology input	no	8 (8)	7.40	7.29	6.82	5.20	8.76	-0.38	0.75
H-E2L	TUH + HEEDB	84.9 / 58.2	16 ep (4 warm-up); bs 512 → 256 at ep 7	yes	16 (12)	7.24	7.02	6.54	4.91	8.49	-0.61	0.76

Table 2.12. Wall time per epoch: data loading and evaluation

The phase-4 runs read about 2.5 M ten-second windows per epoch at random from about 85K recordings. The table gives the median minutes per epoch of the training pass and of the evaluation (val + test, and train-eval when scored), from the training logs. `Exp4noearEEG` trains on the smaller TUH cohort, 1.7 M windows per epoch.

Two fixes made the local runs faster than the cloud ones:

- **`pread` loader** (`labram/data/tuh_datasets.py`, `NPY_READ`):
  - the memory-mapped loader read about 4.4× the needed bytes (page faults, 128 KB readahead, header re-reads);
  - that capped the g5.2xlarge at its 106 MB/s EBS baseline, with the GPU about 35 % busy;
  - one exact `pread` per window makes training GPU-bound.
- **Linear-time window pooling** (`labram/utils/eval_metrics.py::aggregate_windows`):
  - the per-recording pooling scanned every window for every recording, O(windows × recordings);
  - about 2.5 min per call, 4 calls per split;
  - now one pass, 0.16 s.

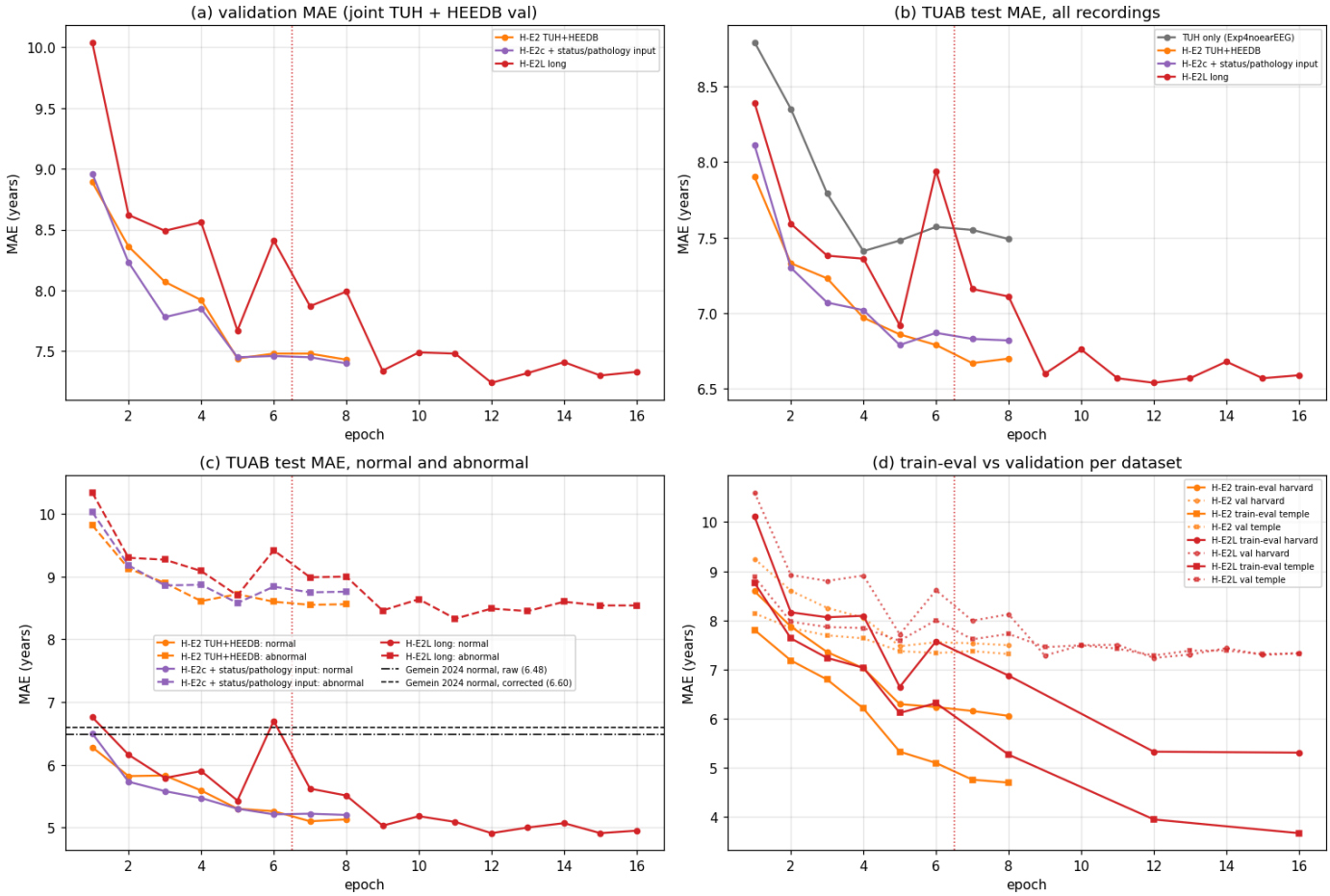
Batch size does not change the speed: the A10G is saturated from batch 256 (775–793 samples/s at batch 256–768, `scripts/benchmark_finetune_speed.py`).

run	epochs	hardware	loader / evaluation	training pass	evaluation	epoch total
Exp4noearEEG	**1–8**	SageMaker g5.4xlarge, 1 A10G	memory map, old pooling	21.73	2.33	24.07
H-E2	**1–8**	SageMaker g5.16xlarge, 1 A10G	memory map, old pooling	68.07	21.45	89.52
H-E2c	**1–8**	local g5.2xlarge, 1 A10G	memory map, old pooling	133.48	33.46	166.94
	1–5	local g5.2xlarge, 1 A10G	memory map, old pooling	138.43	31.30	169.73
H-E2L	**6–6**	local g5.2xlarge, 1 A10G	pread, old pooling	56.62	25.85	82.47
	7–16	local g5.2xlarge, 1 A10G	pread, linear pooling	60.51	7.08	67.58

Figure 2.2. Phase 4 per epoch

MAE (Eq. (0.1), years) after every epoch, from the training logs (`docs/data/heedb_epoch_metrics.csv`). The dotted vertical line marks H-E2L's switch from batch 512 to 256 after epoch 6.

- (a) Validation MAE of the joint runs H-E2, H-E2c and H-E2L, all on the same joint val split.
- (b) TUAB test MAE, all 274 recordings: `Exp4noearEEG` (TUH only) and the joint runs.
- (c) TUAB test MAE of the 149 normal (solid) and 125 abnormal (dashed) recordings. The black lines mark Gemein et al. 2024 on normal recordings: 6.48 raw (dash-dot) and 6.60 after their correction (dashed). Their abnormal 12.85 is above the panel.
- (d) Train-eval (solid) and validation (dotted) MAE per dataset for H-E2 and H-E2L. Train-eval scores a fixed sample of 500 training recordings per dataset like val; H-E2L scores it at epochs 1–5, 8, 12 and 16. The gap between the two shows over-fitting.


(a) validation MAE (joint TUH + HEEDB val)
MAE (years)
epoch
H-E2 TUH+HEEDB
H-E2c + status/pathology input
H-E2L long
(b) TUAB test MAE, all recordings
MAE (years)
epoch
TUH only (Exp4noearEEG)
H-E2 TUH+HEEDB
H-E2c + status/pathology input
H-E2L long
(c) TUAB test MAE, normal and abnormal
MAE (years)
epoch
H-E2 TUH+HEEDB: normal
H-E2 TUH+HEEDB: abnormal
H-E2c + status/pathology input: normal
H-E2c + status/pathology input: abnormal
H-E2L long: normal
H-E2L long: abnormal
Gemein 2024 normal, raw (6.48)
Gemein 2024 normal, corrected (6.60)
(d) train-eval vs validation per dataset
MAE (years)
epoch
H-E2 train-eval harvard
H-E2 val harvard
H-E2 train-eval temple
H-E2 val temple
H-E2L train-eval harvard
H-E2L val harvard
H-E2L train-eval temple
H-E2L val temple

Reading Tables 2.11–2.12 and Figure 2.2.

- **HEEDB data helps TUH.** The TUH-only recipe scores 7.48 on the TUAB test; trained on HEEDB alone (H-E1) it scores 6.90, and on TUH + HEEDB (H-E2) 6.70, with no TUAB test selection (Table 2.11). The gain is largest on abnormal recordings (9.54 → 8.56).
- **TUH transfers poorly to HEEDB.** The TUH-only model scores 9.82 on the HEEDB test patients, against 7.27 once HEEDB is in training.
- **The clinical inputs add nothing at the end.** H-E2c (status + pathology input) learns faster early (Figure 2.2(a)) but ends at TUAB 6.82 against 6.70 for the EEG-only H-E2.
- **Longer training with a lower final learning rate helps.** H-E2L reaches val 7.24, HEEDB test 7.02, TUAB test 6.54 (normal 4.91, abnormal 8.49). Its early epochs at batch 512 / lr 4e-4 were unstable (epoch 6, Figure 2.2(a)); its gains came once the cosine schedule took the learning rate below about 2e-4.
- **Best EEG-only model so far: H-E2L**, TUAB test 6.54. The best phase-2/3 models score 7.10–7.48 (Tables 2.5–2.7). On normal recordings it is 4.91 raw, against 6.48 raw / 6.60 corrected for Gemein et al. 2024 (Table 3.5).
- **Over-fitting grows with the longer schedule.** Train-eval falls much faster than val, most on the Temple part (Figure 2.2(d)), so the epoch is chosen on val.
- **Speed.** With the `pread` loader and the linear-time pooling, a local epoch takes 68 min on one A10G, against 90 min for H-E2 on SageMaker and 167 min locally before the fixes (Table 2.12).

Age prediction by pathology

How good is the age prediction for each kind of pathology? The pathologies are the **18 targets of the E2P pathology classifier** (`multitask_targets_v2.npz` , `dataset_maker/make_multitask_targets.py` , `notebooks/clinical_records_analytics.ipynb` Section 6):

- `status_abnormal` : the EEG report verdict;
- 13 NEDC diagnosis groups: TUH from the clinical sheet; HEEDB from patient ICD-10 codes, except TBI, substance, bipolar, anxiety and schizophrenia, which HEEDB does not code;
- 4 HEEDB-only ICD groups: headache, sleep disorders, intracranial/spinal tumours, movement/cerebellar disorders.

`healthy` is added as a reference row: normal report and no diagnosis flagged.

Recordings: every held-out recording, i.e. TUH val (3,422) + TUAB test (274) + HEEDB val (7,192) + HEEDB test (7,338). The TUAB test set alone has almost no diagnosis positives. Val chose the epoch but is never trained on.

Comparison: a recording is *positive* for a target when its label is 1, and *negative* when it is 0 (unknown labels are left out). Age error grows with age, and diagnoses cluster in older patients (dementia: mean age 72), so the negatives' MAE is re-weighted to the positives' age mix. With

decades d (18–29, 30s, ..., 80+), w_d the share of positives in decade d, and $\mathrm{MAE}_{\mathrm{neg},d}$ the negatives' MAE (Eq. (0.1)) in that decade:

$$\mathrm{MAE}^{\mathrm{std}}_{\mathrm{neg}} = \sum_d w_d \,\mathrm{MAE}_{\mathrm{neg},d}, \qquad \Delta\mathrm{MAE} = \mathrm{MAE}_{\mathrm{pos}} - \mathrm{MAE}^{\mathrm{std}}_{\mathrm{neg}} \tag{2.17}$$

$\Delta\mathrm{MAE} < 0$: the age is predicted **better** for patients with this pathology than for patients of the same ages without it; > 0: **worse**. The 95 % interval comes from 500 bootstrap resamples: positives resampled, negatives resampled within each decade.

The **corrected brain-age gap** answers a different question: does the model see these patients as older? It is the gap after the quadratic age correction of Gemein et al. 2024 (Eq. (3.9)), fitted per dataset on the val recordings with a normal report:

$$\mathrm{BAG}^c_i = \hat{y}_i - p(y_i) - y_i, \qquad \Delta\mathrm{BAG}^c = \overline{\mathrm{BAG}^c}_{\mathrm{pos}} - \overline{\mathrm{BAG}^c}_{\mathrm{neg}} \tag{2.18}$$

Its p-value is a Welch t-test. Rows with fewer than 10 positives are not shown. Source: `scripts/age_by_pathology.py` → `docs/data/age_by_pathology.csv` (aggregates only).

Table 2.13. Age prediction by pathology (best EEG-only phase-4 model)

One row per target, TUH and HEEDB held-out recordings pooled, sorted from best to worst age prediction (ΔMAE). Columns:

- `n pos` : positives;
- `age pos` : their mean age;
- `MAE pos` : positives' MAE (Eq. (0.1));
- `MAE neg (age-matched)` : Eq. (2.17);
- `ΔMAE [95 % CI]` : Eq. (2.17);
- `bias pos` : Eq. (0.3) on the positives, raw;
- `ΔBAGc` , `p` : Eq. (2.18);
- `MAE pos, TUH-only` : `Exp4noearEEG` on the same positives, to show what HEEDB training changed;
- `verdict` : *better* / *worse* when the interval excludes 0, else *same*.

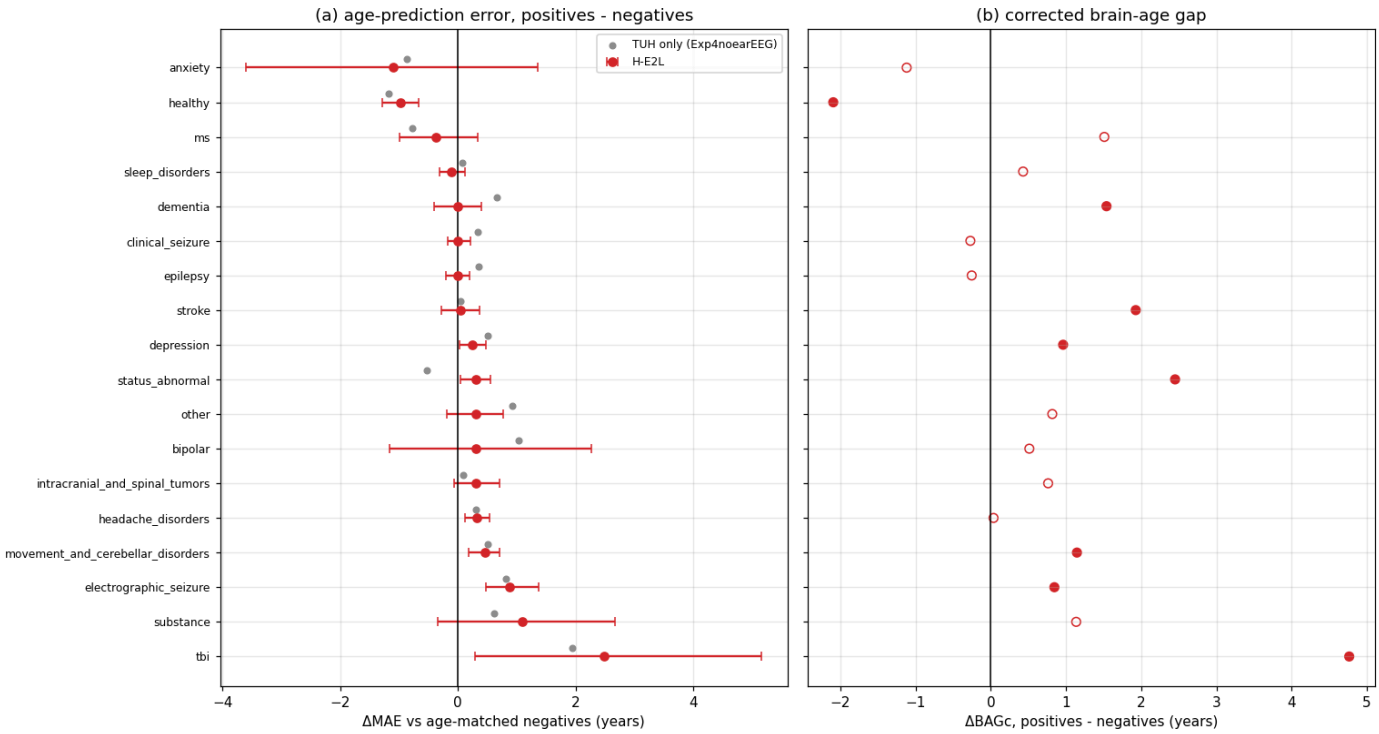
`model: H-E2L`

target	n pos (TUH / HEEDB)	age pos	MAE pos	MAE neg (age-matched)	ΔMAE	95 % CI	bias pos	ΔBAGc	p	MAE pos, TUH-only	ver-dict
anxiety	10 / –	42.00	5.89	7.00	-1.10	[-3.60, +1.36]	1.58	-1.12	5e-01	6.64	same
healthy	1353 / 654	47.10	6.38	7.36	-0.98	[-1.29, -0.67]	0.14	-2.10	2e-33	7.75	better
ms	– / 190	52.98	6.14	6.51	-0.38	[-0.98, +0.33]	1.35	1.51	2e-03	7.49	same
sleep_disorders	– / 2813	57.83	6.83	6.94	-0.11	[-0.32, +0.12]	-1.53	0.43	2e-03	9.90	same
dementia	11 / 1031	71.97	8.55	8.55	0.00	[-0.40, +0.40]	-5.64	1.54	6e-13	13.57	same
clinical_seizure	185 / 4666	49.76	6.99	6.99	0.00	[-0.17, +0.21]	0.47	-0.27	2e-02	9.38	same
epilepsy	46 / 4647	49.89	7.01	7.00	0.01	[-0.20, +0.21]	0.43	-0.25	3e-02	9.43	same
stroke	94 / 1472	65.50	7.58	7.53	0.04	[-0.28, +0.37]	-2.88	1.93	2e-27	11.01	same
depression	– / 2620	55.98	7.18	6.93	0.25	[+0.03, +0.48]	-0.54	0.96	6e-11	9.85	worse
status_abnormal	1801 / 4432	57.70	7.75	7.45	0.30	[+0.05, +0.55]	-0.68	2.45	4e-71	10.02	worse
other	12 / 674	53.38	7.78	7.48	0.31	[-0.19, +0.77]	0.04	0.82	3e-03	11.25	same
bipolar	38 / –	40.68	6.76	6.45	0.31	[-1.16, +2.27]	3.79	0.51	7e-01	7.95	same
intracranial_and_spinal_tumors	– / 797	57.36	7.14	6.83	0.31	[-0.07, +0.70]	-0.96	0.76	1e-03	9.76	same
headache_disorders	– / 3556	54.02	7.21	6.89	0.32	[+0.12, +0.54]	-0.63	0.04	8e-01	9.91	worse
movement_and_cerebellar_disorders	– / 1916	58.98	7.49	7.03	0.46	[+0.19, +0.72]	-1.34	1.15	2e-11	10.74	worse
electrographic_seizure	11 / 1033	53.92	8.10	7.22	0.88	[+0.48, +1.37]	0.00	0.84	5e-04	10.65	worse
substance	113 / –	50.92	8.15	7.07	1.09	[-0.34, +2.67]	1.26	1.14	2e-01	8.29	same
tbi	50 / –	46.10	9.27	6.79	2.48	[+0.29, +5.17]	6.19	4.76	3e-04	9.27	worse

Figure 2.3. Age prediction by pathology

Per target (rows, sorted as in Table 2.13), for the model of Table 2.13:

- (a) ΔMAE (Eq. (2.17), years) with its 95 % bootstrap interval. Left of 0, the age is predicted better than for age-matched patients without the pathology; right of 0, worse. Grey points: the TUH-only `Exp4noearEEG` on the same recordings.
- (b) The corrected brain-age gap difference $\Delta\mathrm{BAG}^{c}$ (Eq. (2.18), years): positive = the model sees the patients as older than their age. Filled marker: Welch $p < 0.001$.



Reading Table 2.13 and Figure 2.3 (H-E2L).

- **Age predicted better than for age-matched patients without the pathology:** healthy (-0.98).
- **Predicted worse:** tbi (+2.48); electrographic_seizure (+0.88); movement_and_cerebellar_disorders (+0.46); headache_disorders (+0.32); status_abnormal (+0.30); depression (+0.25).
- **Within noise (11 targets):** anxiety, ms, sleep_disorders, dementia, clinical_seizure, epilepsy, stroke, other, bipolar, intracranial_and_spinal_tumors, substance. For these, once age is matched, the error equals that of patients without the pathology.
- **Raw MAE is misleading.** Dementia has the highest raw MAE (8.55) because its patients are old (mean 72 years); against age-matched negatives it is +0.00.
- **Older-looking EEG (ΔBAGc > 0, p < 0.001):** tbi +4.8; status_abnormal +2.5; stroke +1.9; dementia +1.5; movement_and_cerebellar_disorders +1.1; depression +1.0; electrographic_seizure +0.8. **Younger-looking:** healthy -2.1. Brain-age theory expects disease to look older; epilepsy and clinical seizure, the largest groups, do not shift the gap.
- **HEEDB training lowers the error on every group** (`MAE pos` against the TUH-only column), most for dementia (5.0 years), other (3.5 years), stroke (3.4 years), movement_and_cerebellar_disorders (3.2 years).
- **Small groups** (TBI, substance, bipolar, anxiety: 10–113 positives, TUH only) have wide intervals: tbi +2.48 [+0.29, +5.17]; substance +1.09 [-0.34, +2.67]; bipolar +0.31 [-1.16, +2.27]; anxiety -1.10 [-3.60, +1.36]. Only an interval that excludes 0 (here: tbi) counts, and even then the size of the effect is poorly known.

3. The best experiment

Which run is best. Single-split runs were compared on the TUAB test set itself (Section 0). Only the CV runs have an error bar, and their CV test / TUEG out-of-fold errors were never used for selection. So "best" is decided among the CV runs:

- **Best overall:** `cvctrl` (= Exp4ctrl cross-validated). It has the lowest TUAB test MAE, as a single fold model and as a 5-model ensemble (Table 2.7).
  - `cvaug` adds T1/T2 channel-dropout augmentation and ties with it: the differences are within noise in Table 5.2 (C2).
  - The simpler `cvctrl` is therefore the recommended configuration.
- **Best EEG-only:** `cvbase`**.** Its head sees only the EEG. `cvctrl` and `cvaug` also see the recording's status (normal / abnormal / no label) and 13 diagnosis flags from the clinical sheet. A deployed brain-age model would not have these, and published TUAB results do not use them. `cvbase` is therefore the number to compare with the literature (Table 3.3).

Both are LaBraM-base fine-tuned on `all_both_v3` with the M1 recipe. They differ in three settings (Table 3.2): class-token vs mean pooling, the condition vector, and 10 vs 5 epochs.

What the model computes

Input and target. The raw EEG u (µV) of a 10 s window is scaled to $x = u/100$ and cut into 1 s patches $x_{c,a,t}$:

- channel $c = 1..C$, with $C = 23$ minus the missing T1/T2 of that recording;
- patch $a = 1..10$, sample $t = 1..200$.

The common average reference removes what all channels share (`labram/data/preprocess.py::common_average_reference`):

$$\tilde{x}_{c,a,t} = x_{c,a,t} - \frac{1}{C}\sum_{c'=1}^{C} x_{c',a,t} \tag{3.1}$$

The age y (years) is z-scored with the training windows' mean μ and standard deviation σ (`labram/data/tuh_datasets.py::TUABAgeLoader._age_target`):

$$z = \frac{y - \mu}{\sigma} \tag{3.2}$$

Mixup. Each training batch is blended with a permuted copy of itself, using one $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$ per batch with $\alpha = 0.4$ (`labram/train/mixup.py::mixup_batch`). The condition vector c travels with the target, so it is blended too:

$$\bar{x} = \lambda x + (1-\lambda)\, x_\pi, \qquad \bar{z} = \lambda z + (1-\lambda)\, z_\pi, \qquad \bar{c} = \lambda c + (1-\lambda)\, c_\pi \tag{3.3}$$

Loss. Huber loss (Eq. (2.3)) with $\delta = 1$ on the z-scored residual $r = \hat{z} - \bar{z}$ (`labram/losses/regression.py::build_regression_criterion`):

$$\ell_\delta(r) = \begin{cases} \frac{1}{2} r^2 & |r| \le \delta \\ \delta\,(|r| - \frac{1}{2}\delta) & \text{otherwise} \end{cases} \tag{3.4}$$

Encoder and head. f_θ is the 12-block LaBraM-base transformer, initialized from the released `labram-base.pth` . Its output is $h \in \mathbb{R}^{200}$:

- the final LayerNorm's class token (`cvctrl` , `cvaug`);
- the mean over patch tokens (`cvbase`).

$$h = f_\theta(\tilde{x}) \tag{3.5}$$

The linear head reads h, concatenated with a 32-d projection of the 17-d condition vector $c \in \{0,1\}^{17}$ (3 one-hot status + 14 multi-hot pathology groups, `conditions_v1.npz`) when conditions are on (`labram/models/neural_transformer.py` , `model.session_features=head`):

$$\hat{z} = w^\top \begin{bmatrix} h \\ W_c\, c \end{bmatrix} + b, \qquad W_c \in \mathbb{R}^{32 \times 17} \tag{3.6}$$

Optimizer and schedule. AdamW with weight decay 0.05. Layer-wise lr decay $\rho = 0.65$ gives block l of $L = 12$ the rate $\mathrm{lr} \cdot \rho^{L+1-l}$ (`labram/optim_factory.py::LayerDecayValueAssigner`). The rate warms up linearly, then follows a cosine down to 10^{-6} (`labram/utils/training.py`).

Scoring. A recording's prediction is the de-normalized mean over its evaluation windows W_i (the first 5 minutes after the trim, 30 windows; `evaluation.agg_windows=mean`):

$$\hat{y}_i = \mu + \sigma \, \frac{1}{|W_i|} \sum_{w \in W_i} \hat{z}_w \tag{3.7}$$

The fold ensemble takes the median over the $K = 5$ fold models (`labram/eval/cv_compare.py::ensemble`):

$$\hat{y}_i^{\text{ens}} = \mathrm{median}_{k=1..K}\ \hat{y}_{i,k} \tag{3.8}$$

Table 3.1. The best configuration (`cvctrl`), step by step

One row per stage of the pipeline:

- `setting` : the configuration field;
- `cvctrl` : its value, read from the configuration rebuilt by `rs.resolve_config` (`finetune_tuab_age.json` + the run's overrides, exactly as the trainer builds it);

- `cvbase` : the EEG-only variant's value, where it differs;
- `equation` : where the setting appears above.

stage	what	setting	cvctrl	cvbase	equation
data	**cohort**	data.data_path	all_both_v3_cvcompact		
	5-fold CV, grouped by	cross_validation.split_by	subject		
	CV pool	cross_validation.pool	train_val		
windows	**window length (s)**	data.window_sec	10		
	trim start / end (s)	data.trim_start_sec / data.trim_end_sec	60 / 60		
	train minutes per recording per epoch	data.train_minutes_per_epoch	5		
	eval minutes per recording	data.eval_minutes	5		(3.7)
channels	**channels left out where missing**	channel_drop.channels	["T1", "T2"]		
	T1/T2 dropout probability (augmentation)	channel_drop.prob	0		
input	**common average reference**	labram_plus.common_average_reference	True		(3.1)
	per-patch z-scoring	labram_plus.z_score_patches	False		
encoder	**model**	model.model	labram_base_patch200_200		(3.5)
	initial weights	finetune_checkpoint.finetune	labram-base.pth		
	drop path	model.drop_path	0.1		
	fused attention (speed only)	model.fused_attention	True		
head	**mean pooling (False = class token)**	model.use_mean_pooling	False	True	(3.5)
	condition vector	model.session_features / data.session_features	head / conditions_v1.npz	off /	(3.6)
	condition projection size	model.session_head_dim	32		(3.6)
	output init scale	model.init_scale	0.001		
target	**task**	model.task	regression		(3.2)
loss	**regression loss / δ**	loss.regression_loss / loss.huber_delta	huber / 1		(3.4)
augmentation	**mixup α (on)**	mixup.alpha / mixup.enabled	0.4 / True		(3.3)
optimizer	**AdamW lr / batch**	optimizer.lr / trainer.batch_size	0.0002 / 128		
	weight decay / layer decay	optimizer.weight_decay / optimizer.layer_decay	0.05 / 0.65		
schedule	**epochs / warmup**	trainer.epochs / optimizer.warmup_epochs	10 / 3	5 / 1	
	min lr	optimizer.min_lr	1e-06		
selection	**checkpoint**	–	lowest val MAE (recording level)		
scoring	**window aggregation / case unit**	evaluation.agg_windows / evaluation.agg_case_by	mean / recording		(3.7)

Table 3.2. The three cross-validated configurations

Every configuration field on which `cvbase` , `cvctrl` and `cvaug` disagree (`labram/eval/research_summary.py::config_diff` , bookkeeping fields excluded).

field	cvbase	cvctrl	cvaug
channel_drop.consistency_weight	0	0	0.1
channel_drop.prob	0	0	0.2
data.session_features		conditions_v1.npz	conditions_v1.npz
model.session_features	off	head	head
model.use_mean_pooling	True	False	False
optimizer.warmup_epochs	1	3	3
trainer.epochs	5	10	10

Table 3.3. Results of the best experiment by evaluation set and cohort

`cvctrl` (best overall) and `cvbase` (best EEG-only), mean ± sample sd over the 5 fold models of the MAE (Eq. (0.1)) and the mean R^2 (Eq. (0.2)), per evaluation set and status cohort:

- `CV val` , `CV test` : the fold's own held-out folds, TUAB and TUEG recordings together;
- `TUEG test` : the TUEG recordings of each fold's test fold;

- `TUAB test`: the official test set, $n = 274$.

R^2 is relative to each cohort's own age spread, so it does not compare across cohorts. The last row gives the median ensemble (Eq. (3.8)) on the TUAB test set when the per-recording predictions are available. For reference, the best published TUAB normal-recording result is MAE 6.60, R^2 0.73 (Gemein et al. 2024, trained on TUAB normal recordings only, on its own patient-wise re-split of TUAB).

set	cohort	cvctrl MAE	cvctrl R²	cvbase MAE	cvbase R²
CV val	all	8.09 ± 0.12	0.64	8.16 ± 0.11	0.64
	normal	7.33 ± 0.20	0.68	7.45 ± 0.17	0.68
	abnormal	8.49 ± 0.14	0.60	8.57 ± 0.13	0.59
CV test	all	8.08 ± 0.13	0.64	8.18 ± 0.15	0.63
	normal	7.39 ± 0.18	0.68	7.47 ± 0.14	0.67
	abnormal	8.46 ± 0.11	0.60	8.60 ± 0.15	0.59
TUEG test	all	8.17 ± 0.15	0.63	8.27 ± 0.15	0.62
	normal	7.52 ± 0.19	0.67	7.59 ± 0.16	0.67
	abnormal	8.50 ± 0.10	0.59	8.65 ± 0.13	0.58
TUAB test	all	7.28 ± 0.14	0.71	7.49 ± 0.09	0.70
	normal	5.53 ± 0.08	0.80	5.81 ± 0.10	0.79
	abnormal	9.37 ± 0.25	0.52	9.50 ± 0.11	0.51
TUAB test, ensemble	all	7.10	0.72	7.44	0.70
	normal	5.31	0.82	5.75	0.80
	abnormal	9.23	0.53	9.46	0.52

Table 3.4. Predicted versus true age by subcohort and status

The three cross-validated models, scored per recording on three sets of recordings (`rs.prediction_sets`, from the per-recording CV predictions):

- `TUAB test, ensemble`: the 274 TUAB test recordings, predicted by the median of the 5 fold models (Eq. (3.8)).
- `TUAB out-of-fold`: the TUAB recordings of the CV pool (TUAB train + val), each predicted by the one fold model that never saw its patient.
- `TUEG out-of-fold`: the same for the pool's TUEG recordings that are not in TUAB.

`status` is the recording's normal / abnormal / no-label status (Eq. (1.2)), and `all` is every recording of the set. Per model, the columns are:

- `MAE` (Eq. (0.1), years);
- `bias` = mean(predicted − true) (Eq. (0.3), years): the signed drift. It is negative when the model predicts too young and positive when too old, so closer to 0 is better;
- R^2 (Eq. (0.2));
- the bias slope β (Eq. (0.5)).

All come from `rs.prediction_metrics`. Needs the per-recording CV predictions and the TUH index.

			cvbase				cvctrl				cvaug			
		n	MAE	bias	R²	β	MAE	bias	R²	β	MAE	bias	R²	β
set	status													
TUAB test, ensemble	all	274	7.44	-1.21	0.70	-0.37	7.10	-0.77	0.72	-0.35	7.23	-0.76	0.72	-0.34
	normal	149	5.75	-0.56	0.80	-0.30	5.31	-0.58	0.82	-0.29	5.43	-0.67	0.81	-0.29
	abnormal	125	9.46	-1.99	0.52	-0.50	9.23	-1.00	0.53	-0.50	9.37	-0.88	0.54	-0.49
TUAB out-of-fold	all	2350	7.18	-0.96	0.70	-0.35	7.11	-0.45	0.71	-0.34	7.12	-0.86	0.71	-0.33
	normal	1258	6.55	-0.85	0.70	-0.38	6.37	-0.61	0.71	-0.34	6.41	-1.15	0.71	-0.35
	abnormal	1092	7.91	-1.08	0.63	-0.41	7.97	-0.27	0.63	-0.41	7.93	-0.52	0.63	-0.41
TUEG out-of-fold	all	26685	8.27	-0.29	0.62	-0.42	8.17	0.02	0.63	-0.39	8.14	-0.05	0.63	-0.39
	normal	10197	7.59	-0.24	0.67	-0.38	7.51	-0.30	0.68	-0.36	7.48	-0.45	0.68	-0.36
	abnormal	14298	8.65	-0.48	0.58	-0.45	8.50	0.20	0.59	-0.43	8.50	0.16	0.59	-0.43
	no label	2190	8.99	0.78	0.51	-0.50	9.07	0.32	0.51	-0.50	8.94	0.38	0.51	-0.49

Figure 3.1. Predicted versus true age, per model and subcohort

Predicted age $\hat{y}_i$ against true age y_i (years), one point per recording; the data of Table 3.4. One row per cross-validated model, and two subcohorts per row:

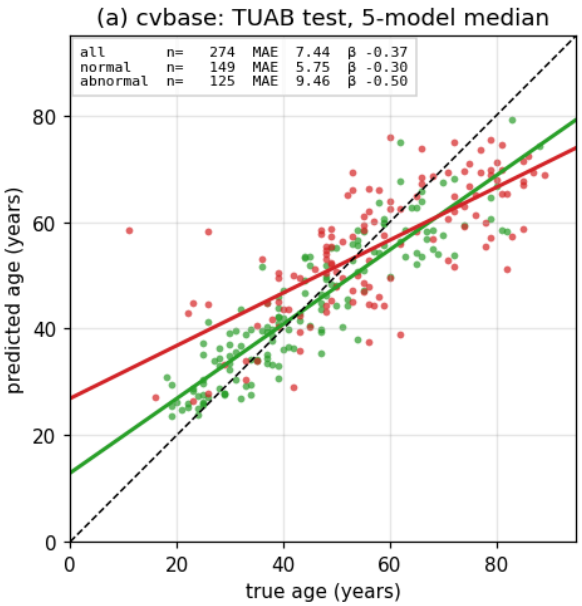
- left column: the 274 TUAB test recordings, predicted by the 5-model median ensemble;
- right column: the TUEG out-of-fold recordings (n = 26,685), each predicted by the fold model that held it out.

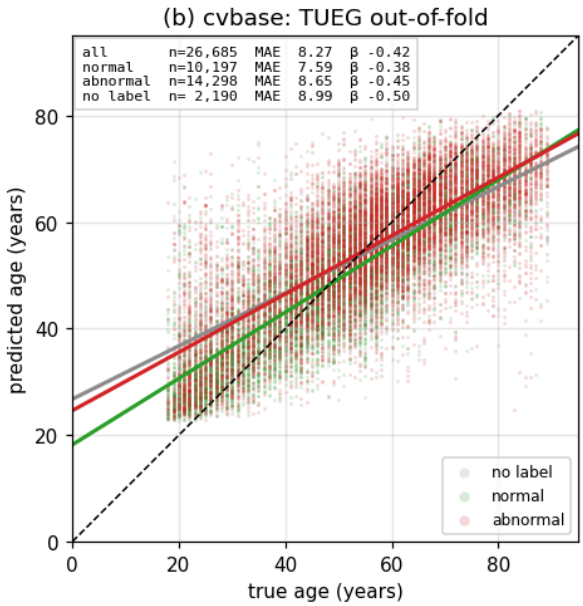
The panels are:

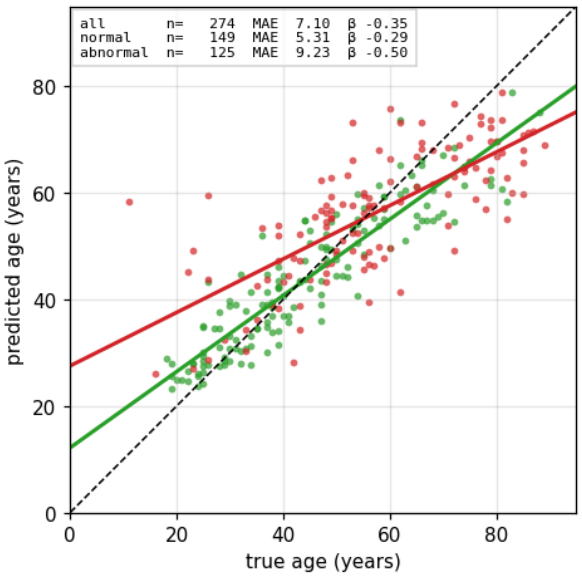
- (a), (b) `cvbase`, the best EEG-only model;
- (c), (d) `cvctrl`, the best overall;
- (e), (f) `cvaug`.

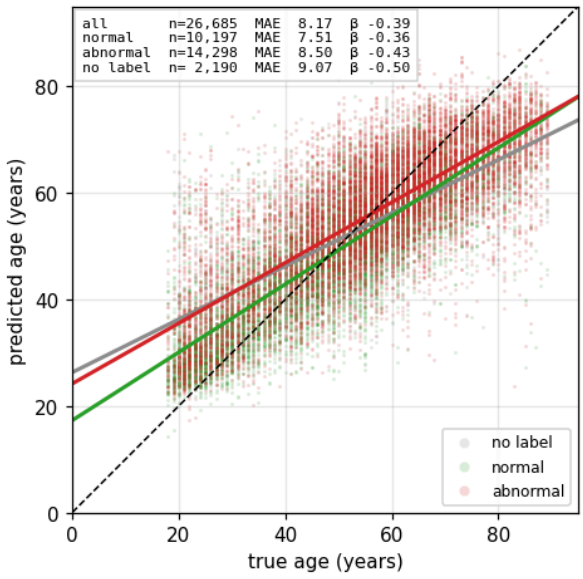
Colour = status (Eq. (1.2)): green normal, red abnormal, grey no label. The no-label group occurs only in TUEG.

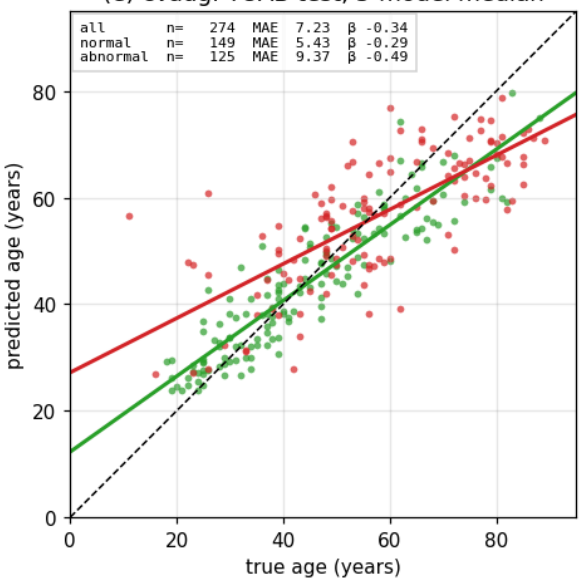
- **Dashed black diagonal:** a perfect prediction, $\hat{y} = y$.
- **Coloured lines:** least-squares fits of $\hat{y}$ on y per status. A fit flatter than the diagonal is the regression to the mean that β measures: the fit's slope is $1 + \beta$ (Eq. (0.5)).
- **Box in each panel:** n, MAE and β per status, and for all recordings together.

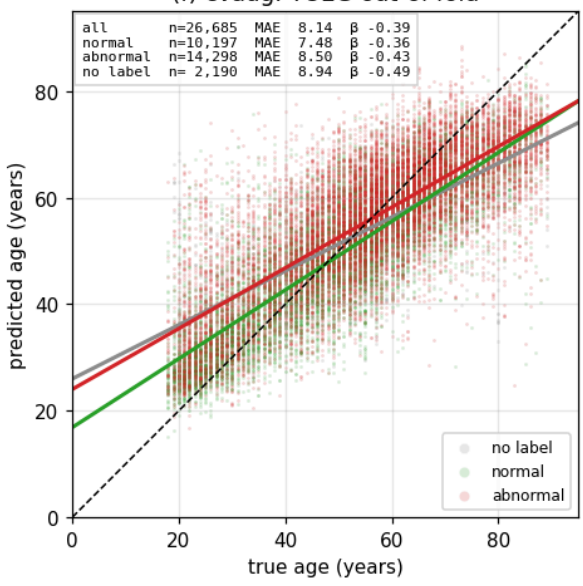

(a) cvbase: TUAB test, 5-model median
all n= 274 MAE 7.44 β -0.37
normal n= 149 MAE 5.75 β -0.30
abnormal n= 125 MAE 9.46 β -0.50
predicted age (years)
true age (years)


(b) cvbase: TUEG out-of-fold
all n=26,685 MAE 8.27 β -0.42
normal n=10,197 MAE 7.59 β -0.38
abnormal n=14,298 MAE 8.65 β -0.45
no label n= 2,190 MAE 8.99 β -0.50
no label
normal
abnormal
predicted age (years)
true age (years)


(c) cvctrl: TUAB test, 5-model median
all n= 274 MAE 7.10 β -0.35
normal n= 149 MAE 5.31 β -0.29
abnormal n= 125 MAE 9.23 β -0.50
predicted age (years)
true age (years)


(d) cvctrl: TUEG out-of-fold
all n=26,685 MAE 8.17 β -0.39
normal n=10,197 MAE 7.51 β -0.36
abnormal n=14,298 MAE 8.50 β -0.43
no label n= 2,190 MAE 9.07 β -0.50
no label
normal
abnormal
predicted age (years)
true age (years)


(e) cvaug: TUAB test, 5-model median
all n= 274 MAE 7.23 β -0.34
normal n= 149 MAE 5.43 β -0.29
abnormal n= 125 MAE 9.37 β -0.49
predicted age (years)
true age (years)


(f) cvaug: TUEG out-of-fold
all n=26,685 MAE 8.14 β -0.39
normal n=10,197 MAE 7.48 β -0.36
abnormal n=14,298 MAE 8.50 β -0.43
no label n= 2,190 MAE 8.94 β -0.49
no label
normal
abnormal
predicted age (years)
true age (years)

Reading Table 3.4 and Figure 3.1.

- **Abnormal recordings are harder in every set.**
  - On TUAB test, their MAE is 3.7–3.9 years above the normal recordings'.
  - In TUEG out-of-fold the gap is about 1 year (0.99–1.06). The no-label recordings are the hardest there (MAE 8.9–9.1).
- **Every model regresses to the mean** ($\beta \approx -0.3$ to -0.5). The fit lines in Figure 3.1 are visibly flatter than the diagonal: old patients are predicted too young and young ones too old. The effect is strongest for the abnormal and no-label recordings ($\beta \approx -0.43$ to -0.50).
- **Bias.** The overall drift is small: within ±1.2 years on TUAB test and ±0.3 on TUEG. But the EEG-only `cvbase` predicts abnormal TUAB test recordings 2.0 years too young on average, against 0.9–1.0 for `cvctrl` / `cvaug`.
- **TUEG is harder than TUAB.** TUEG out-of-fold MAE is 8.1–8.3, against 7.1–7.2 for the TUAB recordings of the same pool. Its normal recordings are worse too: 7.5–7.6 against 6.4–6.6. TUEG's status comes from report text rather than TUAB's curated folders. It also covers a broad clinical population: 38 % of its recordings in the clinical sheet list a sedating drug (Table 1.9).
- **cvctrl and cvaug beat cvbase on the TUAB test ensemble** by 0.2–0.35 years. Their gain is larger on normal recordings (0.3–0.4 years) than on abnormal ones (0.1–0.2).

Table 3.5. TUAB test MAE before and after age-bias correction, and published / classical methods

Every other number in this notebook is before bias correction. That includes the MAEs of Sections 0–5 and Figure 2.1, except the `cal.` columns of Tables 2.5–2.7, which use Eq. (3.11) below. They score the model's raw prediction $\hat{y}_i$. Brain-age studies often also report a *bias-corrected* error. Gemein et al. 2024 does: its 6.60 / 12.85 are after correction, and its raw normal MAE was 6.48. This table puts our CV models on that footing, and lists the other methods evaluated on the TUAB test set.

How a correction is made: three steps, and what each one sees.

1. **Predict.** Each CV experiment's 5 fold models predict every TUAB test recording, and the ensemble is their median (Eq. (3.8)). The TUAB test set is in no fold. No fold model trained on it or chose its epoch on it (the epoch is chosen on the fold's val fold).
2. **Fit, on out-of-fold predictions only.** Each recording of the CV pool (TUAB train + val, TUEG train + val; 29,035 recordings of 13,550 patients) is predicted by the one fold model whose in-pool **test fold** held it out. These are the out-of-fold predictions. They are not the val folds that chose the epoch. The correction is fitted on them with *their* true ages, which is allowed: none of them is a TUAB test recording and none shares a patient with one (Table 3.6). `normal` in a column name means the fit uses only the normal out-of-fold recordings, as Gemein et al. do.
3. **Apply to the TUAB test predictions.** This is where the methods differ:

- **Recalibration** (Eq. (3.11)) and **Cole** (Eq. (3.10)) map each prediction $\hat{y}_i$ to a corrected one using $\hat{y}_i$ alone. They never read a TUAB test true age, so their MAE is a valid test error.
- The **gap correction** (Eq. (3.9)) subtracts the fitted age trend *evaluated at the test recording's true age* y_i. It **does use the TUAB test labels** (Table 3.6: permuting them changes the corrected predictions by more than 25 years). Its MAE is therefore not a prediction error. It is the size of the brain-age gap after its age trend is removed, the quantity Gemein et al. and much of the brain-age literature call "corrected MAE". It is shown only to compare with them, marked †, and is used nowhere else in the notebook.

The three corrections (`rs.bias_corrected_test` , `rs.cole_corrected_test` , `rs.recalibrated_test`):

$$e_i = \hat{y}_i - y_i \approx p(y_i), \qquad \hat{y}_i^{c} = \hat{y}_i - p(y_i) \tag{3.9}$$

where p is a linear or quadratic polynomial fitted by least squares on the out-of-fold pairs (y_j, e_j) and evaluated at the test recording's true age y_i;

$$\hat{y}_i' = (\hat{y}_i - c) \,/\, a, \qquad (a, c) = \arg\min \textstyle\sum_j (\hat{y}_j - a\, y_j - c)^2 \tag{3.10}$$

the inverse of the out-of-fold line $\hat{y} \approx a\, y + c$ (Cole et al.);

$$\hat{y}_i'' = a'\, \hat{y}_i + b', \qquad (a', b') = \arg\min \textstyle\sum_j (y_j - a'\, \hat{y}_j - b')^2 \tag{3.11}$$

the least-squares map from prediction to age over the out-of-fold recordings j. Eq. (3.11) gives the best linear estimate of the age from the prediction. It lowers the error, but it does not remove the age trend of the gap (it shrinks predictions further toward the mean). Eq. (3.10) removes that trend but amplifies the noise of the prediction by $1/a$.

Phase-4 rows (H-E1 … H-E2L, Table 2.11). They are single models, not CV ensembles, so there are no out-of-fold predictions. Their corrections are fitted on each run's own **validation** recordings with a normal report instead (TUH val for the TUAB test; `scripts/brain_age_bias_report.py` , `docs/brain_age_bias_correction.md`). No TUAB test recording or patient is in those sets. The `gap quad, normal †` column is the same Eq. (3.9) quadratic gap correction as above, with the test recording's true age.

Which CV experiments (each name links to its ClearML project; Table 2.7):

- **`cvbase`** , **`cvctrl`** , **`cvaug`** use the scheme with an in-pool test fold: for fold k, test = fold k, val = fold $k+1$, and three folds train. Their out-of-fold predictions, and so the fit set, are the test-fold predictions. These recordings never trained a model and never chose an epoch.
- **`cvaugval`** uses the val-only scheme: val = fold k, the other four folds train, and the test set is TUAB. Its out-of-fold predictions, and so its fit set, are the **val-fold** predictions (`rs.CV_OOF_SPLIT`). They are its best checkpoint's, so the same recordings also chose its

epoch (epoch 7 in four folds, the last epoch, 9, in one). That makes the fit set slightly optimistic, but it still holds no TUAB test recording or patient (Table 3.6).

`gap quad, normal †`, step by step (with `cvctrl`'s numbers):

1. **Fit set.** The out-of-fold predictions of the **normal** recordings of the CV pool: 11,455 recordings of 6,387 patients. None is a TUAB test recording and none shares a patient with one (Table 3.6).
2. **Fit the age trend of the error.** Least squares of the error $e_j = \hat{y}_j - y_j$ on a quadratic in the true age over that set: $p(y) = q_2 y^2 + q_1 y + q_0$. For `cvctrl` this gives $q_2 = -0.00257$, $q_1 = -0.102$, $q_0 = 11.36$. So $p(30) = +6.0$ and $p(75) = -10.8$: even on normal recordings, the model predicts 30-year-olds 6 years too old and 75-year-olds 11 years too young. That is regression to the mean, the trend the correction removes.
3. **Apply it with the test recording's true age**, Eq. (3.9): $\hat{y}_i^c = \hat{y}_i - p(y_i)$. Example: an abnormal TUAB test recording with true age 75 is predicted 70.4 (raw error −4.6). Then $\hat{y}^c = 70.4 - p(75) = 70.4 + 10.8 = 81.2$, a corrected gap of **+6.2 years**: this brain looks 6 years older than the model's prediction for a typical normal 75-year-old.
4. **Score:** MAE (Eq. (0.1)) of the corrected predictions, on all, normal or abnormal TUAB test recordings.

Why this is not a data leak, and why it is still marked †.

- **Nothing about TUAB test is fitted.** The fold models, their epoch choice and p are all fitted without any TUAB test recording or patient (Table 3.6: 0 and 0).
- **The true age is used the way a brain-age study uses it.** In brain-age work the chronological age is always known, also when the model is applied to a new patient. The quantity of interest is the **brain-age gap** $\hat{y} - y$ (does this brain look older than it is?), not an estimate of the age. Step 3 removes from the gap the part that age alone explains. This is standard practice (Smith et al. 2019; de Lange & Cole 2020). **Gemein et al. 2024** do exactly this: a quadratic fitted on normal recordings, applied with the true ages. Their 6.60 (normal) / 12.85 (abnormal) are of this kind, which is why the `gap quad, normal †` column is the one to compare with them.
- **But it is no longer an age prediction.** $\hat{y}^c$ uses y_i, so its MAE measures the remaining gap. It compares only with other gap-corrected numbers (Gemein's), never with a raw MAE or with the leak-free recalibration. Table 3.6 shows this directly: permuting the test ages moves these predictions, and moves no other column.

Columns. TUAB test MAE (Eq. (0.1), years) per model and status:

- `raw`: no correction (as in Table 3.4);
- `recal, normal`: Eq. (3.11), fitted on the normal out-of-fold recordings. No test label;
- `cole, normal`: Eq. (3.10), fitted on the normal out-of-fold recordings. No test label;
- `gap lin, all †` / `gap quad, normal †`: Eq. (3.9), linear on all out-of-fold recordings or quadratic on the normal ones (Gemein et al.'s choice). † = **uses the TUAB test true age**;

- `bias, gap quad normal †`: the mean gap (Eq. (0.3)) after that correction. A positive value means the recordings look older than they are.

Other methods, with each source's own numbers; "–" means the source does not report it:

- **Gemein et al. 2024:** a TCN ensemble trained on TUEG normal recordings, scored on TUAB's evaluation set. Raw normal 6.48; after their quadratic gap correction (Eq. (3.9), so with the test ages), 6.60 normal and 12.85 abnormal.
- **Engemann et al. 2022:** the M/EEG brain-age benchmark. Best model 7.75 on TUAB normal recordings, raw, 10-fold CV on the TUAB normal training recordings (not the official evaluation set).
- **Handcrafted features + ridge** (this repo, `notebooks/eeg_feature_extraction_tuab.ipynb` §10). This is the classical approach of Engemann et al.: filterbank-Riemann, spectral, wavelet and other EEG features with a ridge regression. It is scored raw on **the same 274 TUAB test recordings**, and trained either on TUAB normal recordings only (the standard brain-age setup) or on all TUAB recordings.

model	status	raw	recal, normal	cole, normal	gap lin, all †	gap quad, normal †	bias, gap quad normal †
cvbase	all	7.44	7.28	9.09	6.06	5.61	-0.05
	normal	5.75	5.58	7.77	5.65	4.77	-1.63
	abnormal	9.46	9.31	10.67	6.56	6.60	1.84
cvctrl	all	7.10	6.98	8.86	6.03	5.59	0.39
	normal	5.31	5.19	7.35	5.53	4.62	-1.55
	abnormal	9.23	9.11	10.67	6.61	6.74	2.70
cvaug	all	7.23	7.11	8.81	6.05	5.63	0.57
	normal	5.43	5.30	7.16	5.54	4.55	-1.42
	abnormal	9.37	9.28	10.78	6.67	6.91	2.96
cvaugval	all	7.22	7.04	8.68	6.01	5.60	0.46
	normal	5.36	5.16	7.01	5.40	4.48	-1.49
	abnormal	9.44	9.28	10.68	6.72	6.93	2.79
Gemein et al. 2024 (TCN, TUEG normal)	normal	6.48	–	–	–	6.60	–
	abnormal	–	–	–	–	12.85	–
Engemann et al. 2022 (TUAB normal, 10-fold CV)	normal	7.75	–	–	–	–	–
features + ridge, trained on TUAB normal	all	9.70	–	–	–	–	–
	normal	7.21	–	–	–	–	–
	abnormal	12.67	–	–	–	–	–
features + ridge, trained on all TUAB	all	8.53	–	–	–	–	–
	normal	6.95	–	–	–	–	–
	abnormal	10.41	–	–	–	–	–
H-E1	all	6.90	–	–	–	5.23	0.34
	normal	5.83	–	–	–	4.62	-0.99
	abnormal	8.18	–	–	–	5.95	1.93
H-E2	all	6.70	–	–	–	5.19	0.26
	normal	5.13	–	–	–	4.24	-1.50
	abnormal	8.56	–	–	–	6.32	2.36
H-E2c (+ status/pathology input)	all	6.82	–	–	–	5.23	0.67
	normal	5.20	–	–	–	4.25	-1.23
	abnormal	8.76	–	–	–	6.40	2.95
H-E2L	all	6.54	–	–	–	5.31	0.11
	normal	4.91	–	–	–	4.29	-1.71
	abnormal	8.50	–	–	–	6.52	2.28

Table 3.6. Leakage check: does a correction see the TUAB test labels?

Two checks per correction of Table 3.5 (`rs.correction_leak_check`), over the four CV experiments:

- **Fit set. fit ∩ test recordings** / **fit ∩ test patients**: how many of the out-of-fold recordings the correction is fitted on are TUAB test recordings, or belong to a patient with a TUAB test recording (the patient is the first field of the recording name). Both must be 0. The `normal` fits use a subset of these recordings. The fit set is the whole CV pool, which is also everything the fold models trained on, so 0 shared patients also means no fold model ever saw a TUAB test patient.
- **Application.** The TUAB test true ages are randomly permuted within each experiment and the correction is run again on the same predictions. A correction that never reads the test ages returns identical corrected predictions. `max change (years)` is the largest

difference over all experiments and test recordings, and `uses TUAB test true age` is "yes" when it is above zero.

The `raw` row is the reference: it applies no correction, so it cannot change.

method	fit ∩ test recordings	fit ∩ test patients	max change (years)	uses TUAB test true age
raw	0	0	0.00	no
recal, normal	0	0	0.00	no
cole, normal	0	0	0.00	no
gap lin, all †	0	0	26.52	yes
gap quad, normal †	0	0	26.34	yes

Reading Tables 3.5 and 3.6.

- **No TUAB test recording or patient is in any fit set** (Table 3.6: 0 and 0). The fold models, their epoch choice and every correction fit are independent of the TUAB test set.
- **Only the gap correction reads the test labels**, when it is applied. Permuting the TUAB test ages leaves `recal` and `cole` unchanged (max change 0) but moves the gap-corrected predictions by more than 25 years. Its numbers (†) are brain-age-gap sizes, not test errors.
- **Leak-free, recalibration helps a little.** Eq. (3.11) lowers the MAE from 7.10–7.44 to 6.98–7.28 overall, 5.31–5.75 → 5.16–5.58 on normal and 9.23–9.46 → 9.11–9.31 on abnormal recordings. It is used for the `cal.` columns of Tables 2.5–2.7.
- **The Cole correction makes the error worse** (`cole, normal`): 7.10–7.44 raw becomes 8.68–9.09 overall. Removing the age trend this way spreads out the predictions, and their noise with them.
- **With the test ages (†), the gap shrinks a lot.** A quadratic fitted on normal recordings (Gemein's choice) gives 4.48–4.77 on normal and 6.60–6.93 on abnormal recordings, against Gemein's 6.60 / 12.85 with the same correction. Much of the raw abnormal error is regression to the mean ($\beta \approx -0.5$, Table 3.4), and Eq. (3.9) undoes it with the true age. These numbers compare only with other gap-corrected numbers, never with a raw or recalibrated MAE.
- **Abnormal recordings look older.** After the normal-fit gap correction they keep a positive gap of +1.8 to +3.0 years (`bias, gap quad normal †`), and normal recordings a negative one. This is the brain-age-gap effect Gemein et al. report: −4 years between normal and pathological subjects.
- **Against the other methods, raw on the same test set.** The EEG-only `cvbase` scores 7.44 all / 5.75 normal / 9.46 abnormal. The classical feature-ridge model scores 8.53 / 6.95 / 10.41 when trained on all of TUAB, and 9.70 / 7.21 / 12.67 trained on normal recordings only. Engemann et al.'s best is 7.75 on normal recordings (CV, not the evaluation set). The foundation model trained on TUAB + TUEG is better on every cohort. Against the feature model trained on all of TUAB it is 0.95–1.5 years ahead (smallest on abnormal recordings). Against the normal-only feature model it is 1.5–3.2 years ahead, most on abnormal recordings, where normal-only models fail (12.67 here, 12.85 for Gemein et al.).

Table 3.7. Agreement between experiments on the TUAB test set

How similar are the models' predictions, recording by recording? The comparison covers two kinds of pair:

- every pair of experiment ensembles (cvbase / cvctrl / cvaug: the CV runs with an in-pool test fold; the val-only `cvaugval` is compared in Tables 2.7 and 3.5);
- as a reference, the 10 pairs of fold models *within* each experiment. These pairs share the configuration but differ in training data and seed.

For two predictors a and b on the same recordings, with errors $e_i^a = \hat{y}_i^a - y_i$ (`labram/eval/cv_compare.py::pairwise`):

$$\mathrm{agree}_\tau = \frac{1}{n}\sum_i 1\big[|\hat{y}_i^a - \hat{y}_i^b| \le \tau\big], \qquad \mathrm{sign} = \frac{1}{n}\sum_i 1\big[\operatorname{sgn} e_i^a = \operatorname{sgn} e_i^b\big] \tag{3.12}$$

Columns, per status (all 274, normal 149, abnormal 125):

- `r_pred`, `r_err`: the Pearson correlation of the two models' predictions and of their errors. `r_err` near 1 means they make the same mistakes on the same recordings.
- `agree_2y`, `agree_5y`: agree_τ for τ = 2 and 5 years.
- `sign_agree`: Eq. (3.11, right). The share of recordings where both models err in the same direction (both predict too old, or both too young).
- `dMAE`: $\mathrm{MAE}_a - \mathrm{MAE}_b$ (Eq. (0.4)) for an ensemble pair; the mean $|\Delta\mathrm{MAE}|$ over the 10 pairs for a fold-model row.

status	pair	r_pred	r_err	agree_2y	agree_5y	sign_agree	dMAE
all	cvbase vs cvctrl (ensembles)	0.987	0.974	0.693	0.960	0.912	0.340
	cvbase vs cvaug (ensembles)	0.989	0.977	0.701	0.978	0.912	0.214
	cvctrl vs cvaug (ensembles)	0.994	0.987	0.821	0.989	0.920	-0.126
	cvbase: fold model vs fold model (mean of 10 pairs)	0.982	0.966	0.668	0.934	0.931	0.107
	cvctrl: fold model vs fold model (mean of 10 pairs)	0.969	0.936	0.532	0.876	0.889	0.177
	cvaug: fold model vs fold model (mean of 10 pairs)	0.972	0.939	0.520	0.886	0.872	0.208
normal	cvbase vs cvctrl (ensembles)	0.992	0.975	0.805	0.980	0.893	0.439
	cvbase vs cvaug (ensembles)	0.992	0.976	0.819	0.993	0.899	0.319
	cvctrl vs cvaug (ensembles)	0.995	0.984	0.899	1.000	0.886	-0.120
	cvbase: fold model vs fold model (mean of 10 pairs)	0.989	0.969	0.759	0.982	0.914	0.108
	cvctrl: fold model vs fold model (mean of 10 pairs)	0.981	0.939	0.621	0.938	0.872	0.101
	cvaug: fold model vs fold model (mean of 10 pairs)	0.985	0.949	0.580	0.940	0.851	0.209
abnormal	cvbase vs cvctrl (ensembles)	0.976	0.976	0.560	0.936	0.936	0.222
	cvbase vs cvaug (ensembles)	0.980	0.980	0.560	0.960	0.928	0.088
	cvctrl vs cvaug (ensembles)	0.988	0.988	0.728	0.976	0.960	-0.134
	cvbase: fold model vs fold model (mean of 10 pairs)	0.963	0.964	0.558	0.878	0.950	0.133
	cvctrl: fold model vs fold model (mean of 10 pairs)	0.936	0.936	0.427	0.802	0.909	0.311
	cvaug: fold model vs fold model (mean of 10 pairs)	0.941	0.937	0.450	0.822	0.898	0.214

Figure 3.2. Error correlation between all TUAB-test predictors

The Pearson correlation of the errors $e_i = \hat{y}_i - y_i$ between every pair of the 18 predictors of the TUAB test set: each experiment's 5 fold models (`f0` – `f4`) and its median ensemble (`ens`).

- (a) The 149 normal recordings.
- (b) The 125 abnormal recordings.

The colour scale runs from 0.85 (lighter) to 1 (darker). The thin white lines separate the three experiments.

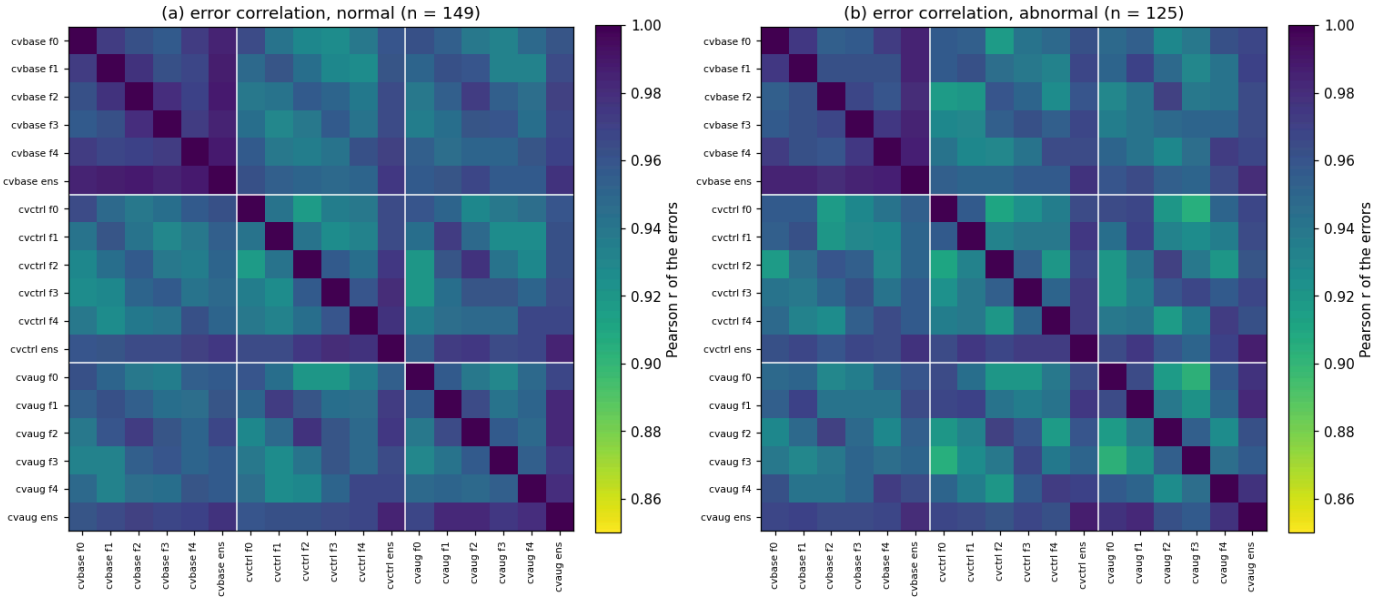
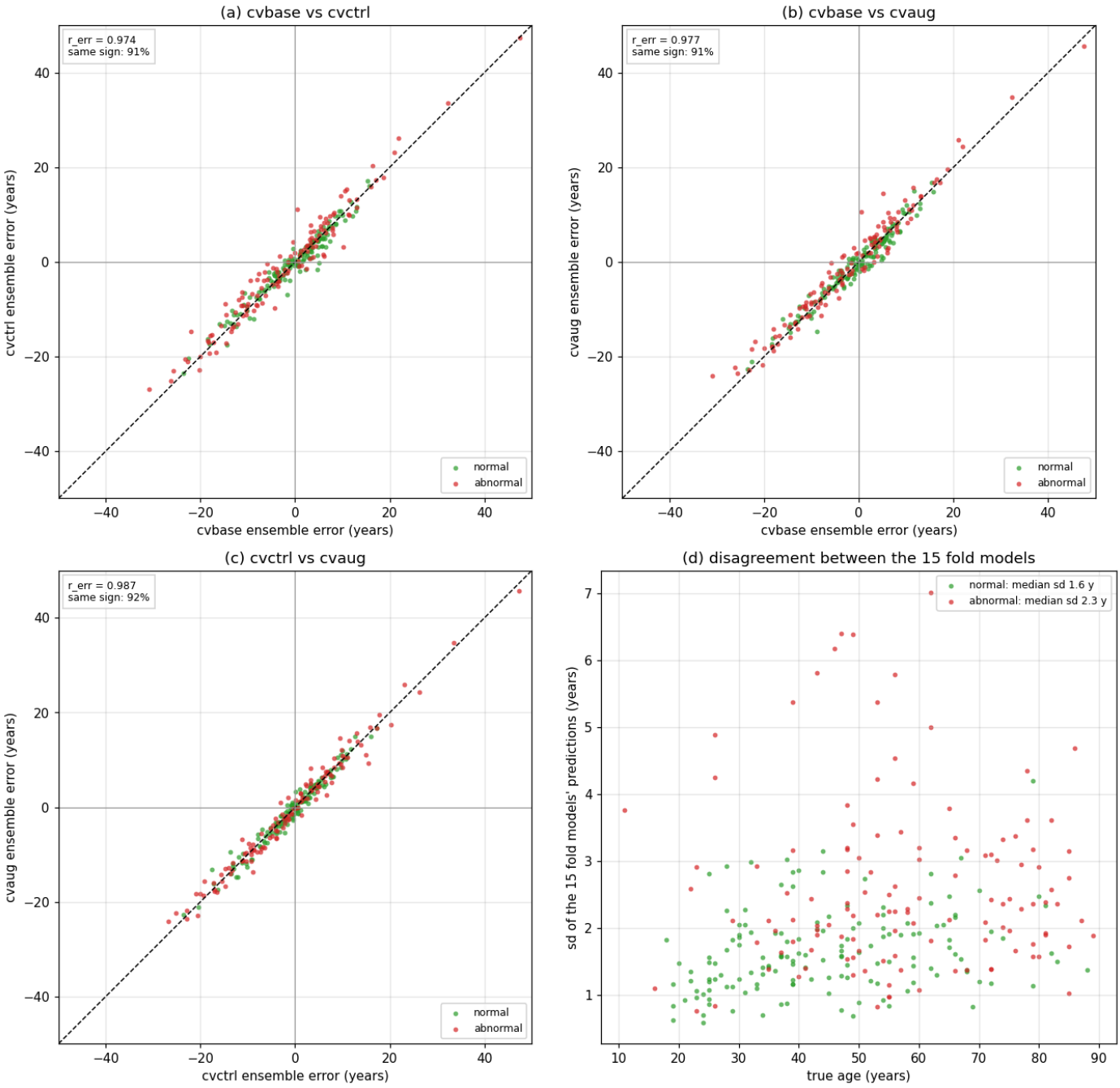


Figure 3.3. Per-recording agreement of the three experiments

TUAB test recordings, coloured by status: green normal, red abnormal. Errors $e = \hat{y} - y$ are in years.

- (a)–(c) The error of one experiment's ensemble against another's, for each pair of experiments. A point on the dashed diagonal means both models made the same error. A point in the upper-right or lower-left quadrant means both erred in the same direction.
- (d) **Disagreement per recording:** the sample standard deviation of the 15 fold models' predictions (3 experiments × 5 folds), against the true age.



Reading Table 3.7 and Figures 3.2 and 3.3.

- **The three experiments make almost the same errors.** Their ensembles' errors correlate at r = 0.974–0.987 (Figure 3.2, Figure 3.3(a)–(c)). On 91–92 % of the recordings both err in the same direction, and on 96–99 % they predict within 5 years of each other. The differences in MAE between them (Table 2.7) come from small shifts, not from different recordings going wrong.
- **The ensembles agree more than single fold models do.** The ensemble pairs agree more closely across experiments (r_err 0.97–0.99, within 2 years on 69–82 % of recordings) than two fold models of the same experiment (r_err 0.94–0.97, 52–67 %). Averaging five models removes much of the fold-to-fold noise, which is why the ensemble beats a single model (Table 5.2, C3).

- **Abnormal recordings: shared errors, more disagreement.** The models agree on the error direction more often on abnormal recordings: 93–96 % same sign, against 89–90 % on normal ones. But their predictions spread more there, with fewer abnormal recordings within 2 years (56–73 %, against 81–90 %). The fold models also disagree more on abnormal recordings (Figure 3.3(d)).
- **The largest errors are shared.** In Figure 3.3(a)–(c), the points far from the origin lie near the diagonal: the recordings that one model gets badly wrong, all models get badly wrong. Of the 20 largest `cvctrl` test errors, 16 are abnormal recordings (46 % of the test set is abnormal), and 17 are patients under 30 or over 70. These recordings are the main lever left for improving the MAE.

4. Changes versus the LaBraM recipe

The reference is **LaBraM as published for TUAB**, `labram/configs/defaults/finetune_tuab_paper.json`:

- LaBraM-base fine-tuned for normal/abnormal classification, the recipe of the paper and the README;
- lr $5 \cdot 10^{-4}$, batch 64, 50 epochs with 5 warmup, layer decay 0.65, drop path 0.1;
- no input re-referencing;
- a 50 Hz notch in preprocessing (`dataset_maker/make_TUAB.py`).

Table 4.1 lists every configuration field the best configuration changes. Table 4.2 lists the changes that live in the data or the code rather than the configuration. Section 5 then tests which of them moved the error.

Table 4.1. Configuration changes: LaBraM TUAB recipe → best configuration

Every field on which `finetune_tuab_paper.json` and `cvctrl` (Table 3.1) differ, computed by `labram/eval/research_summary.py::config_diff`. Excluded:

- bookkeeping fields (paths, logging, infrastructure);
- fields a regression run never reads (`INERT_FOR_REGRESSION`);
- fields of a section that is off in both.

Columns:

- `group` : the part of the pipeline;
- `meaning` : what the change does;
- `evidence` : the rows of Table 5.2 that test it, where one does.

group	field	LaBraM recipe	best (cvctrl)	meaning	evidence
augmentation	**mixup.enabled**	False	True	mixup α 0.4 (Eq. (3.3))	P3
channels	**channel_drop.channels**	[]	["T1", "T2"]	T1/T2 are the channels that may be missing	Q14
	channel_drop.consistency_weight	1	0	0: no consistency loss in cvctrl	Q17, C2
	channel_drop.enabled	False	True	per-sample handling of missing T1/T2 (needed for cohort v3)	Q14
	channel_drop.max_drop	1	2	up to both of T1/T2	Q14
	channel_drop.prob	0.2	0	0: no channel-dropout augmentation in cvctrl	Q17, C2
data	**data.data_format**	pickle	npy	storage only: one float32 npy per recording, windows bit-identical to the pickles	
head	**data.session_features**		conditions_v1.npz	the condition vectors' file (conditions_v1.npz)	Q12, C1
	model.session_features	off	head	status + pathology vector concatenated to the feature (Eq. (3.6))	Q12, C1
	model.use_mean_pooling	True	False	head on the class token instead of the mean patch token	P22, Q13, C1
input	**labram_plus.enabled**	False	True	turns on the LaBraM++ input block (Eq. (3.1))	P1
	labram_plus.z_score_patches	True	False	LaBraM++ block with CAR only, no per-patch z-scoring	P1
optimizer	**optimizer.lr**	0.0005	0.0002	2e-4 at batch 128 = 1e-4 at batch 64, linearly scaled	Q14
	trainer.batch_size	64	128	batch 128 (speed; lr scaled with it)	Q14
protocol	**cross_validation.enabled**	False	True	5-fold patient-grouped cross-validation	C3
schedule	**optimizer.warmup_epochs**	5	3	3 warmup epochs	P2, Q6, C1
	trainer.epochs	50	10	10 epochs instead of 50 (best epoch is early)	P2, Q6, C1
scoring	**evaluation.agg_windows**	none	mean	one prediction per recording = mean of its windows (Eq. (3.7))	
speed	**model.fused_attention**	False	True	fused attention kernel, numerically the same model	
task	**data.dataset**	TUAB	TUAB_AGE	age regression on TUH (TUAB_AGE loader) instead of normal/abnormal	
	model.task	classification	regression	scalar regression head (Huber on z-scored age, Eqs. (3.2), (3.4))	
windows	**data.eval_minutes**	0	5	score each recording on its first 5 minutes after the trim	P26
	data.train_minutes_per_epoch	0	5	5 minutes per recording per epoch, redrawn each epoch (speed)	Q14
	data.trim_end_sec	0	60	drop the last 60 s of every recording	
	data.trim_start_sec	0	60	drop the first 60 s of every recording	

Table 4.2. Changes outside the configuration

Changes in the training data, the preprocessing, the split and the code, with the Table 5.2 rows that test them.

change	LaBraM recipe	this study	evidence
training data	TUAB train, 2,145 recordings (≈1.6k patients)	TUAB + TUEG, 25,613 recordings (≈12k patients, `all_both_v3`)	Q1, Q3
mains interference	50 Hz notch (TUH is recorded at 60 Hz, so the line stays in)	50 + 60 Hz spectrum interpolation	L1
channels	23 fixed channels	23; TUEG recordings missing only T1/T2 kept, with those channels left out per sample	Q14
validation split	TUAB's train split, which leaks 16 patients between train and val	patient-disjoint, seeded; one TUEG patient split shared by all cohorts; TUAB-test patients removed from TUEG	Q11
label	normal/abnormal	age from the EDF header (18–89 in the TUH cohorts)	–
unit of evaluation	10 s window	recording (Eq. (3.7)), first 5 minutes after the trim	P26
per-patient cap	none	at most 10 train/val recordings per patient	–
reported model	one run	median of 5 fold models (Eq. (3.8))	C3
reproducibility	ClearML silently reseeded every tracked run to 1337	fixed: `distributed.seed` is honoured (commit `7b9a5fe`)	–

5. Which changes mattered

Noise floor. Two runs of the same configuration with different seeds do not give the same MAE. The line-removal A/B study ran each of 3 preprocessing recipes r with 3 seeds s, and each run is read at its best-validation epoch. The pooled within-recipe sample standard deviation of a metric m (`labram/eval/research_summary.py::seed_noise`) is

$$\hat{\sigma}_m^2 = \frac{1}{N-R} \sum_r \sum_s \left(m_{r,s} - \bar{m}_r\right)^2 \tag{5.1}$$

with $N = 9$ runs and $R = 3$ recipes.

Significance rule. Each row of Table 5.2 compares a baseline with a variant that changes one thing (or a named bundle), using whichever evidence exists:

- **One run each** (phases 1 and 2). The difference of two single runs has standard deviation $\sqrt{2}\,\hat{\sigma}$, so a change counts only if (`rs.pair_threshold`)

$$|\Delta\mathrm{MAE}| > \tau = 1.96\,\sqrt{2}\,\hat{\sigma}_{\mathrm{test}}. \tag{5.2}$$

- **Three seeds each** (line removal, L1/L2):

$$|\bar{\Delta}| > 1.96\,\hat{\sigma}_{\mathrm{test}}\,\sqrt{2/3}. \tag{5.3}$$

- **Five paired folds** (CV, C1/C2). Fold k gives Δ_k on the same TUAB test recordings, and the 95 % interval is

$$\bar{\Delta} \pm t_{0.975,\,4}\,\frac{s_\Delta}{\sqrt{5}}, \qquad t_{0.975,\,4} = 2.776, \tag{5.4}$$

  where $\bar{\Delta}$ and s_Δ are the mean and sample sd of the Δ_k. The change counts if the interval excludes 0.

- **Ensemble vs single model** (C3). A paired bootstrap over recordings (2,000 resamples, `cv_compare.py::bootstrap_delta_mae`) of the ensemble's absolute error against the fold models' mean absolute error. The change counts if its 95 % interval excludes 0.

The verdict is `improves` / `hurts` / `n.s.` (`rs.verdict`). It is judged on the TUAB test MAE of all 274 recordings, the one set every run shares; the val and cohort columns are context.

Limits.

- $\hat{\sigma}$ is measured on one configuration, the E2 recipe on TUAB (Table 5.1), and assumed to hold for the others.
- A single-split row with $|\Delta|$ just under τ is not evidence of no effect, only of no detectable one.
- Several rows change more than one thing at once; they are marked *bundle*.

Table 5.1. Seed-to-seed spread

The model. All 9 runs train the same model with the same recipe, **E2**: scenario D (`finetune_tuab_age.json`) on the short schedule. Read from each run's ClearML configuration:

- **Encoder:** LaBraM-base (`labram_base_patch200_200` , 12 transformer blocks), initialized from the released `labram-base.pth` , all layers fine-tuned.
- **Input and head:** CAR input (Eq. (3.1)), mean patch-token pooling and a linear head.
- **Loss and optimizer:** Huber $\delta = 1$ on the z-scored age (Eq. (2.3)), no mixup. AdamW with lr 10^{-4}, batch 64, weight decay 0.05, layer decay 0.65 and drop path 0.1.
- **Schedule and evaluation:** 4 epochs with 1 warmup, scored on the first 5 minutes of each recording after the 60 s trims.

What varies. The data is TUAB v3.0.1, on the same recordings and the same subject-disjoint split as phase 1 (Table 1.5). Only two things change:

- **Preprocessing:** the mains-interference removal, 3 variants. A is the 50 Hz notch (LaBraM as published), B a 60 Hz notch, C 50 + 60 Hz spectrum interpolation.
- **Seed:** 3 seeds per variant, which change the head initialization, the batch order, dropout and drop path. Seed 0 ran on a local g5.2xlarge; seeds 1 and 2 ran as SageMaker jobs on ml.g5.2xlarge (`scripts/submit_line_removal_ab.sh`).

Within a variant, the three runs differ only by the seed, which is what Eq. (5.1) pools. Each run is read at its best-validation epoch (`docs/data/line_removal_ab.json`). The table then gives the pooled standard deviation of Eq. (5.1) per metric and the single-pair threshold τ of Eq. (5.2).

Columns:

- `val_mae` : the TUAB validation split.
- `test_mae` , `test_normal_mae` , `test_abnormal_mae` : TUAB test, overall and per cohort.

variant	seed	epoch	val_mae	test_mae	test_normal_mae	test_abnormal_mae
A_notch50	0	3	7.52	8.43	6.62	10.59
	1	3	7.66	8.40	6.60	10.55
	2	3	7.64	8.24	6.41	10.42
B_notch60	0	2	7.46	8.01	6.39	9.96
	1	3	7.55	8.31	6.55	10.41
	2	3	7.54	8.07	6.38	10.08
C_interp5060	0	2	7.46	8.03	6.39	9.99
	1	3	7.56	8.31	6.54	10.42
	2	3	7.56	8.10	6.39	10.13

```
pooled seed sd (years): val_mae 0.061, test_mae 0.138, test_normal_mae 0.100, test_abno
rmal_mae 0.194
single-pair threshold τ on TUAB test (Eq. 5.2): 0.38 years; 3-vs-3-seed threshold (Eq.
5.3): 0.22 years
```

Table 5.2. Effect of each change

One row per tested change. Columns:

- `id` : the row's id, cited in Tables 4.1, 4.2 and 5.3.
- `phase` : 1 (registry, Table 2.2), 2 (Table 2.5), L (line-removal A/B, Table 5.1), C (CV, Table 2.7).

- `baseline → variant`: the two runs compared; `bundle` marks a change of several things at once.
- `Δ val`: Eq. (0.4) on the validation set the two runs share (phase 1: TUAB val; phase 2: TUEG val v2 where both runs were scored on it).
- `Δ test`, `Δ normal`, `Δ abnormal`: Eq. (0.4) on TUAB test, overall and per cohort.
- `noise`: $\pm\tau$ (Eqs. (5.2), (5.3)) or the 95 % interval (Eq. (5.4) / bootstrap).
- `verdict`: the rule above, on `Δ test`.

id	phase	change	baseline → variant	Δ val	Δ test	Δ normal	Δ abnormal	noise	verdict
P1	1	common average reference on the input	B → D	-0.63	-0.61	–	–	± 0.38	improves
P26	1	score the first 5 min instead of whole recordings	A → B	0.11	-0.12	–	–	± 0.38	n.s.
P2	1	4 epochs / 1 warmup instead of 15 / 2	D → E2	-0.09	-0.05	–	–	± 0.38	n.s.
P3	1	mixup α 0.4	E2 → M1	-0.22	0.20	–	–	± 0.38	n.s.
P4	1	C-Mixup (σ 5 y)	E2 → M1c	-0.14	-0.02	–	–	± 0.38	n.s.
P5	1	EMA weights for evaluation	E2 → M2	-0.10	-0.08	–	–	± 0.38	n.s.
P6	1	layer decay 0.5 (stronger)	E2 → F1	0.06	0.11	–	–	± 0.38	n.s.
P7	1	layer decay 0.75 (weaker)	E2 → H1	0.03	0.42	–	–	± 0.38	hurts
P8	1	dropout 0.1	E2 → F2	5.57	6.92	–	–	± 0.38	hurts
P9	1	drop path 0.2	E2 → F3	0.02	0.16	–	–	± 0.38	n.s.
P10	1	weight decay 0.1	E2 → F4	-0.02	-0.01	–	–	± 0.38	n.s.
P11	1	head init scale 1.0	E2 → G1	0.08	0.13	–	–	± 0.38	n.s.
P12	1	LoRA r16 instead of full fine-tuning (bundle)	E2 → M3	0.00	0.99	–	–	± 0.38	hurts
P13	1	L1 loss instead of Huber	B → D2	-0.04	0.41	–	–	± 0.38	hurts
P14	1	Huber δ 2	D → DX2	0.02	0.11	–	–	± 0.38	n.s.
P15	1	soft labels (σ 2.5 y)	E2 → M4	-0.01	0.34	–	–	± 0.38	n.s.
P16	1	age-balanced loss (sqrt-inverse)	E2 → B1	0.15	0.23	–	–	± 0.38	n.s.
P17	1	codebook amplitude reconstruction	D → DX1	-0.05	0.24	–	–	± 0.38	n.s.
P18	1	codebook quantization term	D → DX4	-0.10	0.14	–	–	± 0.38	n.s.
P19	1	30 s windows instead of 10 s	B → C	-0.16	0.44	–	–	± 0.38	hurts
P20	1	random crop	D → E1	0.05	0.26	–	–	± 0.38	n.s.
P21	1	gradient clipping 1.0	D → D-clip1	-0.07	0.06	–	–	± 0.38	n.s.
P22	1	class token instead of mean pooling	D → D-cls	0.13	-0.06	–	–	± 0.38	n.s.
P23	1	[mean ; class token] concatenated	D → D-clsmean	0.03	0.05	–	–	± 0.38	n.s.
P24	1	multi-task head (age + normal/abnormal)	M1 → MT1	0.08	-0.07	0.13	-0.30	± 0.38	n.s.
P25	1	pretrained LaBraM weights instead of training from scratch	S1 → D	-1.02	-1.12	–	–	± 0.38	improves
L1	L	50 + 60 Hz interpolation instead of 50 Hz notch	A_notch50 → C_interp5060	-0.08	-0.21	-0.10	-0.34	± 0.22	n.s.
L2	L	60 Hz notch instead of 50 Hz notch	A_notch50 → B_notch60	-0.09	-0.23	-0.11	-0.37	± 0.22	improves
Q1	2	add every TUEG recording (~10x data)	Control → Exp3	–	-0.97	-0.85	-1.11	± 0.38	improves
Q2	2	train on normal recordings only (TUAB + TUEG) (bundle)	Control → Exp1	-0.62	-0.20	-0.43	0.08	± 0.38	n.s.
Q3	2	add abnormal + unlabelled recordings to normal-only	Exp1 → Exp3	–	-0.77	-0.41	-1.19	± 0.38	improves
Q4	2	TUEG normal only (no TUAB in training)	Exp1 → Exp2	0.07	-0.03	0.07	-0.16	± 0.38	n.s.
Q5	2	10 / 3 epochs on the normal cohort	Exp1 → Exp1b	-0.02	0.08	0.11	0.05	± 0.38	n.s.
Q6	2	10 / 3 epochs on the all cohort	Exp3 → Exp3b	–	0.43	0.60	0.23	± 0.38	hurts
Q7	2	window rejection (±800 µV, flat channel)	Exp3 → Exp3q	–	0.07	0.12	0.03	± 0.38	n.s.
Q8	2	input clip ±800 µV	Exp3 → Exp3c	–	0.02	0.09	-0.06	± 0.38	n.s.
Q9	2	classical features as transformer tokens	Exp1 → Exp1t	–	0.76	1.19	0.26	± 0.38	hurts
Q10	2	classical features only, no EEG encoder	Exp1 → Exp1f	–	0.10	0.22	-0.06	± 0.38	n.s.
Q11	2	shared TUEG patient split (v2)	Exp3 → Exp3v2	–	0.15	-0.11	0.47	± 0.38	n.s.
Q12	2	status + pathology condition vectors	Exp3v2 → Exp3cond	-0.10	-0.14	-0.07	-0.22	± 0.38	n.s.
Q13	2	class token (with conditions)	Exp3cond → Exp3clscond	0.07	-0.03	-0.03	-0.02	± 0.38	n.s.
Q14	2	cohort v3 + missing-T1/T2 handling + speed settings (bundle)	Exp3v2 → Exp4base	-0.14	-0.05	0.16	-0.31	± 0.38	n.s.
Q15	2	compact copy of the cohort (reproduction)	Exp4base → Exp4basec	-0.01	0.06	-0.10	0.24	± 0.38	n.s.
Q16	2	class token + conditions + 10 ep with early stopping (bundle)	Exp4base → Exp4ctrl	0.05	-0.06	-0.27	0.19	± 0.38	n.s.
Q17	2	T1/T2 dropout + consistency (6 vs 10 epochs) (bundle)	Exp4ctrl → Exp4aug	-0.07	-0.01	0.09	-0.14	± 0.38	n.s.
Q18	2	T1/T2 dropped everywhere (21 channels) + cohort v3 (bundle)	Exp3cond → Exp4drop	-0.15	-0.00	-0.13	0.16	± 0.38	n.s.
Q19	2	15 / 5 epochs with early stopping (aug)	Exp4aug → Exp4aug15	0.06	-0.12	-0.06	-0.20	± 0.38	n.s.
C1	C	class token + conditions + 10 / 3 epochs (CV) (bundle)	cvbase → cvctrl	-0.07	-0.21	-0.28	-0.13	[-0.41, -0.01], better in 5/5 folds	improves
C2	C	T1/T2 dropout + consistency λ 0.1 (CV)	cvctrl → cvaug	-0.03	0.04	0.02	0.07	[-0.16, +0.25], better in 2/5 folds	n.s.
C3	C	median of 5 fold models instead of one (cvctrl)	cvctrl single → cvctrl ensemble	–	-0.18	-0.22	-0.13	[-0.27, -0.09] (bootstrap)	improves
C3b	C	median of 5 fold models instead of one (cvbase)	cvbase single → cvbase ensemble	–	-0.05	-0.06	-0.04	[-0.11, +0.01] (bootstrap)	n.s.

Figure 5.1. Effect sizes

What is plotted. One row per tested change (Table 5.2). The point is how much that change moved the TUAB test MAE, Eq. (0.4):

$$\Delta\mathrm{MAE} = \mathrm{MAE}(\text{run with the change}) - \mathrm{MAE}(\text{run without it}),$$

in years, on all 274 test recordings. The two runs are named in brackets in the row label (baseline → variant).

This is a **difference of two errors, not a prediction error**. A point at −0.5 means the change made the model 0.5 years more accurate. It does not mean the model predicts ages 0.5 years too low.

- **Left of 0:** the change lowered the error, so it helped. Further left is better.
- **At 0:** no effect.
- **Right of 0:** the change raised the error, so it hurt.

Bars and colours.

- The horizontal bar is the noise band: how far two runs differing only by chance would land apart. It is $\pm\tau$ (Eq. (5.2)) for single runs and 3-seed pairs, and the 95 % interval (Eq. (5.4) or the bootstrap) for CV and ensemble rows.
- **Green (`improves`):** the whole band lies left of 0, so the gain is larger than seed noise.
- **Red (`hurts`):** the whole band lies right of 0.
- **Grey (`n.s.`):** the band crosses 0. The effect cannot be told apart from noise.

Changes beyond ±2 years (collapsed runs) are drawn at the edge, with their value written next to them. Panels:

- (a) Phase 1, TUAB-only runs.
- (b) Line removal, phase 2 and CV.

The signed drift of the predictions themselves, mean(predicted − true), where closer to 0 is better, is the bias of Eq. (0.3). It is shown per model and status in Table 3.4 (`bias`) and in Figure 3.1.

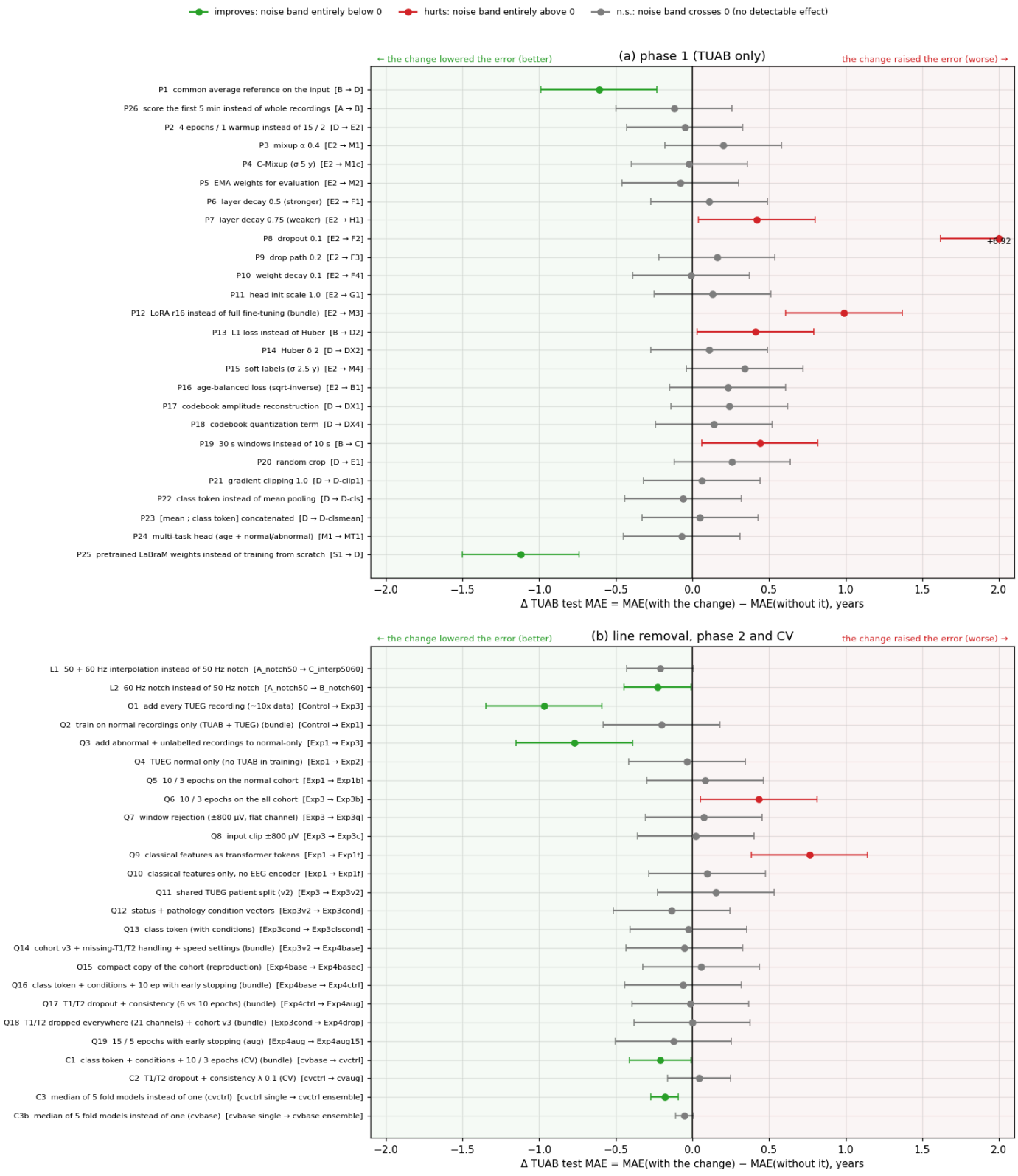


Table 5.3. Changes versus LaBraM: which mattered

The summary the study was after. One row per change from the LaBraM TUAB recipe to the best configuration (Tables 4.1 and 4.2), with:

- `estimate`: the Table 5.2 row that isolates the change best and its Δ test;

- `verdict` : that row's verdict;
- `in best config` : whether the change is part of `cvctrl` .

`not isolated` means no run changes it alone, so its effect is not known.

change vs LaBraM	estimate	Δ test	verdict	in best config
training data: TUAB → TUAB + TUEG (~10x recordings)	Q1: Control → Exp3	-0.97	improves	yes
abnormal and unlabelled recordings in training	Q3: Exp1 → Exp3	-0.77	improves	yes
pretrained LaBraM weights (vs from scratch)	P25: S1 → D	-1.12	improves	yes
common average reference on the input	P1: B → D	-0.61	improves	yes
mains: 50 + 60 Hz interpolation instead of 50 Hz notch	L1: A_notch50 → C_interp5060	-0.21	n.s.	yes
ensemble: median of the 5 fold models	C3: cvctrl single → cvctrl ensemble	-0.18	improves	yes
class token + condition vectors + 10 epochs (bundle)	C1: cvbase → cvctrl	-0.21	improves	yes
condition vectors alone (single split)	Q12: Exp3v2 → Exp3cond	-0.14	n.s.	yes
class-token pooling alone (TUAB / TUH)	P22: D → D-cls	-0.06	n.s.	yes
short schedule (4–10 epochs instead of 15–50)	P2: D → E2	-0.05	n.s.	yes
mixup α 0.4	P3: E2 → M1	0.20	n.s.	yes
cohort v3 + missing-T1/T2 handling + speed settings (bundle)	Q14: Exp3v2 → Exp4base	-0.05	n.s.	yes
5-minute evaluation window	P26: A → B	-0.12	n.s.	yes
T1/T2 channel-dropout augmentation	C2: cvctrl → cvaug	0.04	n.s.	no (cvaug)
longer schedule on the full cohort (10 epochs)	Q6: Exp3 → Exp3b	0.43	hurts	no
dropout 0.1	P8: E2 → F2	6.92	hurts	no
weaker layer decay 0.75	P7: E2 → H1	0.42	hurts	no
LoRA instead of full fine-tuning	P12: E2 → M3	0.99	hurts	no
30 s windows	P19: B → C	0.44	hurts	no
classical features as tokens	Q9: Exp1 → Exp1t	0.76	hurts	no
L1 loss	P13: B → D2	0.41	hurts	no
weight decay, drop path, layer decay 0.5, head init	P10: E2 → F4	-0.01	n.s.	no
loss shape: Huber δ 2, soft labels, age-balanced	P16: E2 → B1	0.23	n.s.	no
codebook regularization	P18: D → DX4	0.14	n.s.	no
signal QC: window rejection / ±800 µV clip	Q7: Exp3 → Exp3q	0.07	n.s.	no
multi-task head	P24: M1 → MT1	-0.07	n.s.	no
trim 60 s at both ends; lr 1e-4 (vs 5e-4); batch 128	–	–	not isolated	yes

6. Conclusions

Generated from the tables above, so they stay in step with the data when the notebook is re-run.

- **Noise floor.** Seed-to-seed sd of the TUAB test MAE is 0.14 years (Table 5.1), so two single runs must differ by more than 0.38 years to count (Eq. (5.2)). Of 51 tested changes, 7 improve, 7 hurt and 37 are within noise (Table 5.2).
- **What improved the error (Table 5.2):** P25 pretrained LaBraM weights instead of training from scratch (-1.12); Q1 add every TUEG recording (~10x data) (-0.97); Q3 add abnormal + unlabelled recordings to normal-only (-0.77); P1 common average reference on the input (-0.61); L2 60 Hz notch instead of 50 Hz notch (-0.23); C1 class token + conditions + 10 / 3 epochs (CV) (bundle) (-0.21); C3 median of 5 fold models instead of one (cvctrl) (-0.18).
- **What hurt:** P13 L1 loss instead of Huber (+0.41); P7 layer decay 0.75 (weaker) (+0.42); Q6 10 / 3 epochs on the all cohort (+0.43); P19 30 s windows instead of 10 s (+0.44); Q9 classical features as transformer tokens (+0.76); P12 LoRA r16 instead of full fine-tuning (bundle) (+0.99); P8 dropout 0.1 (+6.92).
- **Best configuration: `cvctrl`.** TUAB test MAE 7.28 ± 0.14 per fold model (normal 5.53, abnormal 9.37; Table 3.3). Its head is given the status/pathology vector.
- **Best EEG-only: `cvbase`.** 7.49 ± 0.09 per fold model, normal recordings 5.81, against 6.60 for the best published TUAB normal result (Gemein et al. 2024, different split and training data).
- **Ensembles.** The median of the 5 fold models scores 7.10 (`cvctrl`) and 7.44 (`cvbase`) on the TUAB test, normal 5.75 (`cvbase`).
- **Phase 4: adding HEEDB beats every TUH-only model.** H-E2L (TUH + HEEDB, EEG-only, epoch chosen on val) scores 6.54 on the TUAB test (normal 4.91, abnormal 8.49) and 7.02 on the HEEDB test patients, against 7.48 for the same recipe on TUH only (Table 2.11). Its normal-recording error is below the best published result (Gemein et al. 2024: 6.48 raw, 6.60 corrected; different training data). The clinical inputs (H-E2c) do not help once training converges.
- **Most of the gain over LaBraM comes from data.** The largest effects are:
  - more and more varied training recordings (Q1, Q3, and HEEDB in phase 4);
  - pretrained weights (P25);
  - the common average reference (P1). Architecture, loss and regularization changes are within noise or harmful (Table 5.3).
- **The abnormal cohort carries the error.** Abnormal recordings stay about 4 years worse than normal ones in every configuration (Tables 2.5, 3.3).

Caveats.

- **Selection on test:** phases 1 and 2 chose among configurations on the TUAB test set, so their best single-split numbers are optimistic. The CV and TUEG out-of-fold numbers are not.
- **Bundles:** C1, Q14, Q16, Q17 and Q18 change several things at once, so their parts are not separately known.

- **Condition vectors:** `cvctrl` and `cvaug` use the status/pathology vector at test time. Only `cvbase` is comparable with EEG-only brain-age models.
- **Phase 4 is one seed per run, and H-E2L changed batch size and learning rate after epoch 6** (Table 2.11). Its differences to H-E2 (about 0.2 years) are below the 2-run threshold of Eq. (5.2) (0.38 years) for single runs, so they are a trend, not a demonstrated effect. H-E2 vs `Exp4noearEEG` (0.8 years) is above it.
- **Noise floor scope:** it comes from one configuration, the E2 recipe on TUAB (Table 5.1), and is assumed to transfer to the larger cohorts.